

Nonlinear Nanophotonics in Lithium Niobate

by

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Dedicated to my family.

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Biographical Sketch

The author was born in Leshan City, Sichuan Province, China, in 1990. He attended the School of Physics at Peking University and graduated with a Bachelor of Science degree in 2013. With strong interest in optics and photonics, he started his graduate studies at The Institute of Optics, University of Rochester, in the fall of 2013. He conducted his doctoral research under the supervision of Professor Qiang Lin. His research interest was focused on utilizing high-quality nanophotonic devices to develop applications in sensing, nonlinear optics, and quantum optics based on the lithium niobate integrated photonic platform.

List of Publications

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6. H. Jiang, **R. Luo**, H. Liang, X. Chen, Y. Chen, and Q. Lin, "Fast response of photorefraction in lithium niobate microresonators," *Opt. Lett.* **42**, 3267-3270 (2017).
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Conference Presentations

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6. **R. Luo**, Y. He, H. Liang, M. Li, and Q. Lin, "Efficient tuning of second-harmonic generation in a lithium niobate nanophotonic waveguide," Conference on Lasers and Electro-Optics, JTu2A.79, San Jose, CA (2018).
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Abstract

With the rapid development of integrated photonics, multiple material platforms, including silicon, silica, diamond, silicon nitride, aluminum nitride, and gallium arsenide, have been widely studied for photonic applications on a chip scale. Particularly, integrated platforms have attracted considerable attention in nonlinear optics, due to boosted nonlinear optical interactions induced by the tight optical confinement.

Over the past few decades, lithium niobate (LiNbO_3 or LN), with its intriguing material properties, has been extensively studied for nonlinear optics. Recent advances in wafer bonding and etching technology have enabled the fabrication of high-quality LN nanophotonic devices, which have the potential for nonlinear optics with high efficiencies and novel functionalities. This thesis is devoted to the development of nonlinear optical applications based on the LN integrated photonic platform.

This thesis starts with a discussion of the thermo-optic property of LN, which is essential for realizing and tuning phase matching in nonlinear optical processes. The large thermo-optic birefringence of LN enables the demonstration of a self-referenced temperature sensor based on a microdisk resonator, exhibiting both a high sensitivity and a high resolution. The large thermo-optic birefringence is further exploited to demonstrate thermally

tunable second-harmonic generation in an LN nanophotonic waveguide, which simultaneously achieves a large tuning slope and a high conversion efficiency. In order to further increase the efficiency, semi-nonlinear nanophotonic waveguides are proposed and demonstrated as a universal design principle, which offers a large mode overlap by breaking the spatial symmetry of the optical nonlinearity, resulting in extremely efficient optical parametric generation.

Optical microresonators are widely employed to enhance nonlinear optical interactions. This thesis continues on to present applications of LN microresonators in nonlinear optics. First, cavity-enhanced second-harmonic generation and difference-frequency generation are demonstrated in an LN microring resonator with modal phase matching. Then, cyclic phase matching, due to the material anisotropy, is utilized in an X -cut LN microdisk to realize spontaneous parametric down-conversion with an extremely large bandwidth. Finally, Kerr frequency comb generation is demonstrated in an LN microring resonator with an optical quality factor of 2.5 million.

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This thesis work was supervised by a dissertation committee consisting of Professors Qiang Lin (advisor) and Hui Wu of the Department of Electrical and Computer Engineering, and Professors Wayne Knox and Nick Vamivakas of The Institute of Optics.

The fabrication process of nanophotonic devices on LNOI was developed by Dr. Wei C. Jiang, and optimized by Rui Luo, Hanxiao Liang, Yang He, and Mingxiao Li. For chapters 2 and 6, the experimental setup was primarily built by Professor Qiang Lin together with Dr. Jidong Zhang, Dr. Wei C. Jiang, and Dr. Xiyuan Lu, for testing suspended microresonators via evanescent coupling with tapered optical fibers. For chapter 5, the fabrication, characterization, and measurements for cavity-enhanced optical parametric generation were conducted by Rui Luo and Yang He. For chapter 7, the numerical simulations were performed by Yang He, and the fabrication, characterization, and measurements for Kerr frequency comb generation were conducted by Yang He and Rui Luo. Measurements in chapter 2 were done with the help of Haowei Jiang. Measurements in chapters 5 and 7 were done with the help of Jingwei Ling. Measurements in chapter 6 were done with the help of Haowei Jiang and Steven Rogers. All the other work conducted for this dissertation was done by Rui Luo independently.

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Chapter 1

Introduction

1.1 Optical Waveguides and Microresonators

Propagation of light has been an intriguing subject ever since the first sight of sparkling stars in the dark sky by humans. When light travels from one medium into a second medium with a lower refractive index, it can be completely reflected by the boundary. This phenomenon of total internal reflection, explained by Snell's law, gives rise to the guiding of electromagnetic waves in optical waveguides. Confined in these waveguides, optical fields can be conveniently manipulated, which facilitates the development of fundamental sciences and practical applications in optics. The most widely used type of optical waveguides is the optical fiber, which has been the backbone of the optical telecommunication technology in the information era. Optical waveguides are also extensively studied as key components in integrated optics. Being able to confine optical fields at the micro/nanoscale, waveguides on integrate photonic platforms facilitates complex and flexible control of light on a chip scale with an enormous integration density.

By making a waveguide into a closed loop, an optical resonator is produced, inside of which a photon can travel multiple round trips. Resonators offer long propagation with small footprints, and are thus commonly used to accumulate optical path length and boost optical intensity.

1.1.1 Optical Modes of Waveguides

For a straight optical waveguide that is translational symmetric in the propagation direction y , as shown in Fig. 1.1, there can exist optical modes that is unchanged upon propagation. These modes can be described by the wave theory of optics.

From Maxwell's equations, in nonmagnetic, nonconducting, uniform media with no free charges, the electric field vector, $\tilde{\mathbf{E}}(\mathbf{r}, t)$, can be generally described by the following equation

$$\nabla \times [\nabla \times \tilde{\mathbf{E}}(\mathbf{r}, t)] + \frac{1}{c^2} \epsilon(\mathbf{r}, \omega) \frac{\partial^2}{\partial t^2} \tilde{\mathbf{E}}(\mathbf{r}, t) = 0, \quad (1.1)$$

with the boundary condition

$$\mathbf{n} \times \tilde{\mathbf{E}} = 0, \text{ on } \partial\Omega, \quad (1.2)$$

where $\nabla = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})$, $\epsilon(\mathbf{r}, \omega)$ is the relative permittivity tensor, ω is the angular optical frequency, c is the speed of light in vacuum, and $\mathbf{n}$ is the normal vector on the boundary $\partial\Omega$. Since $\nabla \times [\nabla \times \tilde{\mathbf{E}}] = \nabla(\nabla \cdot \tilde{\mathbf{E}}) - \nabla^2 \tilde{\mathbf{E}}$,

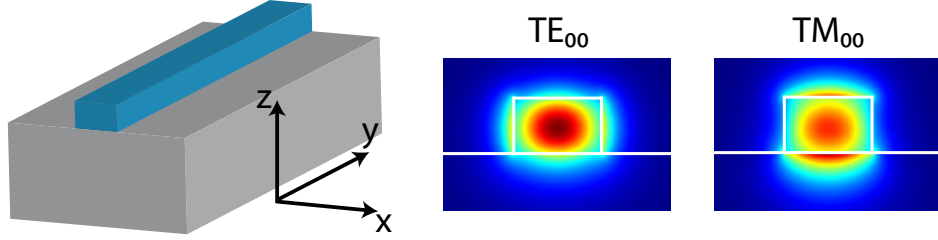


Figure 1.1: Schematic of a nanophotonic waveguide, and mode profiles of the fundamental modes.

and $\nabla \cdot \tilde{\mathbf{D}} \equiv \nabla \cdot (\epsilon \tilde{\mathbf{E}}) = \epsilon(\nabla \cdot \tilde{\mathbf{E}}) = 0$ when the permittivity exhibits no anisotropy, the electric field in isotropic media can be simplified to the basic wave equation

$$\nabla^2 \tilde{\mathbf{E}}(\mathbf{r}, t) - \frac{\epsilon(\mathbf{r}, \omega)}{c^2} \frac{\partial^2}{\partial t^2} \tilde{\mathbf{E}}(\mathbf{r}, t) = 0, \quad (1.3)$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$, and the relative permittivity $\epsilon(\mathbf{r}, \omega)$ is now simply a scalar. Since the waveguide exhibits translational symmetry in the y -direction, the electric field can be written as a superposition of traveling-wave modes,

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \sum_j B_j \mathbf{E}_j(x, z) e^{i\beta_j y - i\omega t}, \quad (1.4)$$

where B_j is the field amplitude, $\mathbf{E}_j(x, z)$ is the electric field profile in the waveguide cross section ($x-z$ plane in Fig. 1.1), and β_j is the propagation constant of the j -th mode. By substituting Eq. (1.4) into Eq. (1.1), the waveguide modes can be shown to satisfy an equation with only two independent spatial variables, x and z , written as

$$-\{\nabla[\nabla \cdot (\mathbf{E}_j e^{i\beta_j y})]\} e^{-i\beta_j y} + \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \beta_j^2 \right) \mathbf{E}_j + \frac{\omega^2}{c^2} \epsilon \mathbf{E}_j = 0, \quad (1.5)$$

where

$$\{\nabla[\nabla \cdot (\mathbf{E}_j e^{i\beta_j y})]\}_x e^{-i\beta_j y} = \frac{\partial^2 E_{jx}}{\partial x^2} + i\beta_j \frac{\partial E_{jy}}{\partial x} + \frac{\partial^2 E_{jz}}{\partial x \partial z}, \quad (1.6)$$

$$\{\nabla[\nabla \cdot (\mathbf{E}_j e^{i\beta_j y})]\}_y e^{-i\beta_j y} = i\beta_j \frac{\partial E_{jx}}{\partial x} - \beta_j^2 E_{jy} + i\beta_j \frac{\partial E_{jz}}{\partial z}, \quad (1.7)$$

$$\{\nabla[\nabla \cdot (\mathbf{E}_j e^{i\beta_j y})]\}_z e^{-i\beta_j y} = \frac{\partial^2 E_{jx}}{\partial x \partial z} + i\beta_j \frac{\partial E_{jy}}{\partial z} + \frac{\partial^2 E_{jz}}{\partial z^2}. \quad (1.8)$$

For isotropic media, Eq. (1.5) is simplified to

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \beta_j^2\right)\mathbf{E}_j(x, z) + \epsilon(x, z, \omega)\frac{\omega^2}{c^2}\mathbf{E}_j(x, z) = 0. \quad (1.9)$$

Generally, Eq. (1.5) or Eq. (1.9) does not have analytical solutions, and numerical methods are often used to solve for the waveguide modes. A common approach is the finite-element method (FEM), which is able to numerically give the effective index, $n_{\text{eff}} = \beta c/\omega$, as well as the mode field profile. In this thesis, a commercially available software, COMSOL Multiphysics, is used for numerical simulations of optical modes with the FEM.

For illustration, Fig. 1.1 shows electric field profiles of the fundamental quasi-transverse-electric mode (TE_{00}) and the fundamental quasi-transverse-magnetic mode (TM_{00}) of a strip waveguide. The polarization primarily lies in the device plane for quasi-TE modes, and lies perpendicular to the device plane for quasi-TM modes. Similar to an optical fiber [1], a waveguide only supports a single mode in each polarization at a certain wavelength, when the

core exhibits a small enough cross section and a small enough index contrast to the cladding/substrate; with an increased cross sectional area of the core and a larger index contrast, at some point the waveguide exhibits multiple modes in each polarization. High-order modes have profiles with multiple field maxima in the x - and/or z -directions.

1.1.2 Optical Modes of Microresonators

Generally, the electric field in an optical microresonator also obeys Eq. (1.1). However, Eq. (1.1) is a three-dimensional differential equation, which usually consumes an enormous amount of computational resources if solved directly with the FEM. Fortunately, in some cases, mode solving can be simplified, as described in the following.

For microresonators exhibiting rotation symmetry, such as a microdisk (see Fig. 1.2, left), a microring (see Fig. 1.2, right), a microtoroid, a microsphere, and a microtube, the electric field of a cavity mode can be represented in the cylindrical coordinates as

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \mathbf{E}(\rho, z)e^{im\phi - i\omega t}, \quad (1.10)$$

where $\mathbf{E}(\rho, z)$ is the electric field profile in the cross section at a certain azimuthal angle ϕ and m is the azimuthal mode order. Thus, similar to the case of a waveguide, Eq. (1.1) reduces to a two-dimensional form in the $\rho - z$ space, which can be solved by the FEM with a reasonable amount

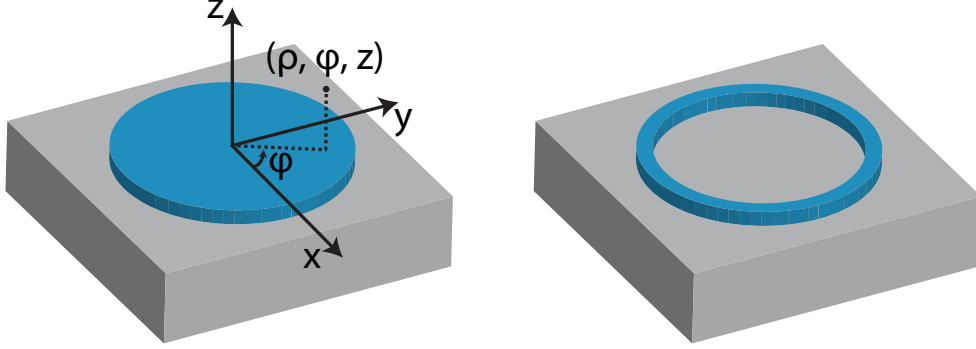


Figure 1.2: Schematic of a microdisk and a microring resonator.

of computational power achievable by common personal computers [2]. In the simple case of isotropic media, optical modes in rotationally symmetric resonators are described by

$$\left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{\partial^2}{\partial z^2}\right) \mathbf{E}(\rho, z) + [\epsilon(\rho, z, \omega) \frac{\omega^2}{c^2} - \frac{m^2}{\rho^2}] \mathbf{E}(\rho, z) = 0. \quad (1.11)$$

For microresonators formed by closed-loop uniform waveguides with large-radius bending, such as microring and racetrack resonators, the electric field in a certain cross section can be approximated by corresponding waveguide modes. When the bending radius becomes smaller, the electric field tends to stay further from the center of curvature, and can be simulated according to Eq. (1.5) or Eq. (1.9), while using a conformal transformation of $\epsilon(\rho, z, \omega)$ to effectively take the bending into account [3].

1.1.3 Optical Loss and Quality Factor

Light experiences loss when propagating in waveguides or microresonators, due to material absorption, scattering, radiation, and external coupling. In the case of a waveguide, the propagation loss is related to the imaginary part of the propagation constant. For a microresonator, the optical power loss can be described by a total decay rate $\kappa_t = \kappa_0 + \kappa_e$, where κ_0 and κ_e are decay rates due to the intrinsic loss and the external coupling loss, respectively. The life time of a photon in the cavity is finite due to these losses, and is written as

$$\tau = \frac{1}{\kappa_t}. \quad (1.12)$$

A figure of merit for the loss of an optical resonator, the quality factor, is defined as the ratio between the optical frequency and the decay rate [4],

$$Q_j = \frac{\omega}{\kappa_j}, (j = 0, e, t), \quad (1.13)$$

with the relation between the loaded, intrinsic, and external quality factor written as

$$\frac{1}{Q_t} = \frac{1}{Q_0} + \frac{1}{Q_e}. \quad (1.14)$$

The higher the loaded quality factor, Q_t , the larger the accumulated phase is within the cavity photon life time.

In a microring resonator, the intrinsic optical quality factor, Q_0 , is closely related to the propagation loss in a straight waveguide with the same cross

section, written as [5]

$$\alpha = \frac{2\pi n_g}{Q_0 \lambda_0}, \quad (1.15)$$

where α is the loss coefficient of optical power, with $\alpha/2$ being the imaginary part of β , n_g is the group index with $n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}$, and λ_0 is the center wavelength of the resonance. Thus, the propagation loss of a waveguide can be experimentally obtained by measuring the intrinsic quality factor of a microring with the same cross section, given a negligible bending loss.

1.2 Lithium Niobate

1.2.1 Material Properties

Lithium niobate (LiNbO_3 or LN) is a compound dielectric, and it has strong electro-optic, nonlinear optical, acousto-optic, piezoelectric, and photorefractive effects, as well as a high mechanical rigidity [6], making it widely used in optical, electrical, and mechanical applications [7–10].

In particular, LN has advantageous properties for nonlinear optics, compared with other common nonlinear photonic materials, as shown in Table 1.1. In terms of linear optical properties, the large bandgap (4 eV) of LN offers a wide transparency window from ultraviolet to mid-infrared ($0.35 - 5 \mu\text{m}$), allowing for optical applications in a large wavelength range. Furthermore, unlike silicon (Si) and gallium arsenide (GaAs) that have narrow bandgaps, LN with a wide bandgap does not suffer from two-photon

Table 1.1: Properties of some nonlinear optical materials

Material	Bandgap (eV)	Refractive index (@1550 nm)	$\chi^{(2)}$ (pm/V)	n_2 (10^{-15} cm ² /W)
Si [11]	1.1	3.48	0	45
SiO ₂ [12]	7.8	1.44	0	0.22
Diamond [13]	5.5	2.39	0	1.3
SiN [14, 15]	4	1.98	0.04	2.5
AlN [16, 17]	6	2.12	5	2.3
LN [12, 18]	4	2.21 (o), 2.14 (e)	54	0.9
GaAs [19, 20]	1.4	3.37	240	160

absorption at wavelengths longer than the visible waveband, allowing for high-power operation of nonlinear optical processes. Meanwhile, LN is a uniaxial crystal exhibiting negative optical birefringence, with $n_o > n_e > 2.1$ in the telecom band, where n_o and n_e are the refractive indices of ordinary and extraordinary light, respectively. Such high refractive indices make LN suitable for tight optical confinement in integrated photonic devices.

Apart from its wide bandgap and high refractive indices, LN also exhibits strong optical nonlinearities. Single-crystalline LN does not exhibit inversion symmetry, and it has an outstanding quadratic nonlinear susceptibility $\chi^{(2)}$, with the largest component of $\chi_{max}^{(2)} = 2d_{33} = 54$ pm/V for interactions between extraordinarily polarized photons. Meanwhile, it also has a large cubic nonlinearity [12], with a Kerr nonlinearity n_2 that is comparable to common integrated nonlinear photonic materials such as diamond [13], silicon nitride (SiN) [14] and aluminum nitride (AlN) [17], all of which have much larger n_2 than silica (SiO₂) [12]. Although Si and GaAs have remarkably

large n_2 , two-photon absorption caused by the small bandgaps limits their application at near-infrared and telecom-band wavelengths.

1.2.2 Optical Platforms

Conventionally, bulk crystals of LN are employed for guiding of optical waves. By using reverse proton exchange (RPE), optical waveguides can be produced with a refractive index contrast of ~ 0.1 between the modified core and the intrinsic substrate, and the optical field is weakly guided with a mode area of $\sim 10 \mu\text{m}^2$ [21]. This platform has been the basis of high-speed LN modulators widely applied for optical telecommunications [9].

LN exhibits a large quadratic nonlinearity, and thus is a good option for quadratic optical parametric generation. To realize substantial optical parametric processes, the phase-matching condition needs to be satisfied [22]. For example, exact phase matching of second-harmonic generation requires an identical effective index, or phase velocity, at the fundamental frequency and its second harmonic. Phase matching can be achieved with different methods, depending on the optical platform [22]. For LN, quasi-phase matching with nonlinear gratings has been a dominant approach. Since LN is a ferroelectric material, the sign of $\chi^{(2)}$ can be permanently flipped by strong electric fields. In order to compensate for the phase mismatch induced by material and geometrical dispersion, $\chi^{(2)}$ can be periodically flipped between positive and negative to produce a nonlinear grating [23]. This technique is called periodic poling, and periodically poled LN (PPLN) has been a mature

platform for various applications in nonlinear and quantum optics through wavelength conversion [8, 24–26].

Recently, due to the advances in wafer bonding technology, single-crystalline LN-on-insulator (LNOI) wafers made from a smart-cut process are commercially available [27]. These wafers have a thin-film LN layer with a thickness of under $1\text{ }\mu\text{m}$, and a silicon oxide layer with a thickness of a few microns, sitting on a substrate made of LN, Si, quartz, or fused silica. In contrast to conventional RPE LN ($\Delta n \sim 0.1$), LNOI offers a much larger index contrast ($\Delta n \sim 0.7$ between LN and SiO_2), and thus enables tight mode confinement with a substantially reduced mode area of $\sim 1\text{ }\mu\text{m}^2$.

Due to its chemical inertia, LN has long been considered hard to etch. Fortunately, recent developments in fabrication technology have led to remarkable improvement in the optical quality of LN nanophotonic devices, opening up numerous applications on the LN integrated photonic platform. For example, by loading a high-index waveguide (e.g., made of SiN or Si) on top of an LNOI wafer, the electro-optic and nonlinear optical properties of LN can be utilized without etching LN thin films [28–30]. In addition, femtosecond laser pulses can be employed to fabricate LN devices through micromachining [31]. In order to achieve integrated photonic circuits with tighter mode confinement and scalable fabrication, it is desired to perform direct etching with lithographically defined masks. Considerable progress has been reported on high-quality etching of LN [32–38], making direct etching a practical and meaningful approach for fabricating LN nanophotonic devices.

1.3 Thesis Outline

This thesis is devoted to studying nonlinear optics in nanophotonic devices on the LN integrated photonic platform. The motivation mainly comes from two factors. First, integrated photonics offers enhanced nonlinear optical effects due to the tight mode confinement, and the LN integrated platform is particularly promising due to the outstanding linear and nonlinear optical properties. Second, nanophotonic devices exhibit multiple degrees of freedom for flexible engineering, which can potentially support novel functionalities utilizing nonlinear optical interactions in LN.

Chapter 2 discusses the thermo-optic property of LN, which is an important control mechanism of nonlinear optical processes. A self-referenced temperature sensor based on an LN microdisk resonator is demonstrated. The large thermo-optic birefringence ($\frac{dn_e}{dT} - \frac{dn_o}{dT}$) of LN is utilized to induce a large differential frequency shift between ordinary and extraordinary cavity modes upon temperature change, which enables a temperature sensor referenced by itself, without the need for external wavelength calibration. The LN microdisk sensor exhibits a temperature sensitivity of 0.834 GHz/K and a resolution of 0.8 mK.

Chapter 3 demonstrates thermal tuning of efficient second-harmonic generation in an LN nanophotonic waveguide by combining the strong quadratic optical nonlinearity with the large thermo-optic birefringence. In type-I second-harmonic generation involving optical modes with orthogonal

polarizations, there exists a large differential thermo-optic response between the effective indices, leading to a phase-matching window sensitively dependent on the device temperature. A tuning slope of 0.84 nm/K and a normalized conversion efficiency of 7.3% $\text{W}^{-1}\text{cm}^{-2}$ are measured for a telecom-band pump.

Chapter 4 describes the path toward extremely efficient optical parametric generation. As a common technique for phase matching on integrated platforms, modal phase matching between different-order optical modes suffers from poor mode overlap, limiting the nonlinear conversion efficiency from being further increased. Semi-nonlinear waveguides are proposed to optimize the mode overlap by breaking the spatial symmetry of the optical nonlinearity. Highly efficient second-harmonic generation is demonstrated in a heterogeneous waveguide made of titanium oxide and LN, exhibiting a conversion efficiency of 2900% $\text{W}^{-1}\text{cm}^{-2}$ in theory and 650% $\text{W}^{-1}\text{cm}^{-2}$ in experiment, well beyond what is achievable by conventional LN nanophotonic waveguides with modal phase matching.

Chapter 5 presents cavity-enhanced optical parametric generation in an LN microring with modal phase matching. By carefully designing a bus waveguide, phase-matched cavity modes can be effectively coupled at both the fundamental wavelength in the telecom band and the second harmonic in the near-infrared, resulting in second-harmonic generation with a conversion efficiency of 1500% W^{-1} . In addition, difference-frequency generation is also observed in the same device.

Chapter 6 demonstrates broadband spontaneous parametric down-conversion in an X -cut LN microdisk. In an optical resonator made of an anisotropic material, a cavity mode can experience sinusoidal oscillation in the local effective index at different azimuthal angles, with the oscillation amplitude determined by the material birefringence. This leads to cyclic phase matching, which is able to induce nonlinear optical interactions in a large spectral range. Spontaneous parametric down-conversion with a bandwidth of over 400 nm is experimentally demonstrated.

Chapter 7 presents preliminary results of Kerr frequency comb generation in an LN microring resonator, due to the cubic optical nonlinearity of LN. The microring exhibits anomalous group velocity dispersion and a loaded optical Q of 2.5×10^6 , which enable Kerr frequency comb generation with an on-chip pump power of only 33 mW in the telecom band. Meanwhile, an up-converted comb is also observed in the near-infrared due to the strong quadratic nonlinearity.

Chapter 2

Self-Referenced Temperature Sensing with Thermo-Optic Birefringence

Nonlinear optical parametric processes rely critically on the phase-matching condition [22]. Thermo-optic effect, i.e., temperature dependence of refractive index ($\frac{dn}{dT}$), has been a dominant control mechanism for phase matching in many nonlinear optical devices [8, 39]. Thus, in order to achieve phase-matched nonlinear optics on a chip scale, understanding thermal responses of nanophotonic devices is a fundamental step. Optical microresonators, especially those with high Q 's or narrow linewidths, make it convenient to study thermo-optic properties of materials, since the temperature-induced index change manifests as a resonance frequency shift that is directly observable.

Externally induced resonance shift has led to a variety of sensing applications based on optical microresonators [40, 41]. In particular, temperature-induced resonance shift has enabled temperature sensing with microresonators on many device and material platforms, from polymer and silica microspheres [42–45], silica microtoroids [46], silica microbubbles

[47], liquid microspheres [48], to silicon microring resonators [49, 50]. As this approach relies on the direct mapping of temperature from the cavity resonance wavelength, its practical implementation requires external wavelength calibration.

An alternative approach is to utilize the temperature-induced differential frequency shift (DFS) between two resonances of the same optical resonator [51]. This approach does not require wavelength calibration of individual cavity resonances, and thus forms an elegant method for self-referenced temperature sensing. Dispersion and birefringence of thermal refraction can introduce temperature-dependent DFS between two-color cavity resonances [52, 53] and two polarization modes [51, 54], respectively, and thus can both be employed for this purpose. The dispersion of thermo-optic effect is generally fairly small, and requires either far separated two-color resonances (e.g., fundamental and second harmonic [53]) or a second mechanism (e.g., optomechanics [52]) to enhance the sensing signal. Thermo-optic birefringence, in contrast, can be quite pronounced for crystalline media [55], and hence conveniently offers a flexible approach for self-referenced temperature sensing, which has been demonstrated recently in magnesium fluoride (MgF_2) resonators [54, 56, 57].

In this chapter, self-referenced temperature sensing is presented in an LN high- Q microresonator, by taking advantage of the strong thermo-optic birefringence of the LN material. The device exhibits a large temperature sensitivity of 0.834 GHz/K, about 1 order of magnitude larger than other

self-referenced sensors [54, 56, 57]. This high sensitivity allows for the measurement of a temperature uncertainty of 0.8 mK. In contrast to bulky optical resonators [54, 56, 57], the demonstrated microscale device has the potential to be integrated in future photonic circuits. Furthermore, this work paves the way toward thermally controlled nonlinear optical applications on the LN integrated photonic platform.

2.1 Thermo-Optic Birefringence of LN

It is well known that uniaxial crystals have birefringence in the refractive index, i.e., $n_e \neq n_o$. In general, they also exhibit birefringence in the thermo-optic coefficient, i.e., $\frac{dn_e}{dT} \neq \frac{dn_o}{dT}$ [55]. For congruent single-crystalline LN, the thermo-optic coefficients of ordinary light (o-light) and extraordinary light (e-light) at the telecom band are given as [58]

$$\frac{dn_o}{dT} = (-0.9 + 3 \times 10^{-3}T) \times 10^{-5}\text{K}^{-1}, \quad (2.1)$$

$$\frac{dn_e}{dT} = (-2.6 + 19.8 \times 10^{-3}T) \times 10^{-5}\text{K}^{-1}, \quad (2.2)$$

where T is temperature in Kelvin. Equations (2.1)-(2.2) are plotted in Fig. 2.1 for a better comparison. At room temperature, the thermo-optic coefficient of o-light is almost 0, while that of e-light is about $3 \times 10^{-5}\text{K}^{-1}$. Also, both thermo-optic coefficients, as well as their difference, increase monotonically with temperature.

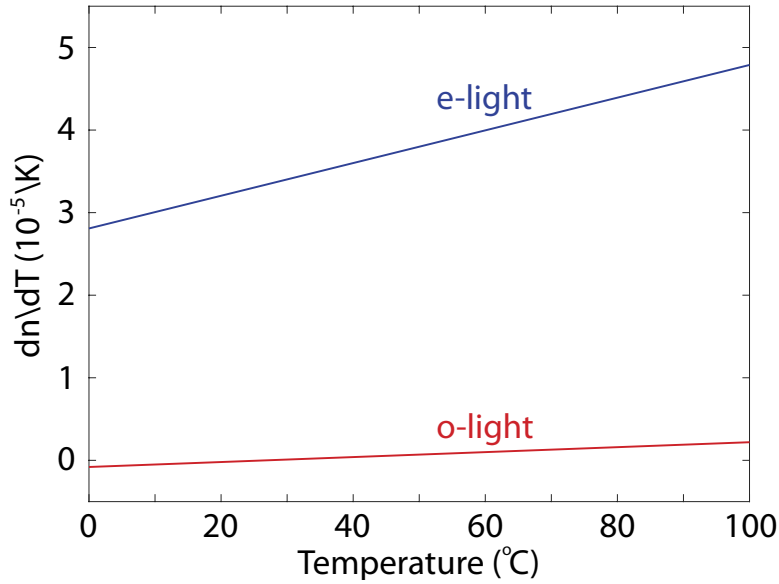


Figure 2.1: Temperature dependence of thermo-optic coefficients of LN in the telecom band.

Interestingly, LN exhibits a remarkably large thermo-optic birefringence among a variety of optical media [55]. For example, Table 2.1 lists the thermo-optic coefficients of three materials, Si, MgF_2 , and LN. Si exhibits a very large thermo-optic coefficient, $\frac{dn}{dT} = 2 \times 10^{-4} K^{-1}$, making it a good option for temperature sensing by directly measuring the thermal response of resonance wavelengths in microresonators [49, 50]. However, since Si is an isotropic material with no thermo-optic birefringence, it is not a good option for self-referenced sensing. On the other hand, both of the other two materials, MgF_2 and LN, exhibit nonzero differences in the thermo-optic coefficients of o-light and e-light, with LN having a remarkably large thermo-optic birefringence, almost 2 orders of magnitude larger than that of MgF_2 .

Table 2.1: Thermo-optic coefficients of some materials in the telecom band ($T = 300$ K)

Material	$\frac{dn_o}{dT}$ (10^{-6} K $^{-1}$)	$\frac{dn_e}{dT}$ (10^{-6} K $^{-1}$)	$ \frac{dn_e}{dT} - \frac{dn_o}{dT} $ (10^{-6} K $^{-1}$)
Si [59]	200	200	0
MgF ₂ [51]	0.6	0.25	0.35
LN [58]	0	33.4	33.4

The strong thermo-optic birefringence of LN implies its potential for sensitive self-referenced temperature sensing. This is particularly enabled by recent advances in the fabrication of LN nanophotonic devices, producing high- Q LN microresonators on a chip-scale platform [31, 35, 37, 38].

2.2 Design of a Microdisk Temperature Sensor

A Z -cut LN microdisk resonator is designed for self-referenced temperature sensing. The microdisk is suspended in air, with a radius of $25\ \mu\text{m}$ and a thickness of $400\ \text{nm}$ (see Fig. 2.2). The device exhibits quasi-TE and quasi-TM cavity modes, with electric fields dominantly in parallel and perpendicular to the device plane, respectively (see Fig. 2.2, insets). For a Z -cut LN microdisk, the optic axis aligns normal to the device plane. Thus, a quasi-TE cavity mode is primarily ordinarily polarized, and a quasi-TM cavity mode is primarily extraordinarily polarized, resulting in a maximal DFS, which is ideal for self-referenced temperature sensing.

To explore the sensing potential of this device, FEM simulations are performed to obtain the thermal response of the cavity resonances,

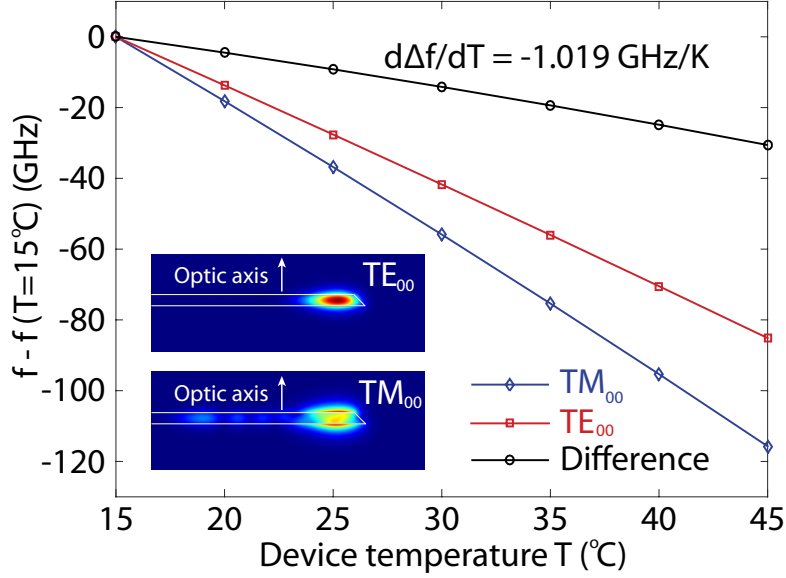


Figure 2.2: Numerically simulated temperature-dependent resonance frequencies of the TM_{00} and the TE_{00} mode of a Z-cut LN microdisk. For simple illustration, the resonance frequencies are plotted in reference to those at 15°C . The black curve shows the frequency difference between the two modes. The device has a radius of $25\ \mu\text{m}$, a thickness of $400\ \text{nm}$, and a slant angle of 45° on the sidewall, introduced by the etching process. Symbols are simulation results and solid lines are linear fittings. The TM_{00} and TE_{00} modes have azimuthal mode numbers of 151 and 189, respectively, with resonance wavelengths both close to $1500\ \text{nm}$. Insets show their mode profiles of the electric field.

considering both thermo-optic [58] and thermal expansion effects [60]. Figure 2.2 shows the simulated temperature dependence of cavity resonance frequencies, f_{TE} and f_{TM} , of the fundamental quasi-TE mode (TE_{00}) and the fundamental quasi-TM mode (TM_{00}) at the telecom band, respectively. In general, both resonances shift toward lower frequencies (longer wavelengths) with increased device temperature, because of thermo-optic and thermal

expansion effects. However, due to the significantly larger thermo-optic effect of e-light, the quasi-TM mode, with a dominant extraordinary polarization, shifts by a larger amount. Thus, the frequency separation between the two modes, $\Delta f = f_{\text{TM}} - f_{\text{TE}}$, depends sensitively on temperature, resulting in a considerable DFS with a slope of $\frac{d\Delta f}{dT} = -1.019$ GHz/K. This implies that a 1-mK drift of the device temperature results in a DFS of about 1 MHz, which can be conveniently resolved by a high- Q microresonator, as shown in the following.

2.3 Fabrication of LN Nanophotonic Devices

With the design and simulation completed, the microdisk temperature sensor is fabricated with a standard top-down process, as shown in Fig. 2.3. The fabrication starts from an LNOI wafer by NANOLN [61], with a 400-nm-thick

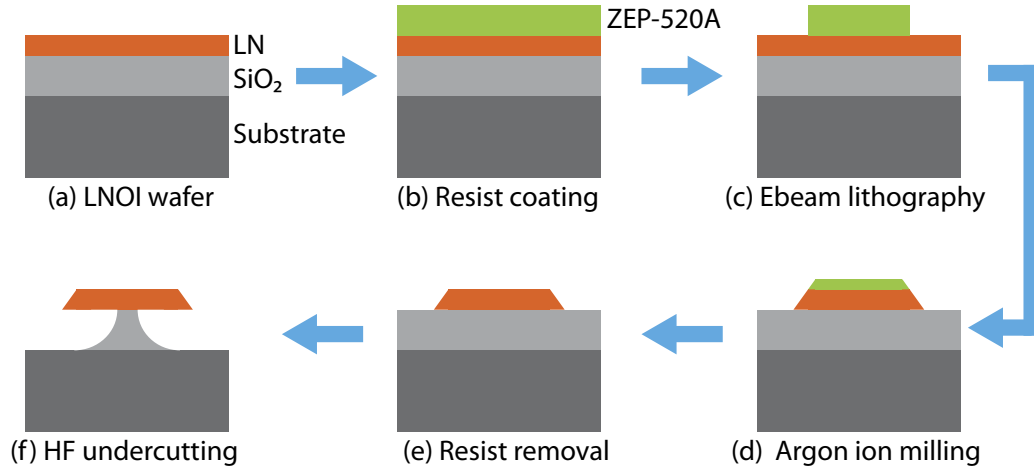


Figure 2.3: Fabrication of LN microdisk resonators.

LN thin film sitting on a 2- μm -thick buried silicon oxide layer and an LN substrate. ZEP520A resist is coated on top of the wafer with double spinning, leaving a resist thickness of about 1 μm . Next, electron-beam lithography is employed for device patterning, and Argon ion milling is used to etch the LN thin film. Then, the wafer is treated with oxygen plasma to remove the residual resist. Finally, the device is immersed in diluted hydrofluoric acid to undercut the oxide, leaving the microdisk suspended in air and supported by a silicon oxide pedestal. A typical representation of the fabricated devices is shown in Fig. 2.4(b). Two suspended forks are fabricated to give mechanical support to a tapered optical fiber used for later testing.

2.4 Experimental Results

To demonstrate self-referenced temperature sensing with the fabricated device, experiments are conducted with the setup shown in Fig. 2.4(a). To probe the temperature-induced DFS, light from a continuous-wave (CW) tunable telecom-band laser is split into two parts. A large portion is coupled to the LN microdisk via evanescent coupling with a tapered optical fiber, to probe the cavity modes. A small portion is coupled to a fiber-based Mach-Zehnder interferometer as a reference that measures the differential frequency, Δf , between two cavity modes with orthogonal polarizations. The device chip is placed on a thermoelectric cooler whose temperature is stabilized by a temperature controller with a feedback loop (Newport 350B,

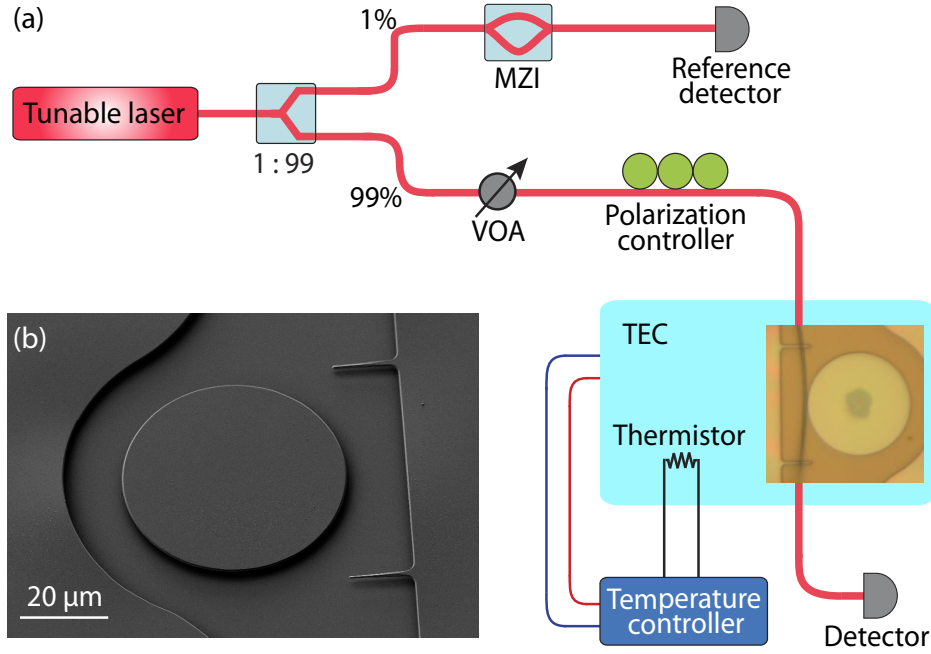


Figure 2.4: (a) Schematic of the experimental setup. MZI: Mach-Zehnder interferometer; VOA: variable optical attenuator; TEC: thermoelectric cooler. (b) Scanning electron microscopy image of a typical fabricated LN microdisk.

with a short-term stability of 1 mK). A variable optical attenuator is used to control the laser power coupled into the microdisk, and a fiber polarization controller is employed for adjustment of the coupled polarization. The device operates in the atmospheric environment.

Two cavity resonances around 1502 nm are selected, one being quasi-TE and the other quasi-TM. Their resonance frequencies are close to each other, so both cavity resonances can lie in the fine tuning range of the tunable laser. Figure 2.5(a) shows the laser-scanned transmission spectra of the two selected cavity resonances. The quasi-TE and quasi-TM modes are

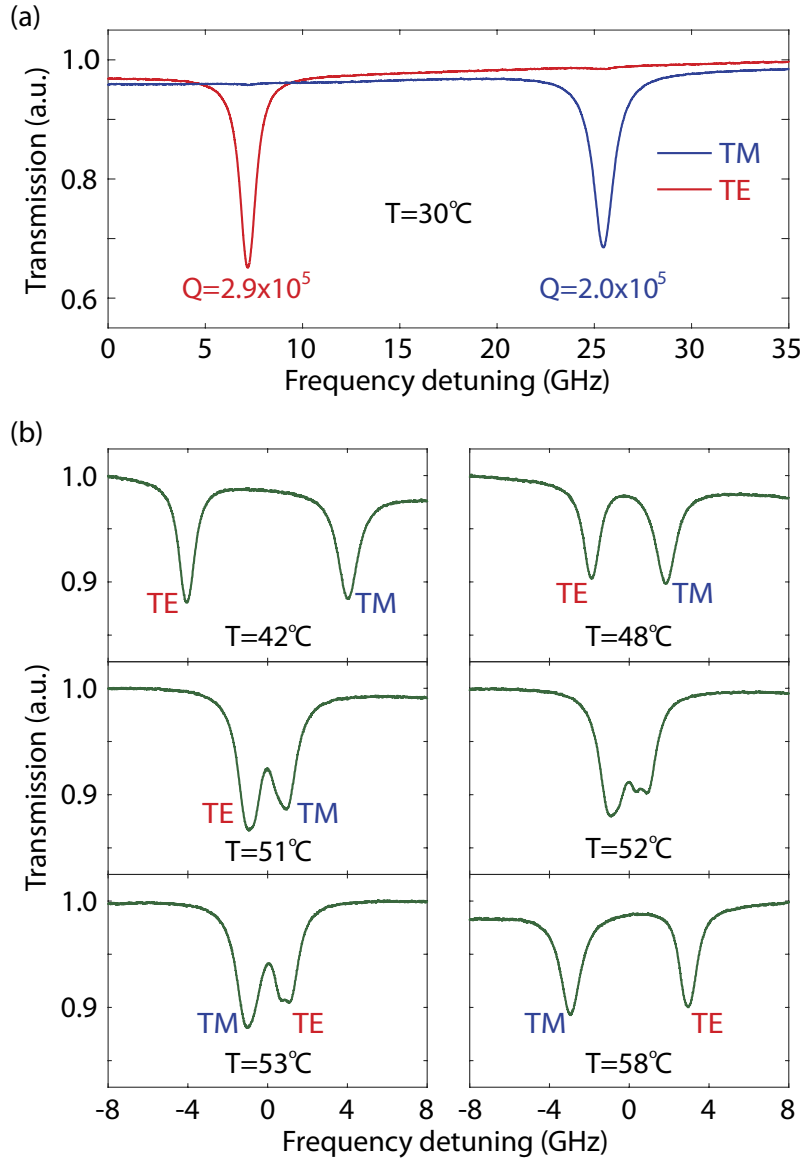


Figure 2.5: (a) Transmission spectra of two orthogonal polarizations, showing a quasi-TE and a quasi-TM resonance, at 30°C . The wavelengths of the two modes are around 1502 nm. (b) Transmission spectra at different temperatures. The polarization of the input light is set such that the two modes have similar coupling depths.

both under-coupled and exhibit loaded optical Q 's of 2.9×10^5 and 2×10^5 , respectively, corresponding to cavity linewidths of 0.7 and 1 GHz. At 30°C, the quasi-TM mode has a resonance frequency 18 GHz higher than the quasi-TE mode. When the device temperature increases, both cavity resonances, f_{TM} and f_{TE} , shift toward longer wavelengths (not shown in Fig. 2.5). However, their frequency separation, $\Delta f = f_{\text{TM}} - f_{\text{TE}}$, decreases until they cross over each other [see Fig. 2.5(b)]. Figure 2.6(a) shows the recorded temperature dependence of Δf . As expected, Δf decreases linearly with increased temperature until about 45°C, above which the coupling between the two modes introduces nonlinear temperature dependence of Δf . In the linear regime where the sensor operates, the device exhibits a temperature sensitivity of $\frac{d\Delta f}{dT} = -0.834$ GHz/K, which agrees with the theoretical expectation shown in Fig. 2.2. The experimentally recorded $\frac{d\Delta f}{dT}$ is about 1 order of magnitude larger than self-referenced sensors based on other material platforms [53, 54, 56, 57], showing the potential of LN devices for self-referenced temperature sensing with a high sensitivity.

To characterize the sensing resolution, the temperature is set at 30°C and an optical power of 1.5 μW is launched into the device to continuously monitor Δf . The optical power is high enough to achieve a large enough signal-to-noise ratio at the detection of the cavity transmission, while low enough to induce negligible nonlinear optical effects. Repeated measurements are conducted at a frequency of 50 Hz, with in total 32768 continuous measurements of Δf . Figure 2.6 (b) shows the Allan deviation of the detected

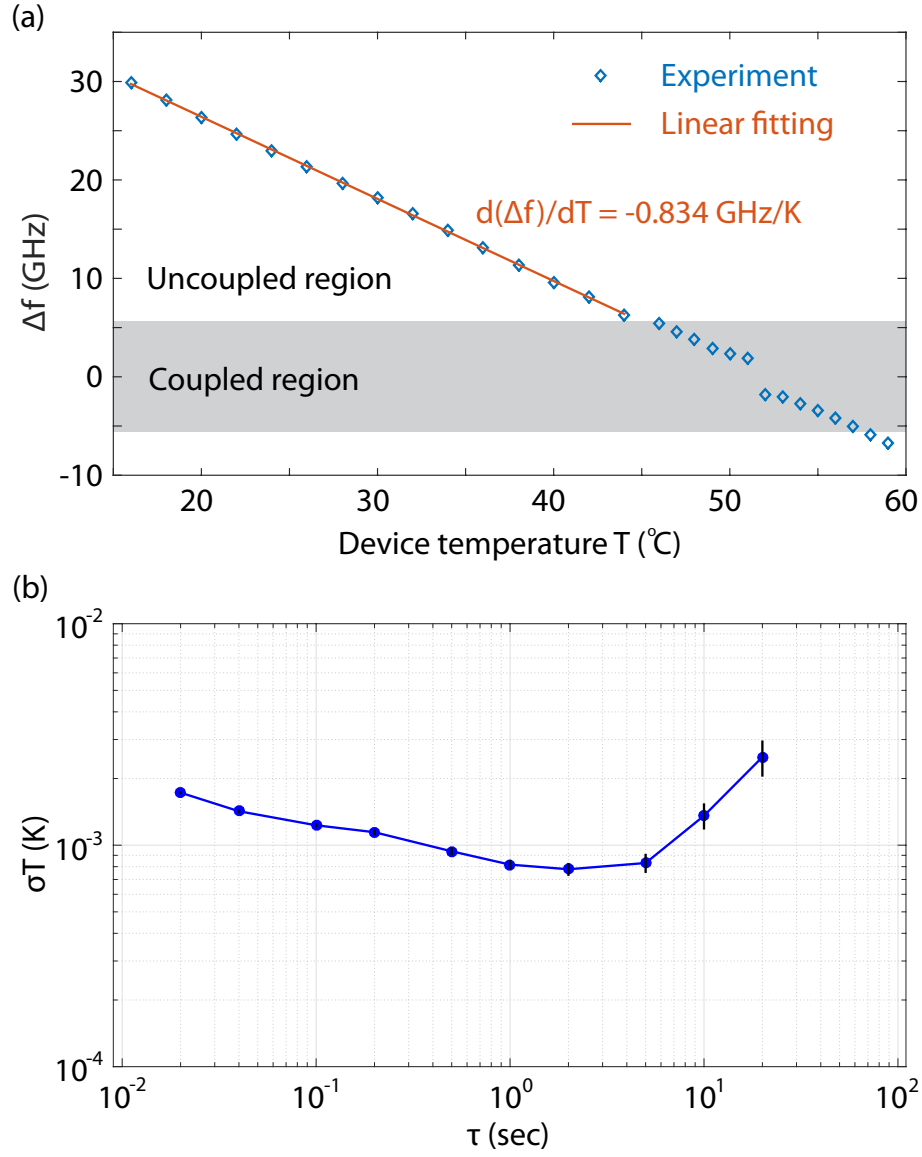


Figure 2.6: (a) Experimentally recorded frequency difference between the two modes, Δf , as a function of temperature. The linear curve is fitted from data in the uncoupled region ($T = 16 \sim 44^\circ\text{C}$). (b) Allan deviation of temperature measurements at 30°C .

temperature as a function of measurement time. The Allan deviation decreases monotonically with increased measurement time, until reaching a minimum of 0.8 mK at an averaging time of 2 seconds. This measurement uncertainty directly corresponds to the short-term stability (1 mK) of the temperature controller employed to stabilize the device temperature.

In principle, the device is expected to offer a sensing resolution much higher than the measured value. The fundamental temperature fluctuation of an optical resonator is given by $\langle \mu^2 \rangle = \frac{k_B T^2}{C \rho V}$ [62], where k_B is the Boltzmann constant, C and ρ are the specific heat capacity and the density of the device material, and V is the volume. For the demonstrated LN microdisk resonator, the fundamental temperature fluctuation is estimated to be $\sqrt{\langle \mu^2 \rangle} \approx 23 \mu\text{K}$. The laser frequency noise is expected to play a minor role for the level of optical Q of the current device [50]. The impact of the laser intensity noise can be easily eliminated by directly monitoring the laser power. Therefore, the sensing resolution of the device is expected to be considerably better than 0.8 mK. Exact characterization of the sensing resolution would require a better temperature stabilization system, which will be left for future exploration.

2.5 Conclusion

In conclusion, self-referenced temperature sensing is demonstrated in a Z -cut LN microdisk resonator, by utilizing the large thermo-optic birefringence

of the single-crystalline LN material. The device exhibits a temperature sensitivity of 0.834 GHz/K, allowing for the measurement of a temperature uncertainty of 0.8 mK. This work shows the potential of the LN integrated photonic platform for microscale temperature sensors with high sensitivity, high precision and low power consumption.

Understanding the thermo-optic property offers a control knob for nonlinear optics in LN nanophotonic devices. The strong thermo-optic birefringence of LN, which induces differential index change between ordinary and extraordinary polarizations, can be applied to achieve a wide-range tuning of phase matching in nonlinear optical interactions. In the next chapter, highly tunable efficient second-harmonic generation is presented in an LN nanophotonic waveguide.

Chapter 3

Thermal Tuning of Efficient Second-Harmonic Generation in an LN Nanophotonic Waveguide

Since their invention in the 1960s [63, 64], lasers have been the backbone of modern optics, playing fundamental roles in various optical sciences and technologies. For a coherent light source, wavelength tunability is one of the most important specifications, underlying crucially many applications including optical communications [65], spectroscopy [66, 67], frequency metrology [68], sensing [69], to name some. However, lasing wavebands of lasers are naturally limited by the gain media. Nonlinear optical parametric processes through the quadratic nonlinearity, such as second-harmonic generation (SHG), sum-frequency generation (SFG), and difference-frequency generation (DFG), with the capability for flexible phase-matching engineering, are among the most prominent approaches to achieve tunable coherent radiation at optical frequencies that are difficult to be obtained by lasers directly [70–72].

LN, with its outstanding nonlinear and linear optical properties, is widely employed for nonlinear wavelength conversion, particularly in RPE PPLN waveguides [8, 23]. In general, a type-0 configuration (with two extraordinarily polarized photons producing one extraordinarily polarized photon, $e + e \rightarrow e$) is employed to achieve a high conversion efficiency, and temperature tuning is a common technique to vary the operation wavelength of a PPLN waveguide. However, the tunability of the type-0 SHG is fairly limited [8, 23, 73], due to the relatively small wavelength dependence of the thermo-optic coefficient (although DFG can exhibit a large tunability with a third wavelength involved). In fact, LN exhibits a remarkable thermo-optic birefringence [58, 74, 75], significantly greater than most other optical media [55]. This characteristic can be utilized to increase the wavelength tunability of SHG in PPLN by employing a type-I configuration ($o + o \rightarrow e$) [8, 23, 76], which, however, severely sacrifices the conversion efficiency due to the significantly weaker nonlinearity compared with a type-0 process.

Over the past decade, a variety of integrated material platforms with $\chi^{(2)}$ nonlinearities have been developed for efficient nonlinear optical parametric processes [15, 39, 77–81], where the tight confinement of optical modes is able to enhance nonlinear optical interactions. Recent advances in the LN integrated platform have inspired the study of nonlinear optics in LN nanophotonic devices [30, 31, 33, 36, 82–87]. This provides an opportunity to achieve a large wavelength tunability by using the type-I configuration, while maintaining a high conversion efficiency.

In this chapter, thermal tuning of efficient SHG is demonstrated in an LN nanophotonic waveguide. SHG is achieved through type-I modal phase matching between orthogonal polarizations. By utilizing the large thermo-optic birefringence of LN, temperature tuning of the SHG wavelength is demonstrated, with a measured tuning slope of 0.84 nm/K for a telecom-band pump, almost 1 order of magnitude higher than type-0 SHG in LN [8, 23, 88]. Meanwhile, the device is designed to exhibit a large mode overlap, resulting in a theoretical normalized conversion efficiency of 22.2% $\text{W}^{-1}\text{cm}^{-2}$, which enables the experimental measurement of a conversion efficiency of 4.7% W^{-1} in a waveguide only 8 mm long, corresponding to a measured normalized conversion efficiency of 7.3% $\text{W}^{-1}\text{cm}^{-2}$. The demonstrated device is of great promise for efficient parametric generation of highly tunable coherent radiation from ultraviolet to mid-infrared.

3.1 Theory of Second-Harmonic Generation in Waveguides

3.1.1 Lossless Case

In order to understand nonlinear optical processes in a nanophotonic waveguide, an analytical model can be derived based on coupled-mode theory. For simplicity, the material anisotropy is not fully accounted for. The derivation is first performed assuming no optical losses, and loss terms are

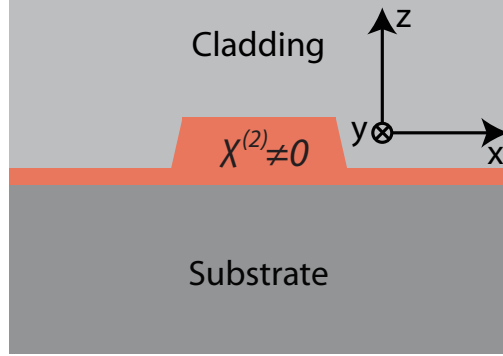


Figure 3.1: Schematic of a nanophotonic waveguide with a quadratic optical nonlinearity.

added in the simplified equations later. Physical quantities are represented in the complex form, and measurable quantities are related to their real parts. For SHG in a lossless straight waveguide with a quadratic optical nonlinearity $\chi^{(2)}$, schematically shown in Fig. 3.1, electric fields at the fundamental frequency (FF) ω_1 and the second harmonic (SH) ω_2 ($= 2\omega_1$) obey driven wave equations considering the nonlinearity, written as [4, 22]

$$\nabla^2 \tilde{\mathbf{E}}_1(\mathbf{r}, t) - \frac{\epsilon(x, z, \omega_1)}{c^2} \frac{\partial^2}{\partial t^2} \tilde{\mathbf{E}}_1(\mathbf{r}, t) = \frac{1}{\epsilon_0 c^2} \frac{\partial^2}{\partial t^2} \tilde{\mathbf{P}}_1^{NL}(\mathbf{r}, t), \quad (3.1)$$

$$\nabla^2 \tilde{\mathbf{E}}_2(\mathbf{r}, t) - \frac{\epsilon(x, z, \omega_2)}{c^2} \frac{\partial^2}{\partial t^2} \tilde{\mathbf{E}}_2(\mathbf{r}, t) = \frac{1}{\epsilon_0 c^2} \frac{\partial^2}{\partial t^2} \tilde{\mathbf{P}}_2^{NL}(\mathbf{r}, t), \quad (3.2)$$

where the nonlinear polarizations are represented as

$$\tilde{\mathbf{P}}_1^{NL}(\mathbf{r}, t) = \epsilon_0 \chi^{(2)}(\omega_1; -\omega_1, \omega_2) \tilde{\mathbf{E}}_1^*(\mathbf{r}, t) \tilde{\mathbf{E}}_2(\mathbf{r}, t), \quad (3.3)$$

$$\tilde{\mathbf{P}}_2^{NL}(\mathbf{r}, t) = \frac{\epsilon_0}{2} \chi^{(2)}(\omega_2; \omega_1, \omega_1) \tilde{\mathbf{E}}_1(\mathbf{r}, t) \tilde{\mathbf{E}}_1(\mathbf{r}, t). \quad (3.4)$$

In Eqs. (3.3)-(3.4), $\chi^{(2)}(\omega_1; -\omega_1, \omega_2) = \chi^{(2)}(\omega_2; \omega_1, \omega_1)$, which are third-rank tensors. Due to the critical phase-matching condition, significant SHG happens only if optical fields at the two frequencies have an identical propagation constant, or phase velocity. Therefore, in the derivation, each frequency is considered propagating in a single optical mode, and the two electric fields can be written as

$$\tilde{\mathbf{E}}_1(\mathbf{r}, t) = B_1(y)\mathbf{E}_1(x, z)e^{i\beta_1 y - i\omega_1 t}, \quad (3.5)$$

$$\tilde{\mathbf{E}}_2(\mathbf{r}, t) = B_2(y)\mathbf{E}_2(x, z)e^{i\beta_2 y - i\omega_2 t}, \quad (3.6)$$

where B_1 (B_2) is the field amplitude and $\mathbf{E}_1$ ($\mathbf{E}_2$) is the mode profile of the FF (SH). Mode profiles $\mathbf{E}_1$ and $\mathbf{E}_2$ satisfy

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \beta_1^2\right)\mathbf{E}_1(x, z) + \frac{\epsilon(x, z, \omega_1)\omega_1^2}{c^2}\mathbf{E}_1(x, z) = 0, \quad (3.7)$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \beta_2^2\right)\mathbf{E}_2(x, z) + \frac{\epsilon(x, z, \omega_2)\omega_2^2}{c^2}\mathbf{E}_2(x, z) = 0. \quad (3.8)$$

By substituting Eqs. (3.5)-(3.6) into Eqs. (3.1)-(3.4), considering Eqs. (3.7)-(3.8) and the slowly-varying-amplitude approximation, $|\frac{d^2 B}{dy^2}| \ll |\beta \frac{dB}{dy}|$, the relation between both amplitudes is obtained as

$$\frac{dB_1}{dy}\mathbf{E}_1 = \frac{i\omega_1^2}{2n_1 c}B_1^*B_2e^{i\Delta\beta y}\chi^{(2)}\mathbf{E}_1^*\mathbf{E}_2, \quad (3.9)$$

$$\frac{dB_2}{dy}\mathbf{E}_2 = \frac{i\omega_2^2}{4n_2 c}B_1^2e^{-i\Delta\beta y}\chi^{(2)}\mathbf{E}_1\mathbf{E}_1, \quad (3.10)$$

where n_1 (n_2) is the effective index of the FF (SH) mode, and $\Delta\beta = \beta_2 - 2\beta_1 = (n_2 - n_1)\frac{2\omega_1}{c}$ is the phase mismatch. By multiplying $\mathbf{E}_1^*$ on both sides of Eq. (3.9), multiplying $\mathbf{E}_2^*$ on both sides of Eq. (3.10), and integrating both equations over the $x - z$ plane, Eqs. (3.9)-(3.10) become

$$\frac{dB_1}{dy} = \frac{i\omega_1^2}{2n_1c} B_1^* B_2 e^{i\Delta\beta y} \frac{\int \chi_{ijk}^{(2)} E_{1i}^* E_{1j}^* E_{2k} dx dz}{\int |\mathbf{E}_1|^2 dx dz}, \quad (3.11)$$

$$\frac{dB_2}{dy} = \frac{i\omega_2^2}{4n_2c} B_1^2 e^{-i\Delta\beta y} \frac{\int \chi_{ijk}^{(2)} E_{2i}^* E_{1j} E_{1k} dx dz}{\int |\mathbf{E}_2|^2 dx dz}, \quad (3.12)$$

where Einstein summation convention has been used.

The above equations can be further simplified to get a better physical picture. For this purpose, normalized field amplitudes can be defines as

$$b_1(y) = B_1(y) \sqrt{\frac{\epsilon_0 c n_1}{2} \int |\mathbf{E}_1|^2 dx dz}, \quad (3.13)$$

$$b_2(y) = B_2(y) \sqrt{\frac{\epsilon_0 c n_2}{2} \int |\mathbf{E}_2|^2 dx dz}, \quad (3.14)$$

such that $|b_1|^2$ and $|b_2|^2$ represent optical powers P_1 and P_2 in the waveguide.

With the coupling coefficient defined as

$$g = \sqrt{\frac{\omega_1^2}{2\epsilon_0 c^3 n_1^2 n_2}} \frac{\int \chi_{ijk}^{(2)} E_{2i}^* E_{1j} E_{1k} dx dz}{\int |\mathbf{E}_1|^2 dx dz \sqrt{\int |\mathbf{E}_2|^2 dx dz}}, \quad (3.15)$$

Eqs. (3.11)-(3.12) become

$$\frac{db_1}{dy} = ig^*b_1^*b_2e^{i\Delta\beta y}, \quad (3.16)$$

$$\frac{db_2}{dy} = igb_1^2e^{-i\Delta\beta y}. \quad (3.17)$$

In the case of no losses and no pump depletion, i.e., $|b_1(y)| = |b_1(0)| \gg |b_2(y)|$, and no SH seed, i.e., $b_2(0) = 0$, the solution to Eq. (3.17) is simply

$$b_2(L) = \int_0^L igb_1(0)^2e^{-i\Delta\beta y}dy = igb_1(0)^2L\text{sinc}\left(\frac{\Delta\beta L}{2}\right)e^{-i\frac{\Delta\beta L}{2}}, \quad (3.18)$$

where $\text{sinc}(x) \equiv \sin(x)/x$. Therefore, the conversion efficiency of a waveguide with a length of L is calculated with

$$\Gamma \equiv \frac{P_2(L)}{P_1(0)^2} = \eta L^2 \text{sinc}^2\left(\frac{\Delta\beta L}{2}\right), \quad (3.19)$$

where $\eta = |g|^2$ represents the normalized conversion efficiency, usually expressed in a unit of ($\% \text{ W}^{-1}\text{cm}^{-2}$). Equation (3.19) clearly shows the importance of phase matching. In order to achieve the maximal efficiency, it is desired to have exact phase matching, i.e., $\Delta\beta = 0$, or $n_1 = n_2$, which gives a critical requirement in the design of waveguides. When phase matching is satisfied, Γ is quadratically dependent on L , under the condition of no propagation losses.

3.1.2 Loss Consideration

In the case of a lossy waveguide, the coupled equations can be written as

$$\frac{db_1}{dy} = ig^*b_1^*b_2e^{i\Delta\beta y} - \frac{\alpha_1}{2}b_1, \quad (3.20)$$

$$\frac{db_2}{dy} = igb_1^2e^{-i\Delta\beta y} - \frac{\alpha_2}{2}b_2, \quad (3.21)$$

where α_1 and α_2 are the loss rates of optical powers. Again, if no pump depletion is considered, i.e., $|b_1(y)| \gg |b_2(y)|$, the above equations become

$$\frac{db_1}{dy} = -\frac{\alpha_1}{2}b_1, \quad (3.22)$$

$$\frac{db_2}{dy} = igb_1^2e^{-i\Delta\beta y} - \frac{\alpha_2}{2}b_2, \quad (3.23)$$

and the solution at $y = L$ is

$$b_1(L) = b_1(0)e^{-\alpha_1 L}, \quad (3.24)$$

$$b_2(L) = igb_1(0)^2e^{-\frac{\alpha_2}{2}L}\frac{e^{(\frac{\alpha_2}{2}-\alpha_1-i\Delta\beta)L}-1}{\frac{\alpha_2}{2}-\alpha_1-i\Delta\beta}, \quad (3.25)$$

resulting in a conversion efficiency of

$$\Gamma \equiv \frac{P_2(L)}{P_1(0)^2} = \eta L^2 e^{-\alpha_2 L} \frac{(e^{\Delta\alpha L} - 1)^2 + 4e^{\Delta\alpha L} \sin^2(\frac{\Delta\beta L}{2})}{(\Delta\alpha L)^2 + (\Delta\beta L)^2}, \quad (3.26)$$

where $\Delta\alpha \equiv \alpha_2/2 - \alpha_1$. In the lossless case, $\alpha_1 = \alpha_2 = \Delta\alpha = 0$, and Eq. (3.26) reduces to Eq. (3.19),

3.1.3 A Simple Form of Normalized Conversion Efficiency

In many cases, there is only one kind of nonlinear medium in a nanophotonic waveguide. Under this condition, in order to obtain a more clear physical picture of the nonlinear optical interaction, the expression of g in Eq. (3.15) can be factorized into a simpler form. The nonlinear effective mode area can be defined as $A_{\text{eff}} \equiv (A_1^2 A_2)^{\frac{1}{3}}$, with

$$A_1 = \frac{(\int_{\text{all}} |\mathbf{E}_1|^2 dx dz)^3}{|\int_{\chi^{(2)}} |\mathbf{E}_1|^2 \mathbf{E}_1 dx dz|^2}, \quad (3.27)$$

$$A_2 = \frac{(\int_{\text{all}} |\mathbf{E}_2|^2 dx dz)^3}{|\int_{\chi^{(2)}} |\mathbf{E}_2|^2 \mathbf{E}_2 dx dz|^2}, \quad (3.28)$$

and the nonlinear mode overlap factor can be defined as

$$\zeta_{ijk} = \frac{\int_{\chi^{(2)}} E_{2i}^* E_{1j} E_{1k} dx dz}{|\int_{\chi^{(2)}} |\mathbf{E}_1|^2 \mathbf{E}_1 dx dz|^{\frac{2}{3}} |\int_{\chi^{(2)}} |\mathbf{E}_2|^2 \mathbf{E}_2 dx dz|^{\frac{1}{3}}}, \quad (3.29)$$

where $\int_{\chi^{(2)}}$ and $\int_{\text{all}}$ denote integration over the nonlinear medium and all space, respectively. As a result, the coupling strength becomes

$$g = \sqrt{\frac{\omega_1^2}{2\epsilon_0 c^3 n_1^2 n_2}} \frac{\chi_{ijk}^{(2)} \zeta_{ijk}}{\sqrt{A_{\text{eff}}}}, \quad (3.30)$$

and the normalized conversion efficiency can simply be represented as

$$\eta = \frac{\omega_1^2}{2\epsilon_0 c^3 n_1^2 n_2} \frac{(\chi_{ijk}^{(2)} \zeta_{ijk})^2}{A_{\text{eff}}}. \quad (3.31)$$

As shown in Eq. (3.31), η is critically dependent on the material nonlinearity, the mode overlap in the nonlinear medium, and the effective mode area.

3.2 Thermal Tuning of Phase Matching

Optical parametric processes depend critically on the phase-matching condition, leading to usually narrow operation bandwidths. In order to achieve parametric generation at different wavelengths, the phase-matching window can be tuned using various physical mechanisms, with thermo-optic effect being a dominant one [8, 39].

In this section, the relation between the phase-matched SHG frequency and the device temperature is derived. Assume at temperature T , phase matching is achieved between ω_1 and ω_2 , i.e.,

$$n_1(\omega_1, T) = n_2(\omega_2, T), \quad (3.32)$$

where $\omega_2 = 2\omega_1$. Now the question is, when the temperature is shifted to $T + \Delta T$, what the new phase-matched frequency $\omega_1 + \Delta\omega$ becomes. In order to answer this question, the effective indices can be expanded to the first

order around T , ω_1 , and ω_2 as

$$n_1(\omega_1 + \Delta\omega_1, T + \Delta T) = n_1(\omega_1, T) + \frac{\partial n_1}{\partial \omega_1} \Delta\omega_1 + \frac{\partial n_1}{\partial T} \Delta T, \quad (3.33)$$

$$n_2(\omega_2 + 2\Delta\omega_1, T + \Delta T) = n_2(\omega_2, T) + \frac{\partial n_2}{\partial \omega_2} (2\Delta\omega_1) + \frac{\partial n_2}{\partial T} \Delta T. \quad (3.34)$$

After the temperature is changed, the phase-matching condition is still satisfied, with

$$n_1(\omega_1 + \Delta\omega_1, T + \Delta T) = n_2(\omega_2 + 2\Delta\omega_1, T + \Delta T). \quad (3.35)$$

By substituting Eqs. (3.33)-(3.34) into Eq. (3.35), the tuning slope is obtained as

$$\frac{\Delta\omega_1}{\Delta T} = -\frac{\frac{\partial n_1}{\partial T} - \frac{\partial n_2}{\partial T}}{\frac{\partial n_1}{\partial \omega_1} - 2\frac{\partial n_2}{\partial \omega_2}}, \quad (3.36)$$

where the numerator is the differential thermal response in the effective indices, and the denominator is directly related to group indices.

3.3 Waveguide Design

According to Eq. (3.36), the thermal tuning slope of SHG is proportional to the differential thermal response in the effective indices of the two modes involved. As shown in Table 3.1, the thermo-optic effect of LN has a much larger birefringence than dispersion, which explains why type-I processes ($o + o \rightarrow e$) are more thermally tunable than type-0 processes ($e + e \rightarrow e$).

Table 3.1: Thermo-optic coefficients of LN at telecom-band and visible wavelengths ($T = 300$ K)

Wavelength (nm)	$\frac{dn_o}{dT}$ (10^{-6} K $^{-1}$)	$\frac{dn_e}{dT}$ (10^{-6} K $^{-1}$)
1523	0	33.4
632	0.4	41.2

* data from Ref. [58]

In particular, by designing for phase matching between an FF mode that is o-light and an SH mode that is e-light, the largest difference in the thermo-optic coefficients can be achieved. In this case, a temperature change results in a considerable variation of the material birefringence, which then shifts the phase-matched wavelength. Meanwhile, in such a type-I process ($o + o \rightarrow e$), the negative birefringence ($n_e < n_o$) of LN helps to compensate for the material dispersion, which offers a larger flexibility in designing for phase matching. The thermal tuning of SHG can be maximized by using a Z -cut LN waveguide, which supports ordinarily and extraordinarily polarized optical modes with high polarization purity.

The geometry of the Z -cut LN waveguide [see Fig. 3.2(a)] is designed such that the fundamental quasi-TE mode ($TE_{00, \text{Tele}}$) in the telecom band is phase matched with a high-order quasi-TM mode ($TM_{20, \text{NIR}}$) in the near-infrared (NIR). Figures 3.2(c) and 3.3(a) present the effective refractive indices of the two modes, simulated by the FEM, showing a phase-matched pump wavelength of $\lambda_{\text{PM}} = 1540$ nm at a temperature of 20°C . As discussed above, LN exhibits a large thermo-optic effect for e-light ($\frac{dn_{e, \text{NIR}}}{dT} \approx 4 \times 10^{-5} \text{K}^{-1}$), while it is negligible for o-light ($\frac{dn_{o, \text{Tele}}}{dT} \approx 0$), at room temperature [58].

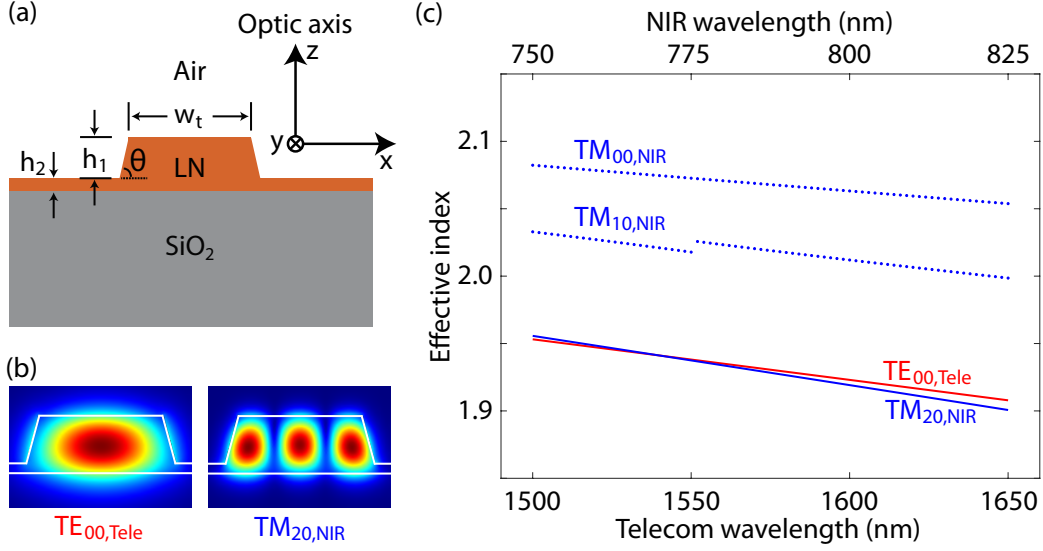


Figure 3.2: (a) Schematic of a Z-cut LN waveguide. (b) Mode profiles of the $TE_{00,Tele}$ mode in the telecom band and the $TM_{20,NIR}$ mode in the NIR. (c) Numerically simulated effective indices as functions of wavelength, of $TE_{00,Tele}$ and $TM_{j0,NIR}$ ($j = 0, 1, 2$), with $w_t = 1200$ nm, $h_1 = 460$ nm, $h_2 = 100$ nm, and $\theta = 75^\circ$, at 20°C . The discontinuity in the curve of $TM_{10,NIR}$ is due to its coupling with $TE_{20,NIR}$ (not shown).

As a result, when the device temperature increases, the effective refractive index of the $TE_{00,Tele}$ mode remains nearly unchanged while that of the $TM_{20,NIR}$ mode increases considerably. Consequently, the phase-matched wavelength moves dramatically toward longer wavelengths. Figure 3.3(b) shows an example of the temperature-induced shift of the phase matching, where λ_{PM} shifts to 1574 nm at a temperature of 70°C . Detailed analysis shows that the phase-matched wavelength depends almost linearly on the device temperature, as shown clearly in Fig. 3.4, with a tuning slope of 0.69 nm/K.

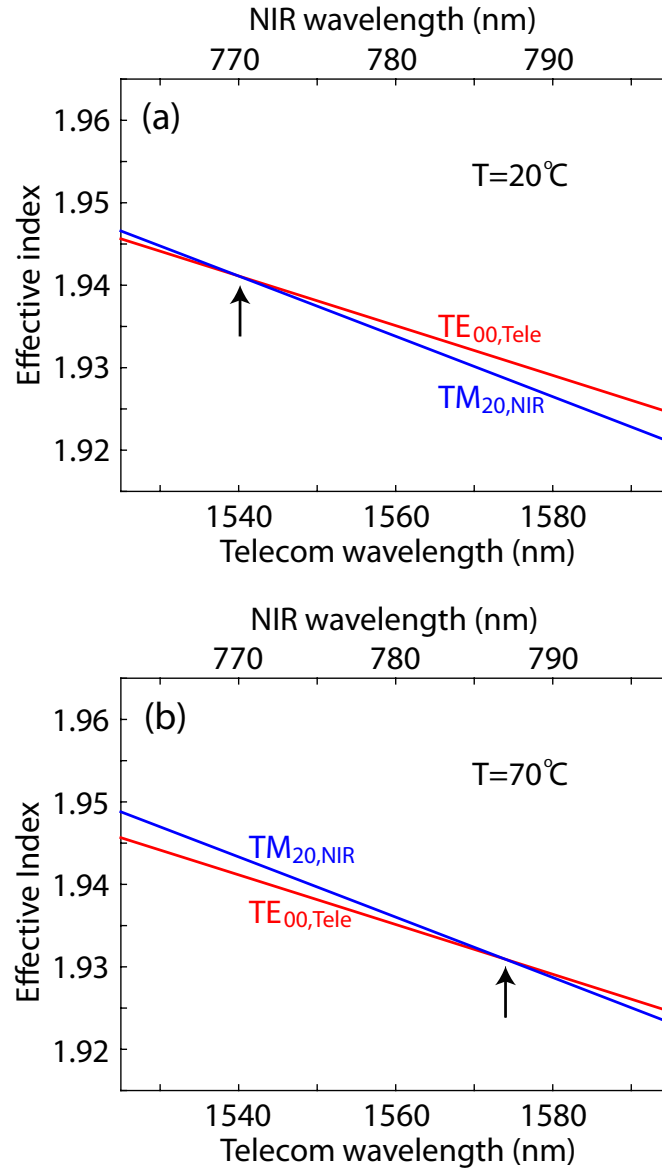


Figure 3.3: Enlargement of the wavelength-dependent effective indices of $\text{TE}_{00,\text{Tele}}$ and $\text{TM}_{20,\text{NIR}}$ at (a) 20°C , and (b) 70°C , with black arrows indicating phase matching. The FEM simulations take into account the temperature and wavelength dependence of the material refractive indices, n_o and n_e [58, 89].

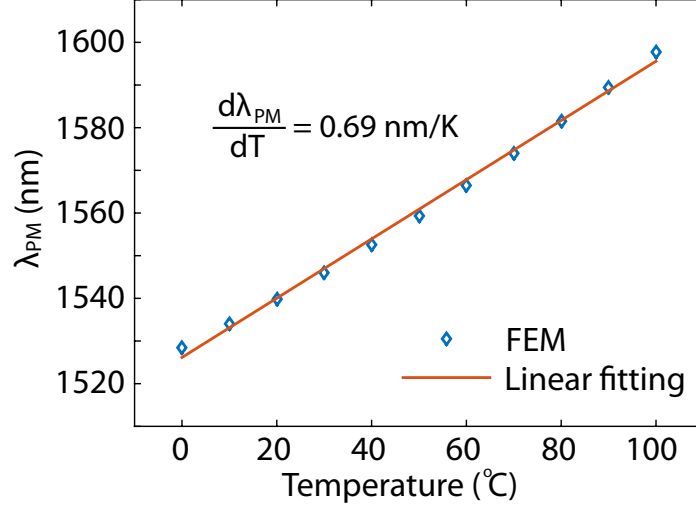


Figure 3.4: Numerically simulated phase-matched pump wavelength, λ_{PM} , as a function of temperature.

Phase matching of the two modes indicates potentially efficient SHG in the designed waveguide. In the current case, only one term contributes significantly to the mode overlap,

$$\zeta = \frac{\int_{\chi^{(2)}} E_{2z}^* E_{1x} E_{1x} dx dz}{|\int_{\chi^{(2)}} |\mathbf{E}_1|^2 \mathbf{E}_1 dx dz|^{\frac{2}{3}} |\int_{\chi^{(2)}} |\mathbf{E}_2|^2 \mathbf{E}_2 dx dz|^{\frac{1}{3}}}, \quad (3.37)$$

and the normalized conversion efficiency is simplified to

$$\eta = \frac{2\omega_1^2}{\epsilon_0 c^3 n_1^2 n_2} \frac{d_{\text{eff}}^2 \zeta^2}{A_{\text{eff}}}, \quad (3.38)$$

where $d_{\text{eff}} = \chi^{(2)}/2$. In this case, $d_{\text{eff}} = d_{31} = 4.3 \text{ pm/V}$ [18]. Numerical simulations show a small effective area of $A_{\text{eff}} = 1.46 \text{ } \mu\text{m}^2$ for the designed waveguide. In particular, the waveguide exhibits a large mode overlap factor

of $\zeta = 0.32$, due to the negative material birefringence that assists phase matching in the type-I process, offering a larger flexibility in optimizing the mode overlap compared with a type-0 process. As a result, the waveguide exhibits a normalized conversion efficiency as high as $\eta = 22.2\% \text{ W}^{-1}\text{cm}^{-2}$. This value is comparable to that of type-0 SHG in typical RPE PPLN [88, 90] and LN nanophotonic waveguides [83] that utilize the maximum component of the $\chi^{(2)}$ nonlinearity ($d_{\text{eff}} = d_{33} = 27 \text{ pm/V}$), although a type-I configuration using d_{31} is employed here. In contrast to those type-0 devices, the designed waveguide is expected to exhibit a significantly larger thermal tuning slope, as experimentally demonstrated in the following.

3.4 Experimental Results

To confirm the simulation results, waveguides are fabricated on a Z -cut LNOI wafer, where the LN thin film has a thickness of about 560 nm, sitting on 2- μm -thick buried oxide. Figure 3.5(b) shows the cross section of a fabricated waveguide, whose geometry is similar to the design [see Fig.3.2(a)]. In particular, as presented in Fig. 3.5(c), the waveguide has a smooth sidewall, implying a low propagation loss. Figure 3.5(a) shows the experimental setup. Pump light from a CW tunable telecom-band laser is coupled via a lensed fiber into the device chip, which is placed on top of a thermoelectric cooler for temperature control. At the waveguide output, the pump light is collected together with the frequency-doubled light by

a second lensed fiber. After being separated from its SH by a 780/1550 wavelength-division multiplexer (WDM), the telecom pump light is directed to an InGaAs detector for characterization, while the generated NIR light is sent to a spectrometer for analysis. A fiber polarization controller is used for optimal coupling from the input lensed fiber to the wanted waveguide mode, and variable optical attenuators are employed to study the power dependence of SHG. The spectrometer is cooled by liquid nitrogen for an improved sensitivity.

In order to quantify the propagation and coupling losses, waveguides with the same cross section, but different lengths, are fabricated, as schematically shown in the inset of Fig. 3.5(d). Since these waveguides share the same coupling and bending losses, by measuring their transmission as a function of length, the propagation loss can be extracted. As shown in Fig. 3.5(d), the propagation loss of the $TE_{00,Tele}$ mode is measured to be 0.54 dB/cm, a small value that represents the state-of-the-art quality of LN nanophotonic waveguides [87, 91, 92]. Together with the overall fiber-to-fiber transmission of a straight waveguide [see Fig. 3.5(e)], a fiber-to-chip coupling loss of about 5 dB/facet is extracted, which mainly comes from the mode mismatch between the lensed fibers and the waveguide.

To demonstrate SHG, a straight waveguide with a length of about 8 mm is employed. The pump laser wavelength is scanned, with the spectrometer recording the generated NIR light during the whole laser scanning period, to measure the efficiency spectrum of SHG. One example is presented in

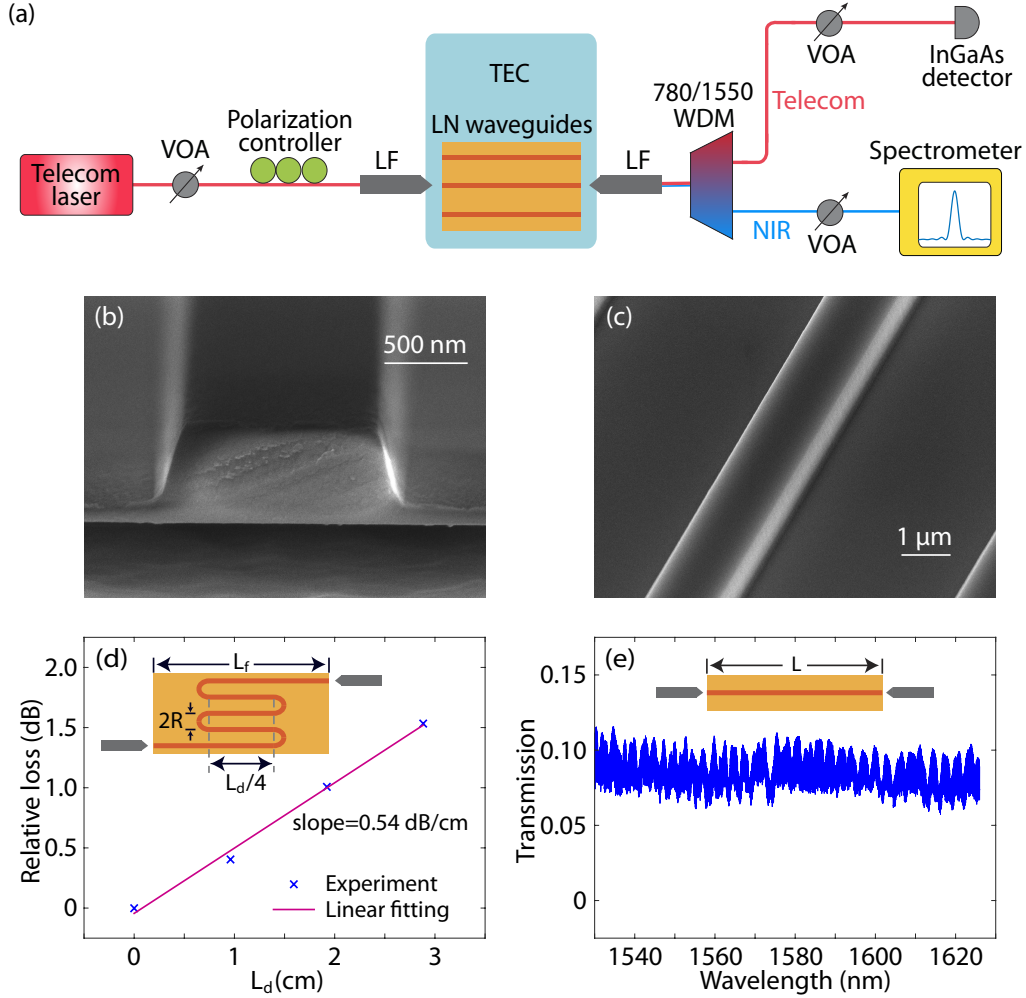


Figure 3.5: (a) Experimental setup for device characterization and SHG measurement. Scanning electron microscopy images showing the waveguide (b) facet, and (c) sidewall. (d) Fiber-to-fiber loss as a function of the differential length L_d , relative to that of $L_d = 0$, for waveguides schematically illustrated in the inset, with L_f and R kept as 8 mm and 100 μm , respectively. (e) Telecom-band transmission spectrum of the TE polarization for a straight waveguide with a length of $L \simeq 8$ mm, which is schematically shown in the inset. VOA: variable optical attenuator; LF: lensed fiber; TEC: thermoelectric cooler; WDM: wavelength-division multiplexer.

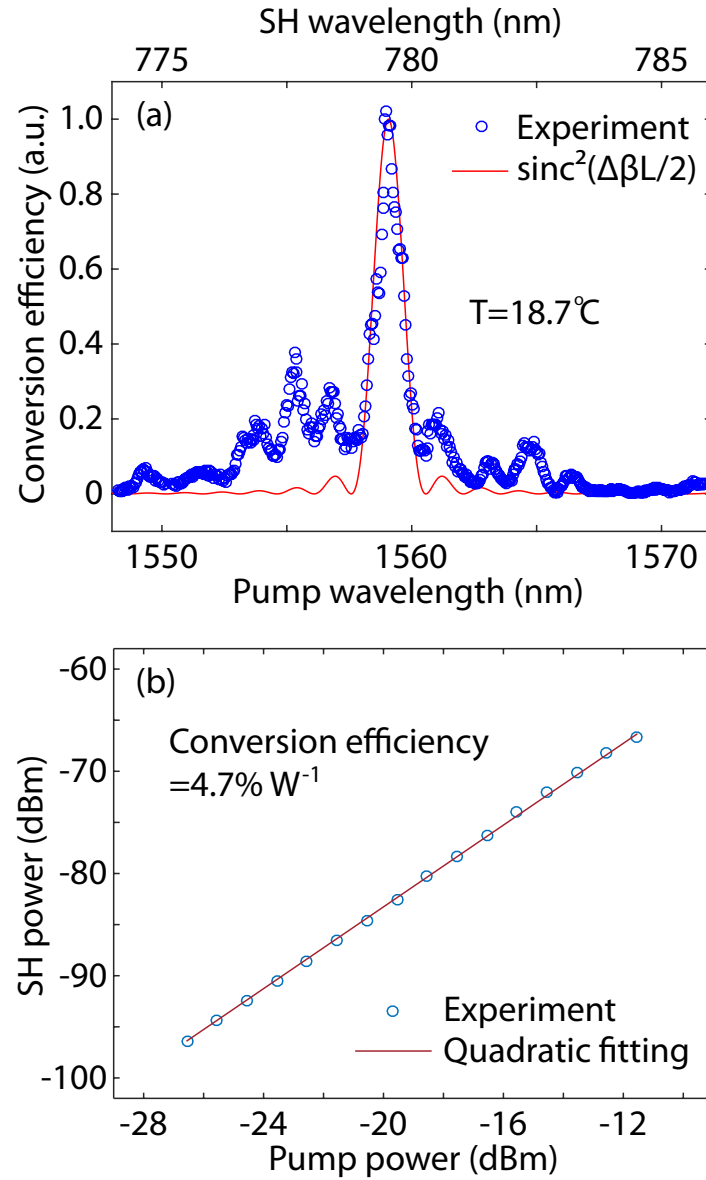


Figure 3.6: SHG from a straight LN nanophotonic waveguide with a length of 8 mm. (a) Conversion efficiency spectrum at $T=18.7^\circ\text{C}$, with the center wavelength of the sinc^2 -function aligned to the measured peak. (b) SH power as a function of FF pump power, with experimental data compared with a quadratic fitting, exhibiting a conversion efficiency of $4.7\% \text{ W}^{-1}$.

Fig. 3.6(a), which shows a phase-matched pump wavelength of 1559 nm at a temperature of 18.7°C. The main lobe of the recorded efficiency spectrum agrees with the theoretical expectation from the function $\text{sinc}^2(\Delta\beta L/2)$. Interestingly, the efficiency spectrum exhibits multiple strong side lobes, which are likely introduced by potential nonuniformity of the waveguide thickness along its total length, since the phase-matching condition is very sensitive to the waveguide geometry.

When the pump wavelength is fixed at 1559.06 nm, where the peak conversion efficiency is located, the SH power is recorded when varying the pump power, as shown in Fig. 3.6(b). The SH shows a quadratic power dependence on the pump, which agrees with the theoretical expectation. By fitting the experimental data, an on-chip conversion efficiency of 4.7% W^{-1} , or a normalized conversion efficiency of 7.3% $\text{W}^{-1}\text{cm}^{-2}$, is obtained.

The recorded SHG efficiency is smaller than the theoretical value given by ηL^2 ($= 14.2\% \text{ W}^{-1}$), primarily due to the nonzero propagation losses. The LN waveguide exhibits a propagation loss of 0.54 dB/cm at the pump wavelength [see Fig. 3.5(d)]. The loss at the SH, however, is challenging to characterize with the current setup, primarily due to the difficulty in effective light coupling into a high-order waveguide mode. As a rough estimate, it is assumed to be 2.16 dB/cm since the propagation loss is dominated by the Rayleigh scattering from the sidewall roughness that scales with the wavelength as $1/\lambda^2$ [93]. As a result, the theoretical conversion efficiency is estimated to be 10.5% W^{-1} for the 8-mm-long waveguide, after these

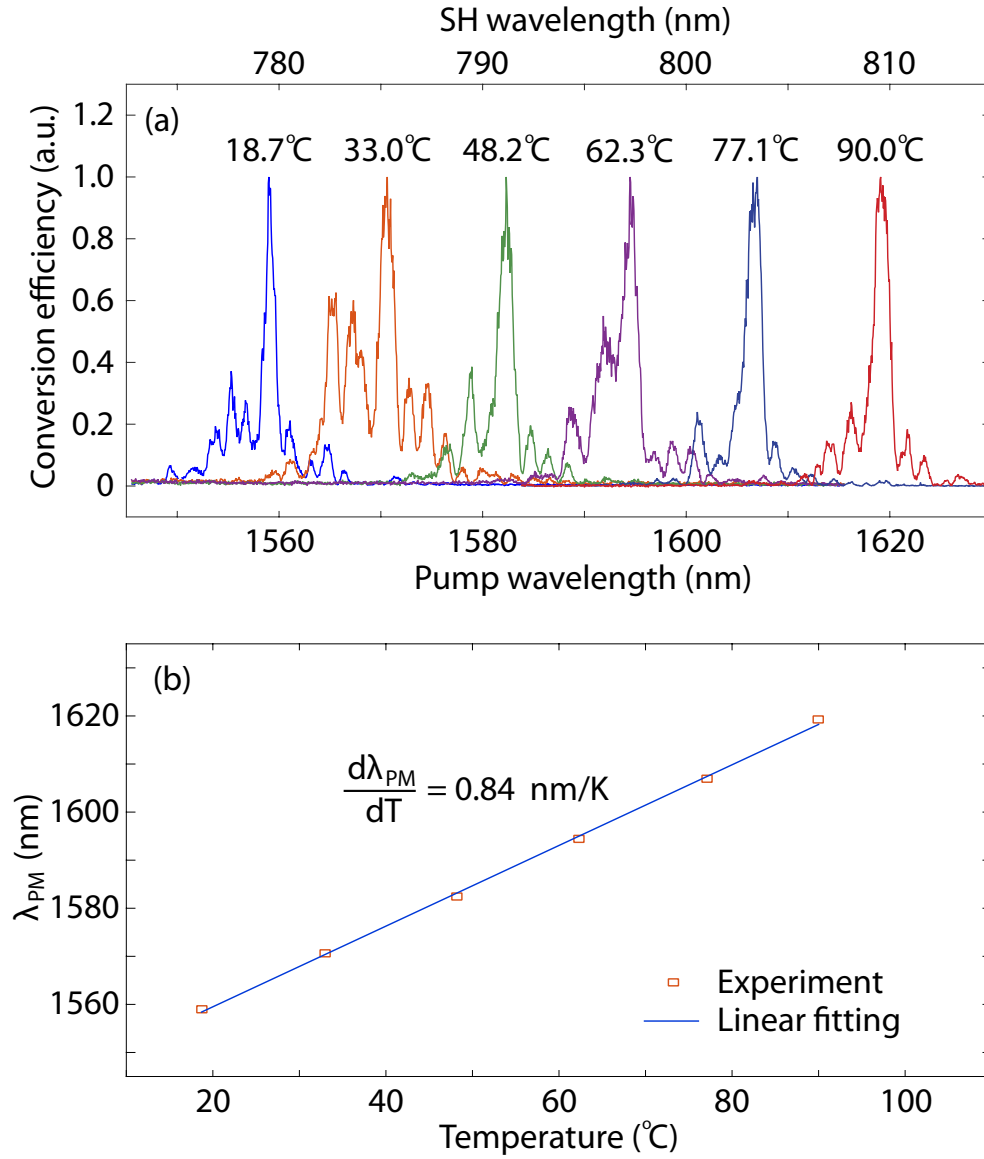


Figure 3.7: Thermal tuning of SHG. (a) Conversion efficiency spectra at different temperatures. Each spectrum is normalized by its peak value for a clear comparison. (b) Measured temperature dependence of the phase-matched pump wavelength λ_{PM} .

propagation losses are taken into account. On the other hand, the potential nonuniformity in the waveguide geometry might also impact the conversion efficiency to a certain extent. In addition, the coupling loss of the high-order SH light is likely underestimated due to its large mode mismatch with the focused fiber modes, leading to a conservative estimation of the recorded conversion efficiency. Therefore, the measured conversion efficiency can be improved in the future by reducing the waveguide sidewall scattering loss (e.g., with an oxide cladding), by increasing the uniformity in the wafer thickness with appropriate chemical mechanical polishing, and by optimizing coupling of the SH (e.g., with a separately designed on-chip coupler [39]).

To demonstrate the spectral tuning capability, the device temperature is varied from 18.7°C to 90.0°C, and the SHG efficiency spectrum is measured at each temperature. The recorded peak efficiencies are similar to that at 18.7°C [see Fig. 3.6(b)]. Figure 3.7(a) presents peak-normalized spectra at different temperatures. It shows that the SHG spectrum shifts toward longer wavelengths when temperature increases. By mapping the phase-matched pump wavelength as a function of temperature, Fig. 3.7(b) is obtained, showing a measured tuning slope of $\frac{d\lambda_{\text{PM}}}{dT} = 0.84 \text{ nm/K}$, almost 1 order of magnitude larger than type-0 SHG in LN [8, 23, 88]. The experimental result agrees with the numerical simulations (see Fig. 3.4). A slightly larger experimental value of the tuning slope is likely due to positive contributions by pyroelectric [34] and thermal expansion effects [60] in the waveguide cross section, which are not taken into consideration in the FEM simulations.

3.5 Conclusion and Discussion

In conclusion, highly tunable efficient SHG is demonstrated in an LN nanophotonic waveguide. The waveguide exhibits a high optical quality with a propagation loss as low as 0.54 dB/cm in the telecom band, which represents the state-of-the-art quality of LN nanophotonic waveguides reported to date [87, 91, 92]. In particular, the strong thermo-optic birefringence of LN is exploited to achieve thermal tuning of the SHG wavelength, with a tuning slope of 0.84 nm/K for a telecom-band pump, significantly higher than that offered by type-0 SHG in LN. At the same time, due to the tight mode confinement and a large spatial mode overlap, the waveguide exhibits a high theoretical normalized conversion efficiency of 22.2% $\text{W}^{-1}\text{cm}^{-2}$ even for the type-I modal phase matching, which is comparable to that of type-0 SHG in typical RPE PPLN and LN nanophotonic waveguides utilizing the largest nonlinear term d_{33} . This waveguide design enables the measurement of an SHG efficiency of 4.7% W^{-1} inside a waveguide only 8 mm long, corresponding to an experimental normalized conversion efficiency of 7.3% $\text{W}^{-1}\text{cm}^{-2}$. This work demonstrates the potential of the LN integrated platform for tunable and efficient on-chip nonlinear wavelength conversion.

In the work presented in this chapter, SHG is achieved through modal phase matching, with different-order modes involved to compensate for the phase mismatch caused by material and waveguide dispersion. This is a common technique that conveniently achieves nonlinear parametric

generation in on-chip devices. However, although advantageous in tight mode confinement, integrated photonic platforms have not yet lived up to their promise in efficient nonlinear optical parametric generation, due to the challenge in simultaneously achieving a large material nonlinearity, a small mode area, and a large mode overlap. In chapter 4, a novel type of nanophotonic waveguides is proposed and experimentally demonstrated. By breaking the spatial symmetry of the optical nonlinearity, optimal mode overlap can be achieved in a waveguide with both a high nonlinearity and a small mode area, resulting in extremely efficient optical parametric generation.

Chapter 4

Highly Efficient Second-Harmonic Generation in Semi-Nonlinear Nanophotonic Waveguides

Quadratic optical parametric processes via a $\chi^{(2)}$ nonlinearity have attracted long-lasting interest ever since the first observation of SHG [22, 94]. Their intriguing capability of creating new light through elastic photon-photon scattering forms a crucial foundation for a wide range of applications from photonic signal processing [24, 95], tunable coherent radiation [71], frequency metrology [96], optical microscopy [97], to quantum information processing [98]. Recently, tremendous interest has been attracted to the development of nonlinear optical parametric processes on chip-scale platforms for enhanced interactions [15, 30, 36, 39, 78, 81–83, 85, 87, 99–110].

The conversion efficiency of an optical parametric process relies essentially on phase matching among the interacting waves to sustain a coherent nonlinear interaction. To date, phase matching on chip-scale platforms is dominantly realized via either domain engineering of the nonlinear

susceptibility [111–113] or refractive-index engineering of the interacting optical modes [111]. Domain engineering is an efficient approach to achieve quasi-phase matching, which, however, is fairly material limited. For example, periodic poling [8, 23, 112–118] and orientation-patterned growth [111, 119–121] are currently implemented only to ferroelectrics and III-V semiconductors, respectively. Domain engineering also has stringent requirement on the domain uniformity [23, 112, 113], which imposes a challenge for applications that require small domain periods [30, 109, 110, 113] or clean parametric noise background [118]. On the other hand, modal index engineering is able to achieve exact phase matching [111], which can be flexibly implemented particularly to an on-chip platform that supports rich guided mode species [15, 36, 39, 78, 81–83, 85, 87, 100–108]. Unfortunately, modal phase matching suffers from mode field mismatch among the interacting optical modes, which severely degrades the nonlinear conversion efficiency [15, 36, 39, 78, 81–83, 85, 87, 100–108]. Although periodic grating structure can be used to assist phase matching while maximizing the mode overlap, intrinsic radiation loss is inevitably introduced [83, 84, 122].

In this chapter, a universal design and operation principle of semi-nonlinear waveguides is proposed for highly efficient optical parametric generation on integrated photonic platforms. The proposed heterogeneous nanophotonic waveguides consist of a nonlinear medium exciting parametric processes and a linear medium assisting exact phase matching, which are able to combine elegantly a large nonlinearity, a strong optical confinement,

and a good spatial mode match together in a single device, resulting in dramatically enhanced nonlinear parametric generation with an extremely high efficiency. To demonstrate the proposed principle, a heterogeneous nanophotonic waveguide, composed of a highly nonlinear $\chi^{(2)}$ material LN and a linear optical material amorphous titanium oxide (TiO_2), is designed. The enhanced nonlinear optical interaction in the device leads to a theoretical normalized conversion efficiency as high as 2900% $\text{W}^{-1}\text{cm}^{-2}$ for SHG, which enables an experimental conversion efficiency of 36.0% W^{-1} in a waveguide only 2.35 mm long, corresponding to an experimentally recorded normalized conversion efficiency of 650% $\text{W}^{-1}\text{cm}^{-2}$. The proposed operation principle can be flexibly applied to any other on-chip quadratic nonlinear photonic platform to support ultrahighly efficient optical parametric interactions, thus opening up a great avenue toward extreme nonlinear and quantum optics with potential for broad applications in energy-efficient photonic signal processing.

4.1 Concept Illustration

In a quadratic nonlinear waveguide, it is well-known that the phase velocities can be exactly matched among different guided modes to support a coherent nonlinear interaction [15, 36, 39, 81–83, 85, 87, 101, 103–108]. When the phase-matching condition is satisfied, the efficiency of SHG depends essentially on the nonlinear susceptibility, the effective mode area, and the spatial mode overlap, as shown in chapter 3 by Eq. (3.31). In particular, Eq. (3.29) shows

that the spatial mode overlap is determined by the relative spatial symmetry between the FF and SH modes. Unfortunately, for a nonlinear waveguide, different-order guided modes generally exhibit distinctive spatial symmetries, leading to a dramatically degraded mode overlap.

To illustrate this problem, a nominal monolithic nanophotonic rib waveguide is considered, as shown in Fig. 4.1(a), with the core material exhibiting a quadratic nonlinearity. An appropriate waveguide geometry will lead to exact phase matching between the fundamental TE_{00} mode at a wavelength λ and a high-order mode (TE_{01} in the example) at the half wavelength $\lambda/2$. However, as shown in Fig. 4.1(b)(c), the electric field (E_z) of the SH mode changes its polarity across the waveguide core while that of the FF mode maintains a single polarity. As a result, the nonlinear parametric interaction in the upper half of the waveguide is out of phase with that in the lower half, and therefore they cancel with each other, leading to a very small net nonlinear effect. It is exactly this modal mismatch of spatial symmetries that severely undermines the nonlinear conversion efficiency on the majority of current nonlinear photonic chips [15, 30, 36, 39, 78, 81–83, 85, 87, 99–110], although they have shown the advantage of strong mode confinement, that is, small mode areas. Consequently, a device has to rely on a long interaction length (or a high optical Q in the case of a resonator) to sustain the nonlinear process [39].

To tackle this challenge, semi-nonlinear waveguides are proposed, which leverage the fact that modal phase matching depends only on the linear

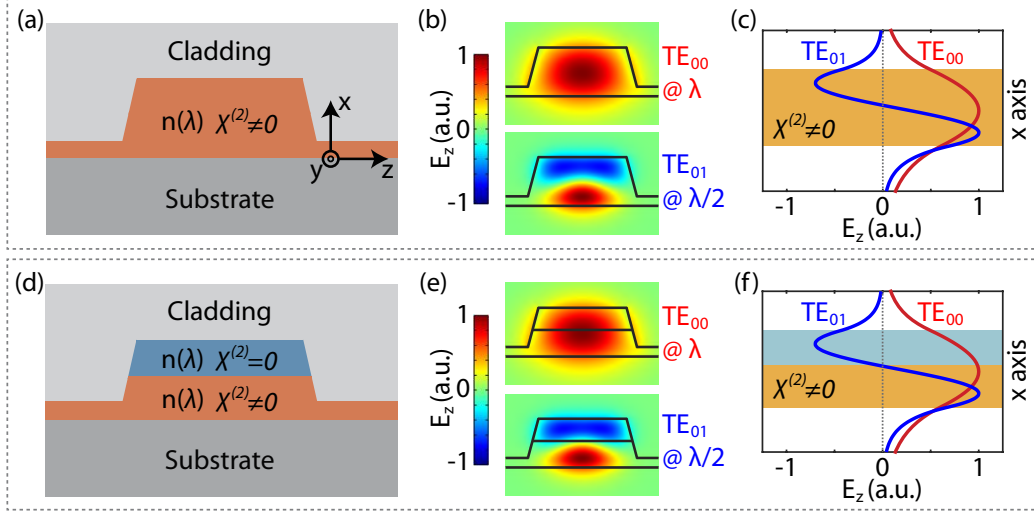


Figure 4.1: Illustration of semi-nonlinear waveguides for highly efficient SHG. (a) Schematic of a monolithic nanophotonic waveguide with a nonvanished $\chi^{(2)}$ nonlinearity. (b) Optical field (E_z) profiles of two phase-matched modes, TE_{00} at the fundamental wavelength λ and TE_{01} at the half wavelength $\lambda/2$. (c) E_z as a function of the vertical position x , at the center of the waveguide, with the orange shaded area indicating the $\chi^{(2)}$ material. (d) Schematic of a semi-nonlinear waveguide composed of two core materials, with both sharing the same linear refractive index $n(\lambda)$, while only the bottom layer has a nonvanished $\chi^{(2)}$ nonlinearity. (e) and (f) are counterparts of (b) and (c), respectively, for the semi-nonlinear waveguide.

properties of the waveguide instead of its nonlinearity. Therefore, by breaking the spatial symmetry of the optical nonlinearity, significant enhancement shall be achieved in the strength of the nonlinear interaction. Figure 4.1(d) shows the schematic of a semi-nonlinear waveguide, which exhibits the same geometry as the monolithic waveguide shown in Fig. 4.1(a), while the top part of the core is replaced by a linear material with the same refractive index and a vanished $\chi^{(2)}$. Since its linear refractive index profile is the same

as that of the monolithic waveguide, the semi-nonlinear waveguide supports an exactly same pair of phase-matched TE_{00} mode at the FF and TE_{01} mode at the SH, with mode profiles identical to those of the monolithic waveguide [see Fig. 4.1(b)(e)]. The thickness of the linear layer is chosen such that, for the SH mode TE_{01} with two opposite polarities, $E_z \approx 0$ at the boundary between the two core materials, resulting in a single polarity inside each of the linear and nonlinear layers. As the top linear layer does not participate in the nonlinear interaction, the parametric process only has a contribution from the bottom nonlinear layer, which will not be canceled. As a result, the net nonlinear effect is remarkably enhanced, manifesting as a large value of the mode overlap factor ζ . The discussion above, for simplicity, has assumed an identical refractive index between the linear and nonlinear layers. In practice, the operation principle can be applied to two core materials with dissimilar refractive indices, since the semi-nonlinear waveguide offers plenty of degrees of freedom for modal-index engineering.

4.2 Waveguide Design

To demonstrate the proposed principle, a semi-nonlinear waveguide is designed, consisting of a single-crystalline LN thin-film layer as the nonlinear medium and an amorphous TiO_2 layer as the linear component. LN exhibits a large $\chi^{(2)}$ nonlinearity and a wide transparency window from ultraviolet to mid-infrared, and is an ideal medium for nonlinear parametric generation

[8, 23, 30, 36, 82, 83, 85, 87, 107–110, 112–118]. TiO_2 is chosen as the linear material for a few reasons. First, TiO_2 deposited by physical vapor deposition is in an amorphous phase with the inversion symmetry leading to a vanished $\chi^{(2)}$. Second, characterization by ellipsometry shows that an amorphous TiO_2 thin film has a relatively large refractive index of 2.137 at 1550 nm and 2.186 at 775 nm, making it suitable for strong optical confinement and flexible dispersion engineering. Third, TiO_2 also has a large bandgap and has been demonstrated for high-quality waveguides from visible to telecom wavelengths [123–125], enabling it to be a low-loss core material that guides both FF and SH light.

The designed TiO_2/LN semi-nonlinear nanophotonic waveguide is schematically shown in Fig. 4.2(a). X -cut LN is employed for its large quadratic nonlinearity in the device plane, ideal for efficient type-0 processes involving quasi-TE modes. Numerical simulations show that the TE_{00} mode at 1550 nm can be phase matched to some quasi-TE modes at 775 nm with appropriate waveguide geometries [see Fig. 4.2(b)]. In particular, as shown in Fig. 4.2(c), a certain waveguide geometry is able to produce exact phase matching between the TE_{00} mode at 1550 nm and the TE_{01} at 775 nm, the latter of which exhibits an optical mode field with two opposite polarities located separately in the linear and nonlinear core layers [see Fig. 4.2(c), insets], a desired property shown in Fig. 4.1.

Detailed simulations show that the waveguide exhibits an overlap factor of $\zeta = 0.66$, a value well beyond what is achievable in monolithic nanophotonic

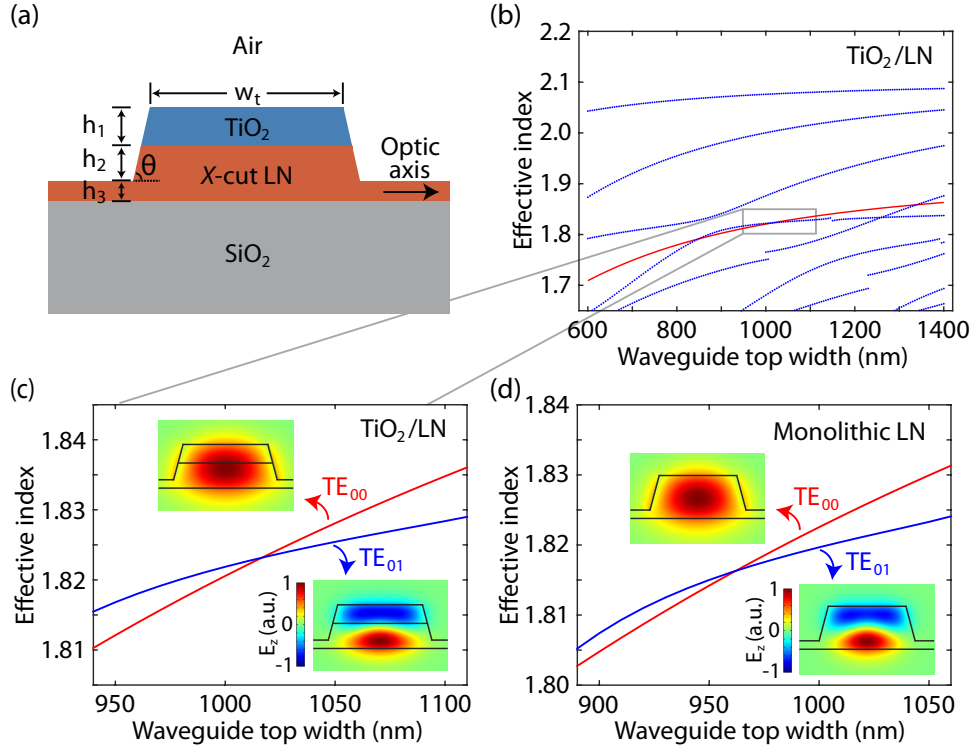


Figure 4.2: Design of a TiO₂/LN semi-nonlinear waveguide for an enhanced SHG efficiency. (a) Schematic of the TiO₂/LN waveguide. (b) Numerically simulated effective indices of the TiO₂/LN waveguide as functions of w_t . Other parameters are set as $h_1 = 220$ nm, $h_2 = 200$ nm, $h_3 = 100$ nm, and $\theta = 75^\circ$. The red solid curve is for the TE₀₀ mode at 1550 nm, and the blue dotted curves are for different quasi-TE modes at 775 nm. Discontinuities in some blue curves are due to coupling with certain quasi-TM modes (not shown). (c) Detailed effective indices of the TiO₂/LN waveguide as functions of w_t , showing modal phase matching between TE₀₀ at 1550 nm and TE₀₁ at 775 nm around $w_t = 1020$ nm. Insets show the field (E_z) profiles of both modes, with a shared color bar. (d) Detailed effective indices as functions of w_t , for a monolithic LN waveguide, showing phase matching between TE₀₀ at 1550 nm and TE₀₁ at 775 nm around $w_t = 960$ nm. Other than a smaller phase-matched w_t due to different material dispersion, the geometry parameters of the monolithic LN waveguide are the same as the TiO₂/LN waveguide in (b) and (c), while with the TiO₂ layer replaced by LN.

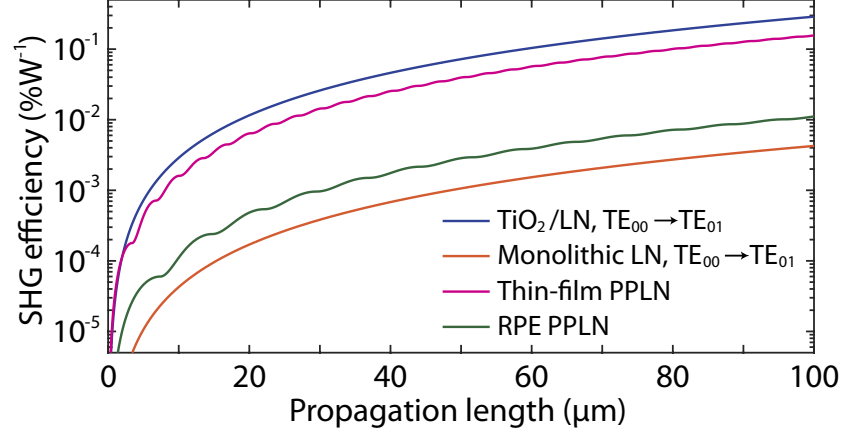


Figure 4.3: Evolution of SHG efficiencies in a TiO_2/LN semi-nonlinear waveguide [see Fig. 4.2(c)], a monolithic LN waveguide [see Fig. 4.2(d)], a thin-film PPLN waveguide (Ref. [30]), and a typical RPE PPLN waveguide (Ref. [116]), assuming no loss and no pump depletion for all types of waveguides, as well as ideally uniform poling for PPLN.

waveguides through modal phase matching [15, 36, 39, 81–83, 85, 87, 101, 103–108]. This large mode overlap factor, together with a small effective mode area of $A_{\text{eff}} = 2.24 \mu\text{m}^2$ and a large quadratic nonlinearity of $d_{\text{eff}} = d_{33} = 27 \text{ pm/V}$, enables the device to exhibit a theoretical normalized conversion efficiency as large as $\eta = 2900\% \text{ W}^{-1}\text{cm}^{-2}$. In contrast, a conventional monolithic LN waveguide with a similar geometry [see Fig. 4.2(d)] can only offer an efficiency of $\eta = 43\% \text{ W}^{-1}\text{cm}^{-2}$, similar to other reported LN nanophotonic waveguides with modal phase matching [83, 107]. Figure 4.3 compares the evolution of the SHG efficiency as the pump light propagates in several different waveguides. It shows clearly that the η value in the semi-nonlinear waveguide is more than 1 order of magnitude larger than that of typical RPE PPLN waveguides [116] and that of monolithic LN nanophotonic

waveguides [83,107], and almost doubles that of PPLN thin films loaded with SiN ridges [30].

4.3 Fabrication of TiO_2/LN Waveguides

To confirm the simulation results, device fabrication is performed. Due to the challenge in high-quality etching of heterogeneous structures, the LN waveguide is made through a standard top-down method [85, 107] and the TiO_2 layer is made via a lift-off approach, with the whole process schematically illustrated in Fig. 4.4(a).

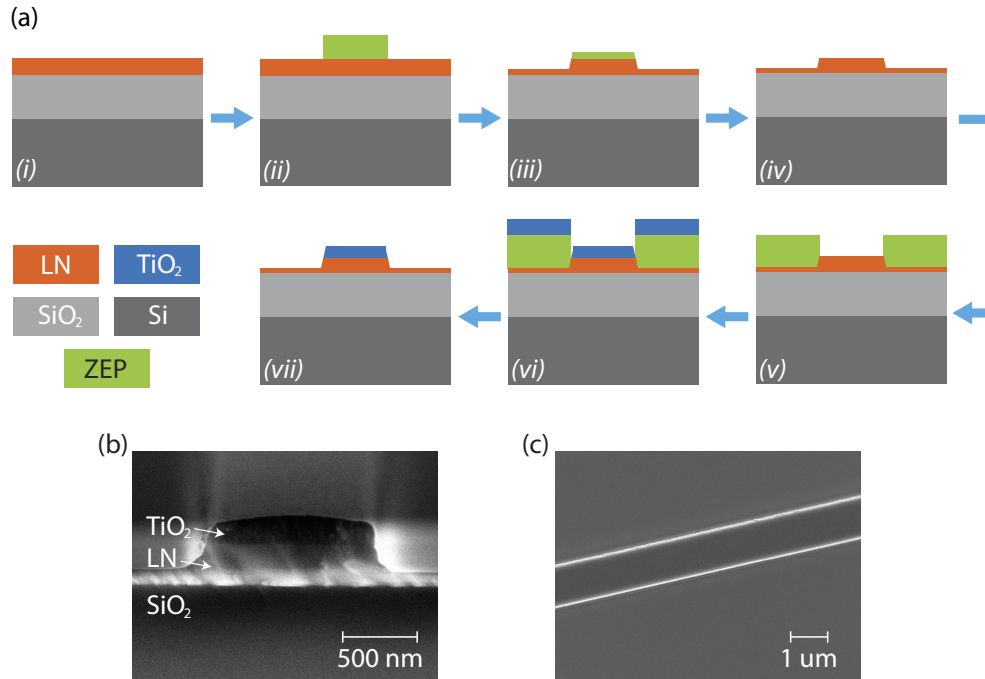


Figure 4.4: Fabrication of TiO_2/LN waveguides. (a) Process flow. Scanning electron microscopy image of the waveguide (b) facet, and (c) top view.

The fabrication process starts from an LNOI wafer with 300 nm of X -cut LN sitting on 2- μm -thick buried oxide and a silicon substrate (*i*). LN waveguides are patterned by electron-beam lithography, with ZEP520A as the resist (*ii*). After LN waveguides are etched with Argon ion milling (*iii*), residual resist is removed with oxygen plasma followed by diluted hydrofluoric acid (*iv*). Then, in order to pattern the TiO_2 layer on top of the etched LN waveguides, aligned electron-beam lithography is employed (*v*), after which physical vapor deposition is used to deposit TiO_2 (*vi*). Next, the wafer is soaked in 1165 resist remover, which removes the resist, leaving the heterogeneous TiO_2 /LN waveguides on the wafer (*vii*). Finally, the device chip is hand-cleaved for light coupling into the waveguides.

Figure 4.4(b) presents the facet of a typical fabricated waveguide, showing the TiO_2 layer landing nicely on top of the etched LN rib waveguide, with a geometry close to the design [see Fig. 4.2(c)]. Figure 4.4(c) presents the top view of the waveguide, which shows a small sidewall roughness, implying a low propagation attenuation.

4.4 Linear Optical Properties

The device is tested with an experimental setup similar to that shown in Fig. 3.5(a). To accurately characterize the waveguide propagation loss, microring resonators are fabricated, with an example shown in Fig. 4.5(a). The radius is 100 μm , which is chosen such that the bending loss is negligible.

Figure 4.5(b) presents the laser-scanned transmission spectrum, which shows clearly the TE_{00} mode family in the telecom band with a free-spectral range of 1.59 nm. Figure 4.5(c) presents the detailed transmission spectrum of a typical resonance with an intrinsic Q of 1.3×10^5 , indicating a waveguide propagation loss of 3.2 dB/cm in the telecom band [5]. For the SH mode around 775 nm, however, the current bus waveguide is very weakly coupled to the microring, making it difficult to measure the quality factor.

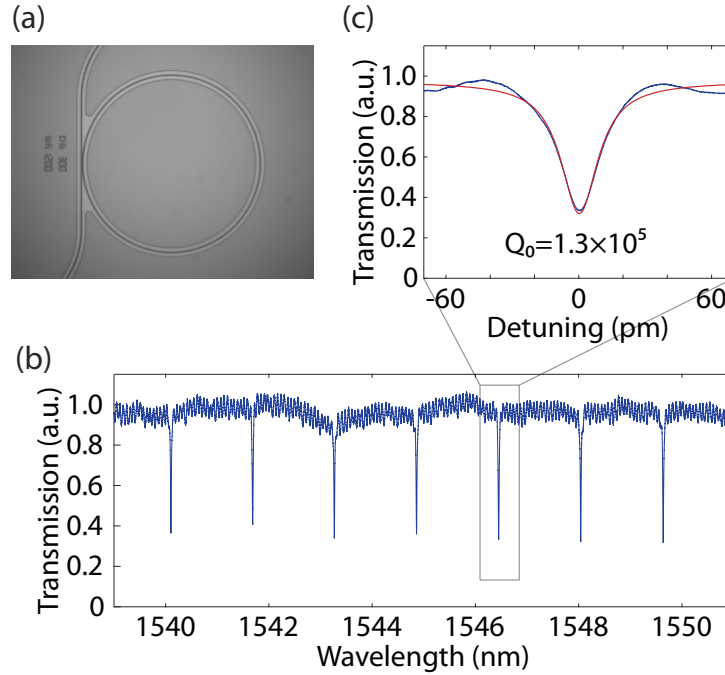


Figure 4.5: Characterization of TiO_2/LN semi-nonlinear waveguides. (a) Optical microscopy image of a TiO_2/LN microring resonator with a radius of 100 μm . (b) Transmission spectrum of the microring resonator in the telecom band for the TE polarization. (c) Detailed transmission spectrum of a typical resonance, with experimental data shown in blue and a fitting curve shown in red.

4.5 Second-Harmonic Generation

To demonstrate efficient SHG, a straight waveguide with a length of about 2.35 mm is employed, with its transmission in the telecom band shown in Fig. 4.6(a) for the TE polarization. From Fig. 4.6(a), a fiber-to-fiber loss of 12.5 dB is extracted. With a propagation loss of 3.2 dB/cm given by the microring resonator [see Fig. 4.5(c)], a fiber-to-chip coupling loss of 5.9 dB/facet is retrieved. By scanning the pump laser wavelength, the efficiency spectrum of SHG is able to be measured. One example is shown in Fig. 4.6(b), which indicates a phase-matched pump wavelength of about 1546 nm. The main lobe of the recorded efficiency spectrum agrees with the theoretical expectation from the sinc^2 -function. The strong side lobes are likely introduced by slight nonuniformity of the waveguide (e.g., potential thickness variations of the core layers).

By fixing the pump wavelength at 1546.10 nm where the peak conversion is located, coherent radiation from its SH at 773.05 nm is observed, shown as a sharp peak in Fig. 4.6(c). The measured SH power shows a quadratic dependence on the pump power [see Fig. 4.6(d)], which agrees with the theoretical expectation. Fitting the experimental data, an on-chip conversion efficiency of 36.0% W^{-1} is obtained [see Fig. 4.6(d)], indicating an experimentally recorded normalized conversion efficiency of 650% $\text{W}^{-1}\text{cm}^{-2}$. This value is about a 4-fold increase compared with the previous record of SHG in LN in the same waveband [30,117].

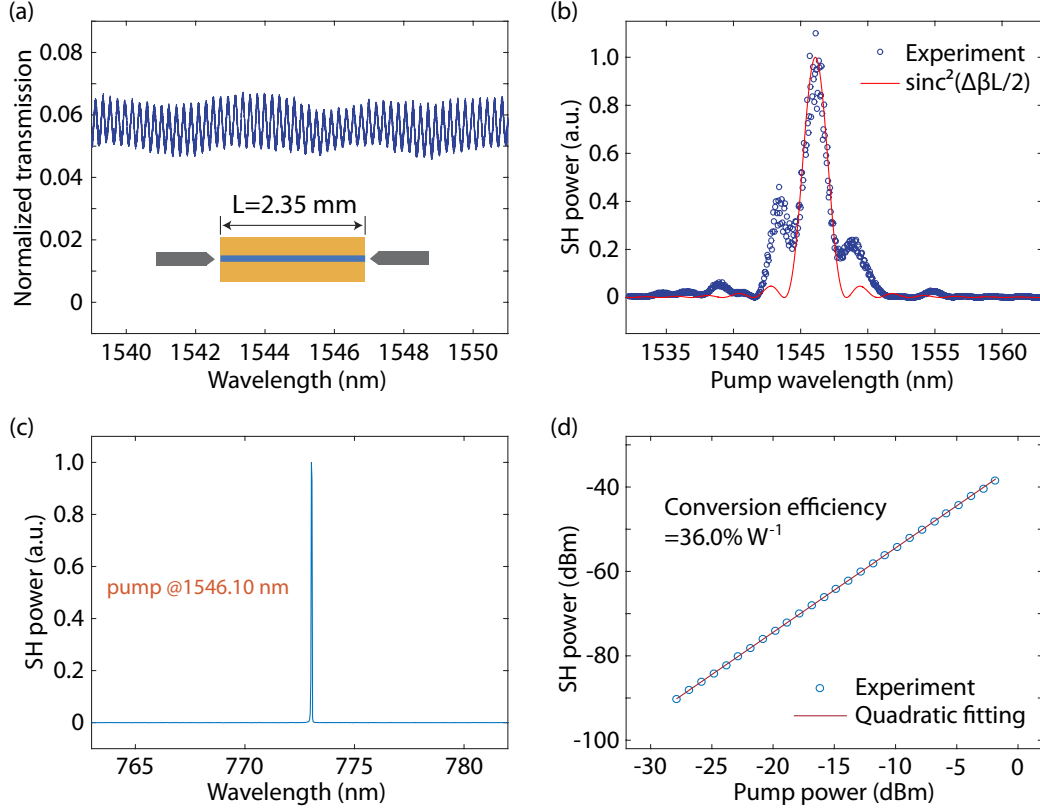


Figure 4.6: Measurement of SHG from a TiO_2/LN semi-nonlinear waveguide. (a) Transmission spectrum of a TiO_2/LN straight waveguide with a length of $L = 2.35$ mm, in the telecom band for the TE polarization. (b) Measured SHG spectrum (blue circles) of the TiO_2/LN waveguide. The red curve shows $\text{sinc}^2(\Delta\beta L/2)$ for comparison. (c) SHG spectrum of the device when it is pumped at a fixed wavelength of 1546.10 nm. (d) Power dependence of the SHG, with experimental data shown in blue and a theoretical fitting shown in red. The theoretical fitting indicates a conversion efficiency of $36.0\% \text{ W}^{-1}$.

The recorded SHG efficiency is smaller than the theoretical value given by ηL^2 ($= 160\% \text{ W}^{-1}$), which is mainly due to the propagation losses of the waveguide. A propagation loss of 3.2 dB/cm has been measured for the FF mode. As a rough estimate, the loss of the SH mode is assumed to be

12.8 dB/cm, since the propagation loss is dominated by Rayleigh scattering from the roughness of the waveguide surface, which scales with wavelength as $1/\lambda^2$ [93]. Consequently, the theoretical conversion efficiency is estimated to be 96.0% W^{-1} for the 2.35-mm-long waveguide, after the propagation losses are taken into account [see Eq. (3.26)]. Meanwhile, the coupling loss at the SH carried in a high-order waveguide mode is fairly underestimated, due to the large mode mismatch between the waveguide and the receiving lensed fiber, leading to a conservative estimation of the measured conversion efficiency. In addition, the slight waveguide nonuniformity could also impact the recorded conversion efficiency to a certain extent.

All these technical issues can be addressed with further optimization. On the one hand, unlike grating-assisted waveguides [83, 84, 122], the scattering loss in the current waveguide with translational symmetry is not physically fundamental, and can be mitigated by optimized fabrication. For example, chemical-mechanical polishing can be employed to increase the surface smoothness, as well as the waveguide uniformity [87]. On the other hand, a single-mode on-chip coupler can be utilized to minimize the facet coupling loss, especially at the SH [39]. Therefore, it is expected the measured conversion efficiency can be increased considerably in the near future.

4.6 Conclusion and Discussion

In conclusion, a universal design and operation principle is proposed and demonstrated for quadratic parametric processes on integrated photonic platforms, to achieve exact phase matching with a large nonlinearity, a small mode area, and a large mode overlap factor. With this principle, a TiO_2/LN semi-nonlinear waveguide is designed to offer a theoretical conversion efficiency as high as $2900\% \text{ W}^{-1}\text{cm}^{-2}$, more than 1 order of magnitude higher than those achievable in RPE PPLN and monolithic on-chip LN waveguides [83, 107, 116]. An SHG efficiency of $36.0\% \text{ W}^{-1}$ is measured with a waveguide only 2.35 mm long, corresponding to a normalized efficiency of $650\% \text{ W}^{-1}\text{cm}^{-2}$.

The normalized conversion efficiency of this device is comparable with recently reported thin-film PPLN waveguides [30, 110]. However, the demonstrated approach is free from the complicated periodic poling process. Domain engineering has been regarded as the holy grail for quadratic nonlinear photonics [111–113], which, however, is material selective. The proposed approach, instead, can be universally applied to any on-chip quadratic nonlinear platforms, including for example, certain dielectrics [15, 81], group IV [99, 100], III-V [39, 104, 106], and II-VI [111] semiconductor chips to which domain engineering is challenging to apply. Therefore, the proposed approach is promising for extreme nonlinear and quantum optics on various photonic platforms.

Nonlinear optical phenomena are closely related to the optical intensity. For example, as discussed in chapters 3 and 4, in an SHG process inside a waveguide, the generated SH power is quadratically dependent on the FF pump power. Thus, in order to further increase the nonlinear conversion efficiency on integrated photonic platforms, one can employ optical microresonators, which are able to largely enhance nonlinear optical interactions by boosting the optical intensity at resonance frequencies. In the following chapter, cavity-enhanced optical parametric generation is demonstrated in an LN microring resonator.

Chapter 5

Cavity-Enhanced Optical Parametric Generation in an LN Microring

Among all the integrated nonlinear photonic platforms, LN stands out due to its wide transparency window and strong quadratic optical nonlinearity, making it promising for integrated nonlinear photonic applications with low power consumption. To date, a wide variety of nanophotonic systems, including waveguides [30, 36, 83, 84, 107, 126, 127], microdisks [35, 82, 85, 108, 128, 129], microrings [130, 131], and photonic crystal cavities [132–134], have been studied for optical parametric processes in LN. In particular, cavity-enhanced nonlinear wavelength conversion has been demonstrated in doubly or triply resonant LN microresonators through a number of techniques including modal phase matching [35, 108, 128, 131], cyclic phase matching [82, 85, 129], and quasi-phase matching [130]. However, the potential of the LN integrated platform has not yet been fully explored, and LN microresonators have demonstrated only moderate efficiencies far from what the material can provide, due to unoptimized optical quality, mode selection, phase matching and/or light extraction.

In this chapter, cavity-enhanced optical parametric generation is presented in a high- Q Z -cut LN microring resonator through exact modal phase matching. The device exhibits optical Q 's of about 10^5 for the designed cavity modes in the 1550- and 780-nm bands, and both modes are well coupled to a single bus waveguide, conveniently enabling the measurement of SHG with a conversion efficiency of 1500% W^{-1} . In addition, by pumping into the mode in the 780-nm band, difference-frequency generation (DFG) in the telecom band is also observed. This work demonstrates the potential of modal-phase-matched LN microresonators for efficiency-enhanced optical parametric generation.

5.1 Theory of Second-Harmonic Generation in Optical Cavities

5.1.1 Coupled-Mode Equations

The derivation of the SHG efficiency in an optical cavity is similar to that in a waveguide, while here the optical modes are considered in the three-dimensional space, and mode amplitudes are time-dependent. Again, for simplicity, the material anisotropy is not fully accounted for. The derivation is first performed to obtain the coupling strength between two cavity modes assuming no optical losses, and loss terms and laser-cavity detunings are added in the simplified equations later.

Electric fields at optical frequencies ω_1 and $\omega_2(= 2\omega_1)$, which are around two cavity resonances, can be written as

$$\tilde{\mathbf{E}}_1(\mathbf{r}, t) = A_1(t)\mathbf{E}_1(\mathbf{r})e^{-i\omega_{10}t}, \quad (5.1)$$

$$\tilde{\mathbf{E}}_2(\mathbf{r}, t) = A_2(t)\mathbf{E}_2(\mathbf{r})e^{-i\omega_{20}t}, \quad (5.2)$$

where ω_{10} and ω_{20} are the resonance frequencies that are close to phase matching, i.e., $\Delta\omega \equiv \omega_{20} - 2\omega_{10} \ll \min(\omega_{\text{FSR},1}, \omega_{\text{FSR},2})$, with $\omega_{\text{FSR},1}$ and $\omega_{\text{FSR},2}$ being the free-spectral ranges; $\mathbf{E}_1(\mathbf{r})$ and $\mathbf{E}_2(\mathbf{r})$ are the mode profiles in the three-dimensional cavity; and $A_1(t)$ and $A_2(t)$ are the field amplitudes. By substituting Eqs. (3.3)-(3.4) and Eqs. (5.1)-(5.2) into the driven wave equations, Eqs. (3.1)-(3.2), the relation between the coupled field amplitudes is obtained as

$$\frac{dA_1}{dt}\epsilon(\mathbf{r}, \omega_{10})\mathbf{E}_1(\mathbf{r}) = \frac{i\omega_{10}}{2}A_1^*A_2e^{-i\Delta\omega t}\chi^{(2)}\mathbf{E}_1^*(\mathbf{r})\mathbf{E}_2(\mathbf{r}), \quad (5.3)$$

$$\frac{dA_2}{dt}\epsilon(\mathbf{r}, \omega_{20})\mathbf{E}_2(\mathbf{r}) = \frac{i\omega_{20}}{4}A_1^2e^{i\Delta\omega t}\chi^{(2)}\mathbf{E}_1(\mathbf{r})\mathbf{E}_1(\mathbf{r}), \quad (5.4)$$

where slowly evolving amplitudes, $|\frac{d^2A_j}{dt^2}| \ll |\omega_{j0}\frac{dA_j}{dt}|$, and weak dispersion, $\epsilon(\mathbf{r}, \omega_j) \approx \epsilon(\mathbf{r}, \omega_{j0})$ ($j = 1, 2$), have been assumed. Also, it has been considered that the mode profiles, $\mathbf{E}_1(\mathbf{r})$ and $\mathbf{E}_2(\mathbf{r})$, satisfy

$$\nabla^2\mathbf{E}_1(\mathbf{r}) + \epsilon(\mathbf{r}, \omega_1)\frac{\omega_{10}^2}{c^2}\mathbf{E}_1(\mathbf{r}) = 0, \quad (5.5)$$

$$\nabla^2\mathbf{E}_2(\mathbf{r}) + \epsilon(\mathbf{r}, \omega_{20})\frac{\omega_{20}^2}{c^2}\mathbf{E}_2(\mathbf{r}) = 0. \quad (5.6)$$

By multiplying $\mathbf{E}_1^*(\mathbf{r})$ on both sides of Eq. (5.3), multiplying $\mathbf{E}_2^*(\mathbf{r})$ on both sides of Eq. (5.4), and integrating both equations over all three-dimensional space, Eqs. (5.3)-(5.4) become

$$\frac{dA_1}{dt} = \frac{i\omega_{10}}{2} A_1^* A_2 e^{-i\Delta\omega t} \frac{\int \chi_{ijk}^{(2)} E_{1i}^* E_{1j}^* E_{2k} d^3x}{\int \epsilon(\mathbf{r}, \omega_{10}) |\mathbf{E}_1|^2 d^3x}, \quad (5.7)$$

$$\frac{dA_2}{dt} = \frac{i\omega_{20}}{4} A_1^2 e^{i\Delta\omega t} \frac{\int \chi_{ijk}^{(2)} E_{2i}^* E_{1j} E_{1k} d^3x}{\int \epsilon(\mathbf{r}, \omega_{20}) |\mathbf{E}_2|^2 d^3x}, \quad (5.8)$$

where Einstein summation convention has been used. The amplitudes can be normalized as

$$a'_1(t) = A_1(t) \sqrt{\frac{\epsilon_0}{2} \int \epsilon(\mathbf{r}, \omega_{10}) |\mathbf{E}_1(\mathbf{r})|^2 d^3x}, \quad (5.9)$$

$$a'_2(t) = A_2(t) \sqrt{\frac{\epsilon_0}{2} \int \epsilon(\mathbf{r}, \omega_{20}) |\mathbf{E}_2(\mathbf{r})|^2 d^3x}, \quad (5.10)$$

such that $|a'_1|^2$ and $|a'_2|^2$ represent intracavity optical energies at the FF and SH modes, respectively.

The amplitudes can be further normalized as

$$a_1(t) = \frac{a'_1(t)}{\sqrt{\hbar\omega_{10}}}, \quad (5.11)$$

$$a_2(t) = \frac{a'_2(t)}{\sqrt{\hbar\omega_{20}}}, \quad (5.12)$$

such that $|a_1|^2$ and $|a_2|^2$ represent intracavity photon numbers, with

$$\frac{da_1}{dt} = \frac{i\omega_{10}}{2} \sqrt{\frac{2\hbar\omega_{20}}{\epsilon_0}} a_1^* a_2 e^{-i\Delta\omega t} \frac{\int \chi_{ijk}^{(2)} E_{1i}^* E_{1j}^* E_{2k} d^3x}{\int \epsilon(\mathbf{r}, \omega_{10}) |\mathbf{E}_1|^2 d^3x \sqrt{\int \epsilon(\mathbf{r}, \omega_{20}) |\mathbf{E}_2|^2 d^3x}}, \quad (5.13)$$

$$\frac{da_2}{dt} = \frac{i\omega_{10}}{4} \sqrt{\frac{2\hbar\omega_{20}}{\epsilon_0}} a_1^2 e^{i\Delta\omega t} \frac{\int \chi_{ijk}^{(2)} E_{2i}^* E_{1j} E_{1k} d^3x}{\int \epsilon(\mathbf{r}, \omega_{10}) |\mathbf{E}_1|^2 d^3x \sqrt{\int \epsilon(\mathbf{r}, \omega_{20}) |\mathbf{E}_2|^2 d^3x}}, \quad (5.14)$$

Similar to what has been done for waveguides, the above equations can be simplified when there is only one nonlinear medium in the system. In this case, the effective mode volume can be defined as $V_{\text{eff}} \equiv (V_1^2 V_2)^{\frac{1}{3}}$, with

$$V_j = \frac{[\int_{\text{all}} \epsilon(\mathbf{r}, \omega_j) |\mathbf{E}_j|^2 d^3x]^3}{|\int_{\chi^{(2)}} \epsilon(\mathbf{r}, \omega_j)^{\frac{3}{2}} |\mathbf{E}_j|^2 \mathbf{E}_j d^3x|^2}, \quad (j = 1, 2), \quad (5.15)$$

and the mode overlap factor is written as

$$\zeta_{ijk} = \frac{\int_{\chi^{(2)}} E_{2i}^* E_{1j} E_{1k} d^3x}{|\int_{\chi^{(2)}} |\mathbf{E}_1|^2 \mathbf{E}_1 d^3x|^{\frac{2}{3}} |\int_{\chi^{(2)}} |\mathbf{E}_2|^2 \mathbf{E}_2 d^3x|^{\frac{1}{3}}}}, \quad (5.16)$$

where $\int_{\chi^{(2)}}$ and $\int_{\text{all}}$ denote integration over the nonlinear medium and all space, respectively. As a result, the single-photon coupling strength is written as

$$\gamma = \sqrt{\frac{\hbar\omega_{10}\omega_{20}}{8\epsilon_0\tilde{\epsilon}_1^2\tilde{\epsilon}_2}} \frac{\chi_{ijk}^{(2)} \zeta_{ijk}}{\sqrt{V_{\text{eff}}}}, \quad (5.17)$$

where $\tilde{\epsilon}_1$ and $\tilde{\epsilon}_2$ are the relative permittivity of the nonlinear medium at

ω_{10} and ω_{20} , respectively. Equation (5.17) is a general form of the coupling strength for SHG in an optical cavity.

When the mode overlap factor has a dominant contribution from a single term, e.g., $\int_{\chi^{(2)}} E_{2z}^* E_{1z}^2 d^3x$ due to $\chi_{333}^{(2)}$, it can be simplified to

$$\zeta = \frac{\int_{\chi^{(2)}} E_{2z}^* E_{1z}^2 d^3x}{|\int_{\chi^{(2)}} |\mathbf{E}_1|^2 \mathbf{E}_1 d^3x|^{\frac{2}{3}} |\int_{\chi^{(2)}} |\mathbf{E}_2|^2 \mathbf{E}_2 d^3x|^{\frac{1}{3}}}. \quad (5.18)$$

For a rotationally symmetric microring with a relatively large radius, the transverse mode profiles can be approximated by those of the corresponding waveguide modes, $\mathbf{E}_1^{(\text{wg})}$ and $\mathbf{E}_2^{(\text{wg})}$; thus, the overlap factor can be further represented as that of the waveguide, i.e.,

$$\zeta = \zeta^{(\text{wg})} = \frac{\int_{\chi^{(2)}} (E_{2z}^{(\text{wg})})^* (E_{1z}^{(\text{wg})})^2 dxdz}{|\int_{\chi^{(2)}} |\mathbf{E}_1^{(\text{wg})}|^2 \mathbf{E}_1^{(\text{wg})} dxdz|^{\frac{2}{3}} |\int_{\chi^{(2)}} |\mathbf{E}_2^{(\text{wg})}|^2 \mathbf{E}_2^{(\text{wg})} dxdz|^{\frac{1}{3}}}. \quad (5.19)$$

Similarly, the effective volume of the cavity can be represented as the product of the waveguide effective area $A_{\text{eff}}^{(\text{wg})}$ and the cavity length L , i.e., $V_{\text{eff}} = A_{\text{eff}}^{(\text{wg})} L$, where $A_{\text{eff}}^{(\text{wg})} \equiv [(A_1^{(\text{wg})})^2 A_2^{(\text{wg})}]^{\frac{1}{3}}$, with

$$A_j^{(\text{wg})} = \frac{[\int_{\text{all}} \epsilon(\mathbf{r}, \omega_{j0}) |\mathbf{E}_j^{(\text{wg})}|^2 dxdz]^3}{|\int_{\chi^{(2)}} \epsilon(\mathbf{r}, \omega_{j0})^{\frac{3}{2}} |\mathbf{E}_j^{(\text{wg})}|^2 \mathbf{E}_j^{(\text{wg})} dxdz|^2}, \quad (j = 1, 2). \quad (5.20)$$

Therefore, the coupling strength can then be written as

$$\gamma = \sqrt{\frac{\hbar \omega_{10}^2 \omega_{20}}{2 \epsilon_0 \tilde{\epsilon}_1^2 \tilde{\epsilon}_2}} \frac{d_{\text{eff}} \zeta^{(\text{wg})}}{\sqrt{A_{\text{eff}}^{(\text{wg})} L}}, \quad (5.21)$$

where $d_{\text{eff}} = d_{33} = \chi_{333}^{(2)}/2$. Note that Eq. (5.21) only contains basic physical quantities of the cavity, and information about the optical modes is from the corresponding waveguide. As a result, the coupled-mode equations can be simplified to

$$\frac{da_1}{dt} = 2i\gamma^* e^{-i\Delta\omega t} a_1^* a_2, \quad (5.22)$$

$$\frac{da_2}{dt} = i\gamma e^{i\Delta\omega t} a_1^2. \quad (5.23)$$

When the two cavity modes have finite linewidths and external pump fields, the driven coupled-mode equations can be written as

$$\frac{da_1}{dt} = -\frac{\kappa_{1t}}{2} a_1 + 2i\gamma^* e^{-i\Delta\omega t} a_1^* a_2 + i\sqrt{\kappa_{1e}} F_1 e^{-i\delta_1 t}, \quad (5.24)$$

$$\frac{da_2}{dt} = -\frac{\kappa_{2t}}{2} a_2 + i\gamma e^{i\Delta\omega t} a_1^2 + i\sqrt{\kappa_{2e}} F_2 e^{-i\delta_2 t}, \quad (5.25)$$

where κ_{1t} (κ_{2t}) is the total cavity loss rate, κ_{1e} (κ_{2e}) is the external coupling rate, and $\delta_1 \equiv \omega_1 - \omega_{10}$ ($\delta_2 \equiv \omega_2 - \omega_{20}$) is the laser-cavity detuning of the FF (SH), and F_1 (F_2) is the amplitude of the driving field at ω_1 (ω_2). In an SHG process, only the FF mode is externally pumped, leading to

$$\frac{da_1}{dt} = -\frac{\kappa_{1t}}{2} a_1 + 2i\gamma^* e^{-i\Delta\omega t} a_1^* a_2 + i\sqrt{\kappa_{1e}} F_1 e^{-i\delta_1 t}, \quad (5.26)$$

$$\frac{da_2}{dt} = -\frac{\kappa_{2t}}{2} a_2 + i\gamma e^{i\Delta\omega t} a_1^2. \quad (5.27)$$

In order to remove the slowly oscillating factors, $e^{\pm i\Delta\omega t}$ and $e^{-i\delta_1 t}$, Eqs. (5.26)-

(5.27) can be written in a rotating reference frame ($a_1 \rightarrow a_1 e^{-i\delta_1 t}$, $a_2 \rightarrow a_2 e^{-i\delta_2 t}$), which yields

$$\frac{da_1}{dt} = (i\delta_1 - \frac{\kappa_{1t}}{2})a_1 + 2i\gamma^* a_1^* a_2 + i\sqrt{\kappa_{1e}}F_1, \quad (5.28)$$

$$\frac{da_2}{dt} = [i(2\delta_1 - \Delta\omega) - \frac{\kappa_{2t}}{2}]a_2 + i\gamma a_1^2, \quad (5.29)$$

where the energy conservation law, $\omega_2 = \omega_{20} + \delta_2 = 2(\omega_{10} + \delta_1) = 2\omega_1$, has been applied to obtain $\delta_2 = 2\delta_1 - \Delta\omega$. In this case, an amplitude can be defined as $G_2 = i\sqrt{\kappa_{2e}}a_2$, with $|G_2|^2$ representing the flux of SH photons at the cavity output.

5.1.2 Without Pump Depletion

In the weak-conversion regime without pump depletion, i.e., $|a_1| \gg |a_2|$, the cross term in Eq. (5.28) is negligible, and Eq. (5.28) can be simplified to

$$\frac{da_1}{dt} = (i\delta_1 - \frac{\kappa_{1t}}{2})a_1 + i\sqrt{\kappa_{1e}}F_1. \quad (5.30)$$

For a CW pump, the steady state requires $da_1/dt = da_2/dt = 0$ in Eqs. (5.29)-(5.30), and the normalized amplitudes are written as

$$a_1 = \frac{i\sqrt{\kappa_{1e}}F_1}{(\kappa_{1t}/2) - i\delta_1}, \quad (5.31)$$

$$a_2 = \frac{i\gamma a_1^2}{(\kappa_{2t}/2) - i(2\delta_1 - \Delta\omega)}. \quad (5.32)$$

Thus, the relation between the pump photon flux, $N_1 = |F_1|^2$, and the output SH photon flux, $N_2 = |G_2|^2$, is obtained as

$$N_2 = N_1^2 |\gamma|^2 \frac{\kappa_{1e}^2}{[(\kappa_{1t}/2)^2 + \delta_1^2]^2} \frac{\kappa_{2e}}{(\kappa_{2t}/2)^2 + (2\delta_1 - \Delta\omega)^2}. \quad (5.33)$$

With the input FF power, $P_1 = N_1 \hbar \omega_1$, and the output SH power, $P_2 = N_2 \hbar \omega_2$, the conversion efficiency can be calculated as

$$\Gamma \equiv \frac{P_2}{P_1^2} = \frac{|\gamma|^2 \omega_2}{\hbar \omega_1^2} \frac{\kappa_{1e}^2}{[(\kappa_{1t}/2)^2 + \delta_1^2]^2} \frac{\kappa_{2e}}{(\kappa_{2t}/2)^2 + (2\delta_1 - \Delta\omega)^2}. \quad (5.34)$$

In the case of exact phase matching and on-resonance pumping, i.e., $\Delta\omega = \delta_1 = 0$, Eq. (5.34) can be written with regard to quality factors, $Q_{t(e)} = \omega / \kappa_{t(e)}$, as follows:

$$\Gamma = \frac{64 |\gamma|^2}{\hbar \omega_1^4} \frac{Q_{1t}^4 Q_{2t}^2}{Q_{1e}^2 Q_{2e}}. \quad (5.35)$$

5.1.3 With Pump Depletion

When the pump is depleted, the cross term in Eq. (5.28) cannot be neglected, and in the steady state,

$$(i\delta_1 - \frac{\kappa_{1t}}{2})a_1 + 2i\gamma^* e^{-i\Delta\omega t} a_1^* a_2 + i\sqrt{\kappa_{1e}} F_1 = 0, \quad (5.36)$$

$$a_2 = \frac{i\gamma a_1^2}{(\kappa_{2t}/2) - i\delta_2}. \quad (5.37)$$

Eq. (5.37) can be substituted into Eq. (5.36), leading to

$$[(i\delta_1 - \frac{\kappa_{1t}}{2}) - \frac{2|\gamma|^2 e^{-i\Delta\omega t} |a_1|^2}{(\kappa_{2t}/2) - i\delta_2}]a_1 + i\sqrt{\kappa_{1e}}F_1 = 0. \quad (5.38)$$

In the case of perfect phase matching and on-resonance pumping, *i.e.* $\Delta\omega = \delta_1 = \delta_2 = 0$, $|a_1|^2$ and $|a_2|^2$ can be solved with

$$(\frac{\kappa_{1t}}{2} + \frac{4}{\kappa_{2t}}|\gamma|^2|a_1|^2)|a_1|^2 = \kappa_{1e}|F_1|^2, \quad (5.39)$$

$$|a_2|^2 = \frac{4}{\kappa_{2t}}|\gamma|^2|a_1|^2. \quad (5.40)$$

5.2 Design and Characterization

In order to achieve modal phase matching in a microresonator, numerical simulations are performed with a Z -cut LN thin film, whose optic axis lies vertically, showing no anisotropy of refractive index in the device plane. To utilize the largest nonlinear term d_{33} , phase matching is achieved between the fundamental mode TM_{00} at 1550 nm and a high-order mode TM_{20} at 775 nm. For simplicity, effective indices of optical modes are simulated in a straight waveguide, as a guideline for microring resonators with a relatively large radius, which is 50 μm in the current study. Figure 5.1(a) presents the simulation result, which shows that for a waveguide thickness of 600 nm, modal phase matching happens for TM_{00} at 1550 nm and TM_{20} at 775 nm [see Fig. 5.1(b)], when the waveguide top width is about 690 nm. For a

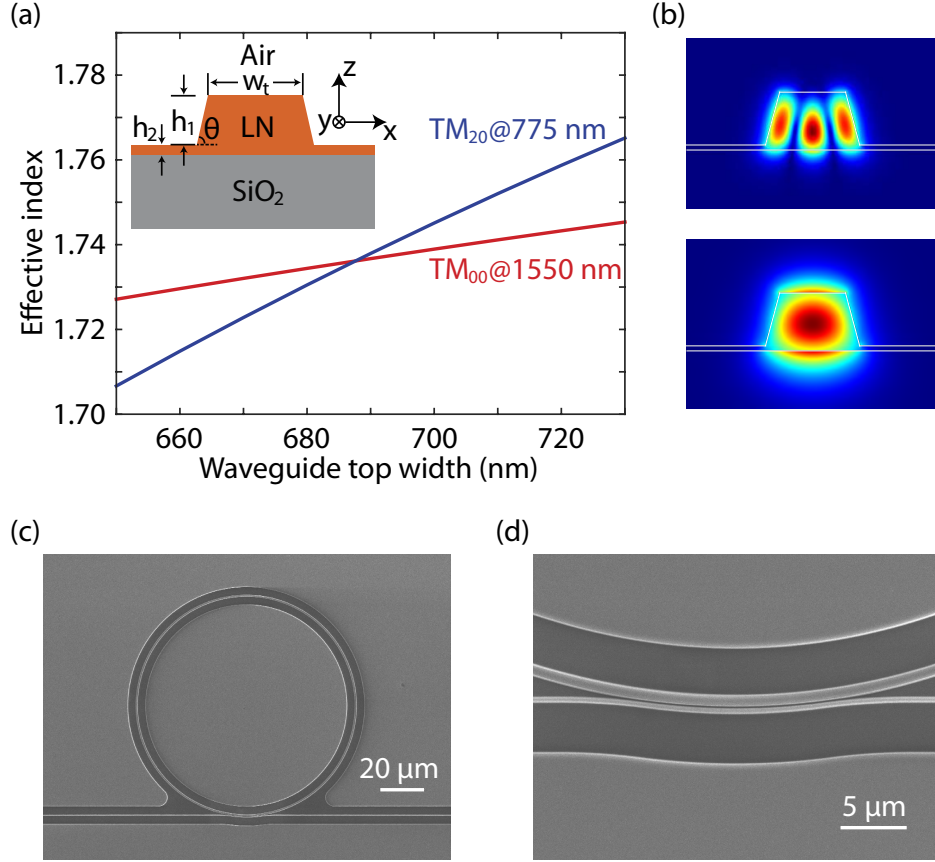


Figure 5.1: (a) Numerically simulated effective indices of the TM_{20} mode at 775 nm and the TM_{00} mode at 1550 nm in a straight waveguide, as functions of waveguide top width w_t . Other geometry parameters are $h_1 = 550$ nm, $h_2 = 50$ nm, and $\theta = 75^\circ$. (b) Mode profiles of TM_{20} at 775 nm and TM_{00} at 1550 nm. (c) Scanning electron microscopy image of the LN microring. (d) Enlargement of the bus-ring coupling region.

microring resonator with the same cross section, since the Z-cut LN thin film is isotropic in the device plane, the phase-matching condition is consistently satisfied at any azimuthal angle, which is expected to produce strong SHG as the phase-matched FF light travels around the cavity.

The fabrication starts from a *Z*-cut LNOI wafer, with a 600-nm-thick LN thin film sitting on a 3- μ m-thick buried oxide layer and a silicon substrate, and the process is similar to that for microdisk resonators (see Fig. 2.3), except for an extra amorphous-silicon hard mask that is helpful for etching features with high aspect ratios [134]. Figure 5.1(c) shows a fabricated microring resonator, coupled to a pulley waveguide [135,136], and Fig. 5.1(d) gives a closer look at the coupling region. Later device characterization shows that a bus waveguide top width of approximately 200 nm, a gap (measured at the top surface of the LN thin film) of approximately 350 nm, and a coupling length of approximately 20 μ m are able to give good coupling for both the FF mode and the SH mode.

After fabricating the device, experiments are conducted to characterize its linear optical properties and demonstrate nonlinear parametric generation, with the setup shown in Fig. 5.2. Two CW tunable lasers are used, one in the telecom band around 1550 nm, the other in the NIR around 780 nm. Light from both lasers is combined by a 780/1550 WDM, and launched into the on-chip bus waveguide via a lensed fiber. The bus waveguide couples light at both wavebands into and out of the microring, inside of which nonlinear optical parametric processes take place. A second lensed fiber is used to collect output light from the chip, and a second 780/1550 WDM is utilized to separate light at the two wavebands. At the 1550 port of the WDM, a long-pass filter that passes light with a wavelength over 1100 nm is used to eliminate residual NIR light, and the telecom light is further split into two

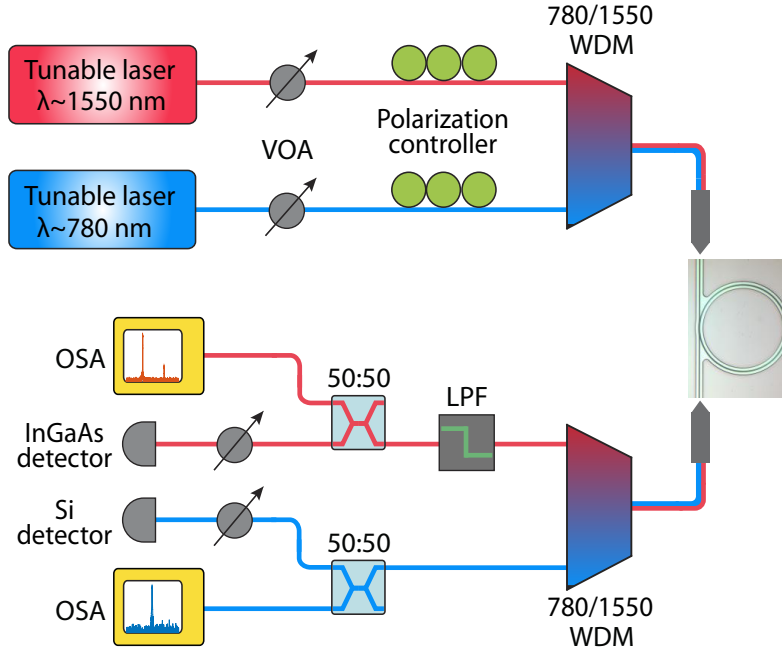


Figure 5.2: Schematic of the experimental setup for device characterization and optical parametric generation. VOA: variable optical attenuator; WDM: wavelength-division multiplexer; LPF: long-pass filter; OSA: optical spectrum analyzer.

paths, one to an InGaAs detector for characterization, and the other to an OSA for spectral analysis of DFG; at the 780 port, the NIR light is also split into two paths, one to a silicon detector for characterization, and the other to an OSA for detection of SHG. Variable optical attenuators are employed to study power-dependent properties, and polarization controllers are used for optimal coupling of the wanted polarization. The lensed fibers are made from 780HP fibers that are single-mode at both wavebands. The 780HP lensed fibers are kept straight to avoid the large bending loss at the telecom band, and are spliced with SMF28 fibers that are connected to the WDMs.

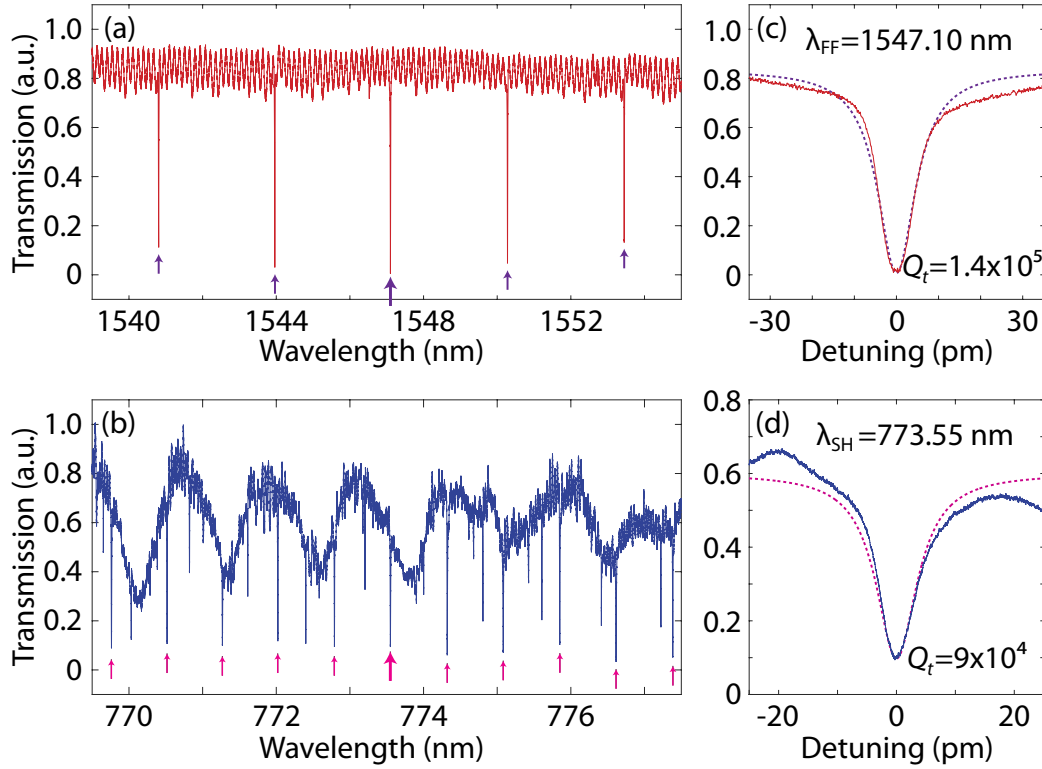


Figure 5.3: Transmission spectra of an LN microring for the TM polarization, near (a) 1550 nm, and (b) 780 nm. TM_{00} modes around 1550 nm and TM_{20} modes around 780 nm are indicated by purple and magenta arrows, respectively, with two big arrows showing the phase-matched mode pair. (c),(d) Detailed transmission spectra of the phase-matched modes, with experimental data shown in solid curves and fittings shown in dashed curves.

In order to obtain the linear optical properties of the microring resonator, both lasers are wavelength-scanned to measure the transmission spectra near both 1550 and 780 nm, as shown in Fig. 5.3(a)(b). The device exhibits a single TM mode family near 1550 nm, and the mode at 1547.10 nm, which is the FF mode for modal-phase-matched SHG, is almost critically coupled, with a coupling depth of approximately 99% and a loaded optical Q of

1.4×10^5 [see Fig. 5.3(c)]. On the other hand, the SH mode at 773.55 nm is under-coupled, with a coupling depth of approximately 83% and a loaded optical Q of 9×10^4 [see Fig. 5.3(d)]. To achieve exact phase matching, the FF and SH resonances are fine-tuned by controlling the temperature with a thermoelectric cooler under the device chip. The fiber-to-chip coupling losses are about 6.9 and 11.4 dB/facet for the FF and SH modes, respectively. These phase-matched resonances with high optical Q 's, together with the large nonlinearity in the designed type-0 process using $d_{\text{eff}} = d_{33} = 27$ pm/V, indicate efficient optical parametric generation with cavity enhancement.

5.3 Optical Parametric Generation

To study SHG in the microring resonator, pump light is launched into the FF mode at 1547.10 nm, and strong scattering of NIR light generated from the resonator is observed by an optical microscope, with an example shown in the inset of Fig. 5.4. By varying the pump power, the power dependence of the SHG is obtained, as shown in Fig. 5.4. The experimental data exhibit a quadratic relation between the generated SH power and the FF pump power, which is the signature of SHG in the regime without pump depletion. The measured conversion efficiency is $1500\% \text{ W}^{-1}$. This efficiency is more than 1 order of magnitude higher than those in many other LN microresonators [35, 82, 85, 108, 130–134]. It is even comparable with a recent study of cyclic phase matching in an X -cut microdisk exhibiting an ultrahigh

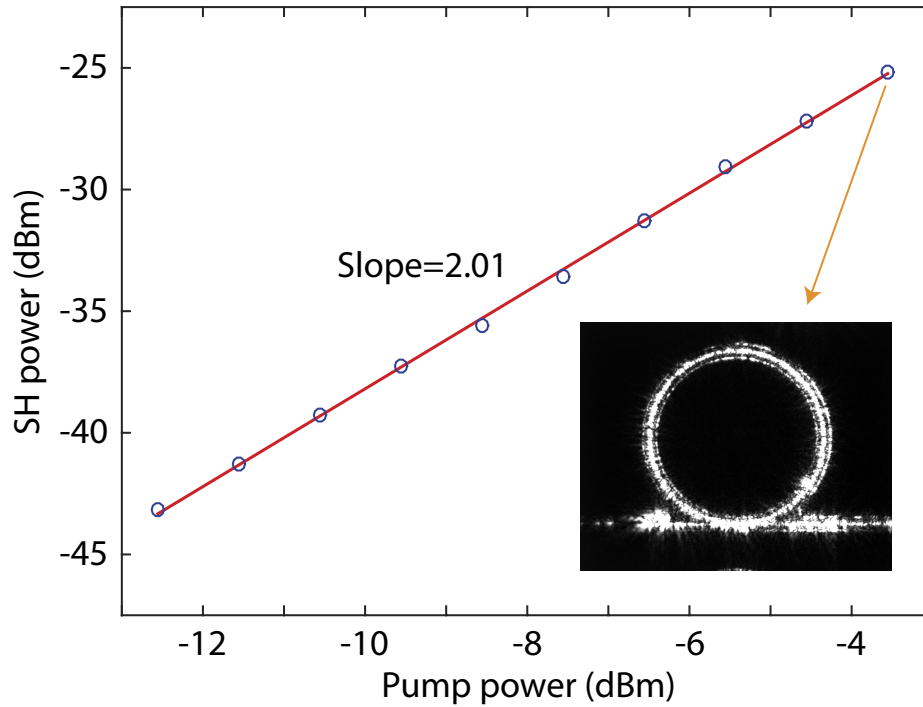


Figure 5.4: Power dependence of SHG, showing a quadratic relation between the SH power and the pump power. The measured conversion efficiency is $1500\% \text{ W}^{-1}$. The inset shows an optical image of generated SH light scattered from the microring when the pump power is $440 \mu\text{W}$.

Q of about 10^7 [129], 2 orders of magnitude higher than that of the currently studied microring resonator, due to the use of modal phase matching that offers a major advantage. With future optimization of the optical Q 's, by using a thicker LN film and an oxide cladding to reduce the sidewall scattering loss, a further increase in the conversion efficiency is expected.

The measured efficient SHG validates phase matching in the microring, and also indicates its capability for other optical parametric processes. In order to explore this, optical power is launched at both the SH mode, and one

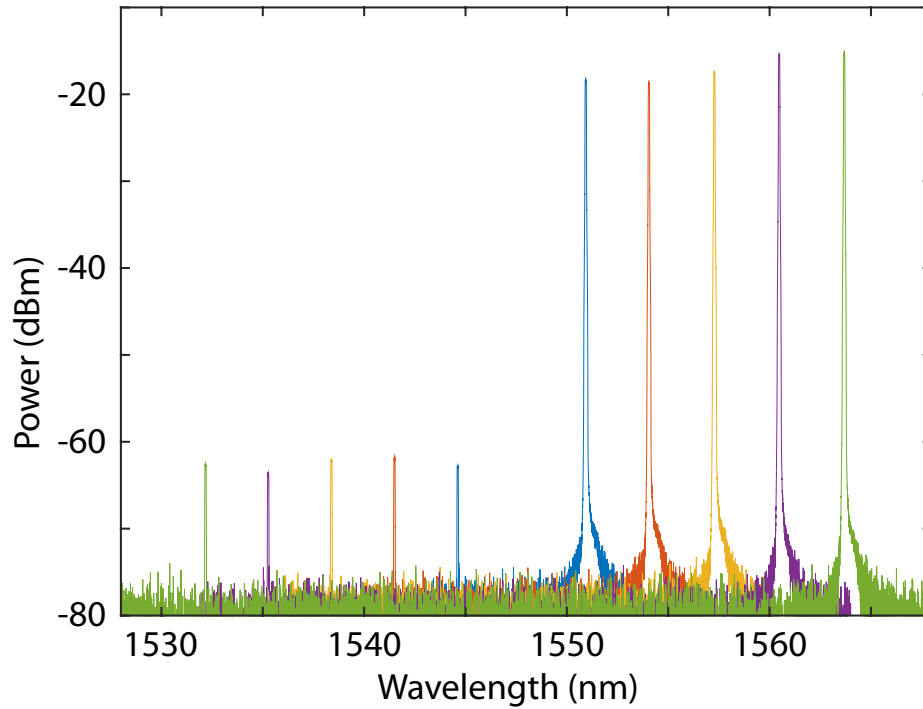


Figure 5.5: Recorded DFG spectra, when pumping at both the SH mode in the NIR and one of the five nearest modes with longer wavelengths than the FF mode in the telecom band. The pump power at the SH mode is $6.6 \mu\text{W}$.

of the modes near the FF mode. Figure 5.5 presents the recorded spectra in the telecom band. With only $6.6 \mu\text{W}$ of on-chip power at the SH mode in the NIR, long-wavelength telecom light is able to be coherently converted into shorter wavelengths through DFG. The long-wavelength seed power launched on chip is $105 \mu\text{W}$, and the generated power at the difference frequencies is about 480 pW , indicating a conversion rate of about -53 dB .

5.4 Discussion

With Eq. (5.35), the SHG efficiency in the LN microring is calculated to be $\Gamma \approx 30000\% \text{ W}^{-1}$. Thus, there is a difference of more than 1 order of magnitude between the theoretical prediction and our experimental result. The main reason for this discrepancy is likely nonuniformity of the microring at different azimuthal angles. By simulation, a change of 1 nm in the waveguide width, for example, will lead to a shift of about -3 nm in the phase-matched pump wavelength of SHG. Considering the small linewidths of our cavity modes, which are only 11 pm for the FF mode and 9 pm for the SH mode, the local phase-matching window at some azimuthal angle is easily shifted out of the cavity resonances due to fabrication imperfections. In the current work, relevant fabrication imperfections include the noncircular beam spot in electron-beam lithography, the proximity effect in lithography and etching, and the nonuniformity in the thickness of the LN thin film, all of which could prevent the fabricated microring from exhibiting perfect rotational symmetry.

The discrepancy between theory and experiment can be further explained analytically. For the two cavity modes in the microring resonator, the resonance frequencies are determined by the overall resonant condition of

the resonator (related to the total phase shift per round trip), given as

$$\frac{\omega_{10}}{c} \bar{n}_1(\omega_{10}) L \equiv \frac{\omega_{10}}{c} \int_0^{2\pi} n_1(\omega_{10}, \phi) R d\phi = 2m_1\pi, \quad (5.41)$$

$$\frac{\omega_{20}}{c} \bar{n}_2(\omega_{20}) L \equiv \frac{\omega_{20}}{c} \int_0^{2\pi} n_2(\omega_{20}, \phi) R d\phi = 2m_2\pi, \quad (5.42)$$

where L is the cavity length, $\bar{n}_1$ and $\bar{n}_2$ are the effective indices averaged over the cavity, and m_1 and m_2 are the azimuthal mode orders, with $m_2 = 2m_1$. Even with fabrication imperfections, global phase matching can be conveniently achieved between the two cavity modes, i.e., $\omega_{20} = 2\omega_{10}$ and $\bar{n}_2(\omega_{20}) = \bar{n}_1(\omega_{10})$.

Despite the global phase matching, the measured SHG efficiency can be lower than the theoretical prediction, due to imperfect rotation symmetry of the fabricated device. The SHG efficiency is determined by the mode overlap between the two cavity modes over the whole cavity,

$$\zeta \propto \int_{\chi^{(2)}} E_{2z}^* E_{1z}^2 d^3x = \xi \int_{\chi^{(2)}} (E_{2z}^{(T)})^* (E_{1z}^{(T)})^2 d\rho dz, \quad (5.43)$$

where $\mathbf{E}_1^{(T)}(\rho, z)$ and $\mathbf{E}_2^{(T)}(\rho, z)$ are transverse mode profiles of the cavity modes; and ξ is an integrated phase mismatch factor written as

$$\xi = \frac{1}{L} \int_0^{2\pi} e^{i\frac{\omega_{20}}{c} [n_2(\omega_{20}, \phi) - n_1(\omega_{10}, \phi)] R \phi} R d\phi. \quad (5.44)$$

In the ideal case with $n_1(\omega_{10}, \phi) = n_2(\omega_{20}, \phi) = \text{const}$ at any azimuthal

angle, ξ has a maximal value of 1, leading to the maximal conversion efficiency calculated by Eq. (5.35). However, the ϕ -dependent variations of $n_1(\omega_{10}, \phi)$ and $n_2(\omega_{20}, \phi)$, which are introduced by fabrication imperfections, can decrease the value of ξ , resulting in a degraded conversion efficiency.

5.5 Conclusion

In conclusion, cavity-enhanced optical parametric generation is demonstrated in an LN microring resonator with modal phase matching. A single bus waveguide is utilized to conveniently couple the FF and SH modes, both exhibiting coupling depths of over 80% and loaded optical Q 's of around 10^5 , resulting in a measured conversion efficiency of 1500% W^{-1} for SHG. In addition, DFG is also observed in the telecom band. This work represents an important step toward ultrahighly efficient optical parametric generation in photonic circuits based on the LN integrated platform.

In this chapter, a Z -cut LN microring is presented to achieve parametric generation through modal phase matching. Microresonators can also be produced based on X -cut LN, whose optic axis lies in the device plane. Interestingly, this type of microresonators offer a unique phase-matching scheme, cyclic phase matching, which can lead to unconventional nonlinear optical phenomena. In the next chapter, broadband spontaneous parametric down-conversion is demonstrated in an X -cut LN microdisk resonator.

Chapter 6

Broadband Spontaneous Parametric Down-Conversion with Cyclic Phase Matching

Quadratic nonlinear optical processes generally exhibit limited spectral bandwidths due to the material dispersion of nonlinear media that seriously impacts the phase-matching condition, which hinders the development of practical applications. This issue remains for on-chip devices, including microresonators with the benefit of cavity enhancement. Quadratic nonlinear optics enhanced by optical resonators are studied via kinds of phase-matching techniques [137, 138], including modal phase matching that is discussed in chapter 5. However, the precise wavelength matching between narrow-linewidth cavity modes imposes a serious technical challenge in practice.

Interestingly, in an optical cavity made of an anisotropic material, the effective index of a cavity mode can oscillate sinusoidally as the optical wave travels around. This leads to a technique called cyclic phase matching, which is able to achieve quadratic nonlinear optical processes with a large

tolerance in design and fabrication. Over the past few years, cyclic phase matching has been used for SHG in optical resonators made of various uniaxial crystals [82, 137–139].

In this chapter, broadband spontaneous parametric down-conversion (SPDC) is demonstrated in a high- Q X -cut LN microdisk resonator, whose rich mode species are able to support broadband nonlinear optical interactions through cyclic phase matching. The recorded SPDC spectrum infers a total bandwidth over 400 nm, which is significantly beyond conventional dispersion-engineering and phase-matching approaches [15, 30, 35, 39, 82–84, 104, 105, 109, 140–142]. The remarkably broadband SPDC among multiple cavity mode families is promising for integrated quantum photonic applications that could potentially utilize the frequency degree of freedom for information coding and processing [143, 144].

6.1 Concept Illustration

The physical mechanism of cyclic phase matching is illustrated in Fig. 6.1. The device is an anisotropic cavity with the optic axis lying in the device plane (e.g., an X -cut LN microresonator). As shown in Fig. 6.1(a), for a quasi-TE mode whose polarization lies dominantly in the device plane, the experienced material refractive index oscillates between n_o and n_e as light travels around the cavity. As a result, the local effective index oscillates sinusoidally in every round trip, with the oscillating amplitude determined

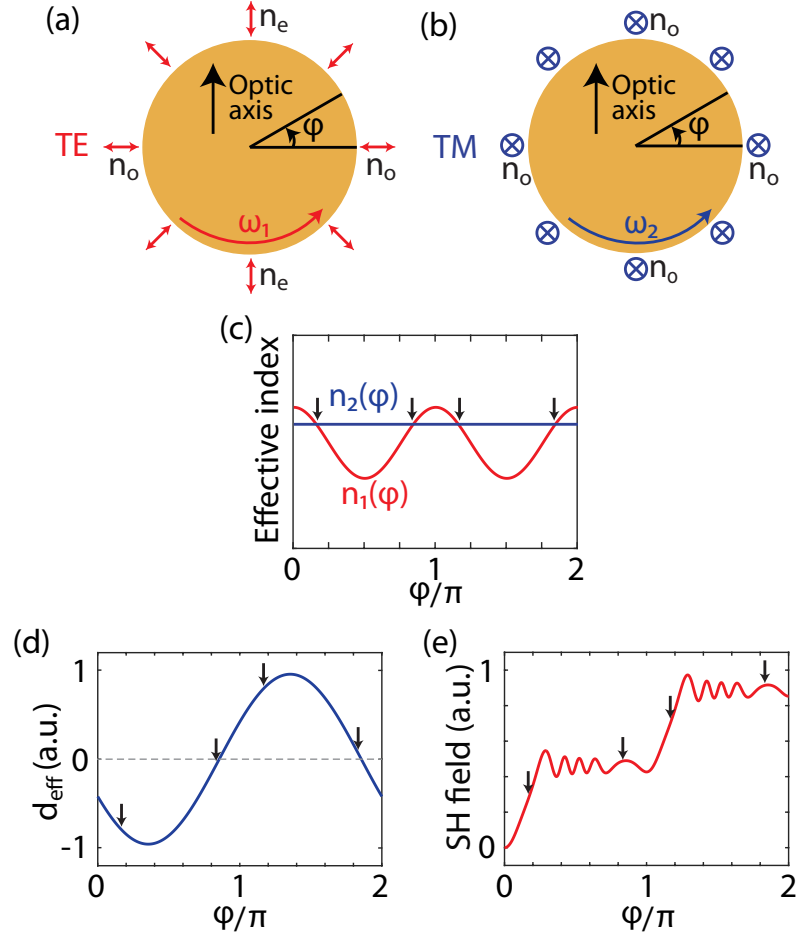


Figure 6.1: Illustration of SHG through cyclic phase matching in an anisotropic optical cavity. Angle-dependent polarization state of (a) a quasi-TE mode at ω_1 , and (b) a quasi-TM mode at $\omega_2 (= 2\omega_1)$. (c) Angle-dependent effective indices of the two modes, with black arrows indicating local phase matching. (d) Effective nonlinear coefficient as a function of the azimuthal angle. (e) SH field amplitude as a function of the azimuthal angle, showing nonvanished build-up around phase-matched angles.

by the material birefringence, as shown in Fig. 6.1(c). In contrast, a quasi-TM mode keeps experiencing n_o [see Fig. 6.1(b)], thus it exhibits a constant effective index, as shown in Fig. 6.1(c). As a result, SHG can take

place through a special phase-matching scheme called cyclic phase matching [82, 137, 139], with perfect phase matching between a quasi-TE mode and a quasi-TM mode satisfied at four azimuthal angles of the microresonator, as shown Fig. 6.1(c). In the case of an X -cut LN microresonator, for doubly resonant modes with $\omega_2 = 2\omega_1$, the effective index of the quasi-TE FF mode can be calculated by

$$\frac{1}{n_1(\phi)^2} = \frac{\cos^2(\phi)}{n_{1o}^2} + \frac{\sin^2(\phi)}{n_{1e}^2}, \quad (6.1)$$

where n_{1o} and n_{1e} are the local effective indices of the FF mode at $\phi = 0$ and $\phi = \pi/2$, respectively. Also, the nonlinearity is dependent on the azimuthal angle, written as [145]

$$d_{\text{eff}}(\phi) = d_{31}\sin(\phi) - d_{22}\cos(\phi), \quad (6.2)$$

with $d_{33} = -4.3$ pm/V and $d_{22} = 2.1$ pm/V [18]. SHG in the microresonator can be modeled by unfolding the cavity into a waveguide, with an expression similar to Eq. (3.17), while considering a modulated nonlinearity and phase mismatch [82, 137, 139], i.e.,

$$\frac{1}{R} \frac{db_2}{d\phi} = ig(\phi)b_1^2 e^{-i \int_0^\phi \Delta\beta(\phi') R d\phi'}, \quad (6.3)$$

where the angle-dependent coupling strength is written as

$$g(\phi) = \sqrt{\frac{2\omega_1^2}{\epsilon_0 c^3 [n_1(\phi)]^2 n_2}} \frac{d_{\text{eff}}(\phi) \zeta}{\sqrt{A_{\text{eff}}}}. \quad (6.4)$$

As shown in Eq. (6.3), the nonlinear optical interaction has contributions mainly from stationary phases, where $\Delta\beta = 0$, or $n_2 = n_1$. As shown in Fig. 6.1(e), the SH field builds up near locally phase-matched angles, with the growth proportional to the local effective nonlinear coefficient [see Fig. 6.1(d)], while oscillating in phase-mismatched regions with no net growth. In one round trip, the SH field experiences nonvanished gain, which will be repeated for doubly resonant modes during the photon lifetime, leading to optical frequency doubling at the cavity output.

Things become more interesting for SPDC. Figure 6.2 illustrates SPDC through cyclic phase matching. As shown in Fig. 6.2(a), for a quasi-TE mode at ω_1 and a quasi-TM mode at ω_3 , in the degenerate case where the two down-converted photons are identical, the phase-matching condition is exactly the same as that in an SHG process [see Fig. 6.1], and a single tone will be generated in the down-conversion spectrum. In a nondegenerate case shown in Fig. 6.2(b), the two down-converted photons have different frequencies, with ω_1 in a quasi-TE mode and ω_2 in a quasi-TM mode for simple illustration. The phase-matching condition, or momentum conversation, now involves three optical frequencies, in the form of $n_3\omega_3 = n_1\omega_1 + n_2\omega_2$, and it can still be easily satisfied at four azimuthal angles, resulting in two tones

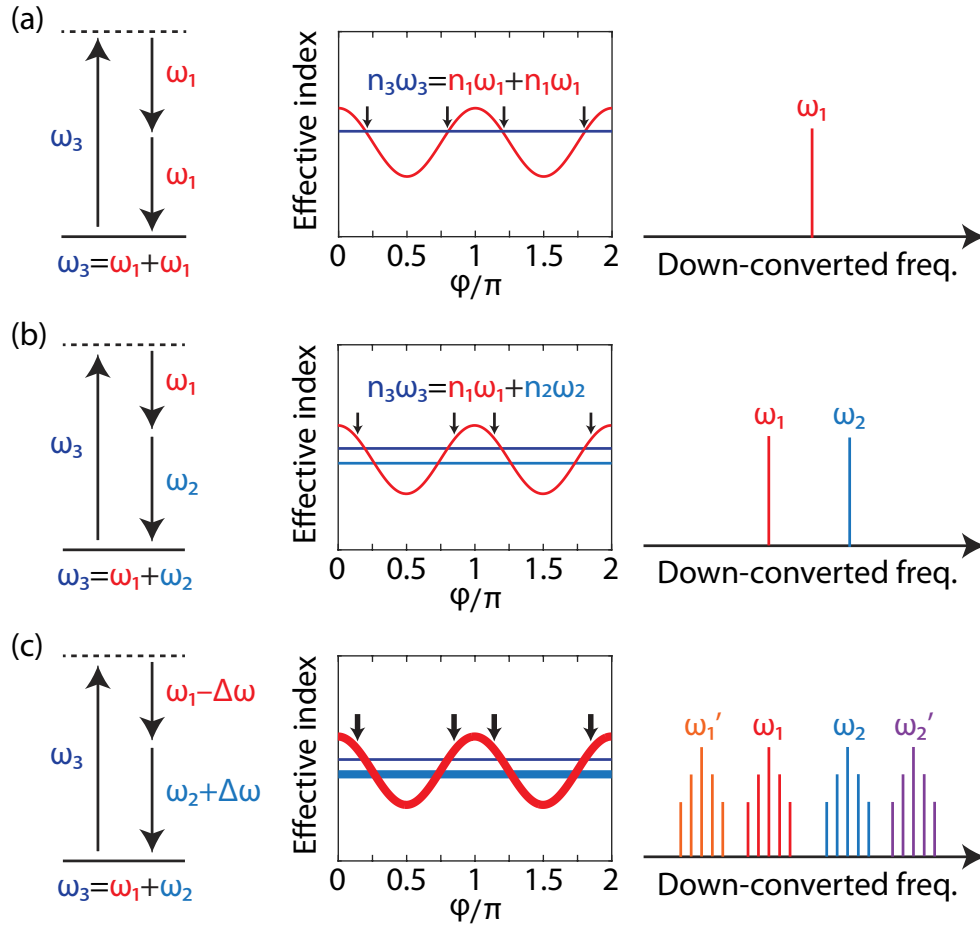


Figure 6.2: Illustration of SPDC through cyclic phase matching, in the case of (a) degenerate SPDC, (b) nondegenerated SPDC with one pair of down-converted modes, and (c) nondegenerated SPDC with multiple pairs of down-converted modes. Black arrows in the effective index plots indicate phase-matched azimuthal angles. In (c), emission lines around ω_1 , ω_2 , ω_1' and ω_2' correspond to cavity modes in four different mode families.

in the down-conversion spectrum. Actually, as shown in Fig. 6.2(c), the nondegenerate phase-matching condition is so flexible that it can be satisfied by modes with other azimuthal mode orders, whose effective indices make

quasi-continua, represented by thick curves in the middle plot of Fig. 6.2(c). As a result, sub-combs can be generated around ω_1 and ω_2 . Furthermore, phase matching can even be satisfied by modes in a second pair of mode families around two new frequencies, ω'_1 and ω'_2 , resulting in a broadband SPDC spectrum, as shown in the right plot of Fig. 6.2(c). For simplicity, only one polarization combination of interacting waves has been discussed. Cyclic phase matching can also work if two or three of the interacting waves are quasi-TE modes, adding more flexibility to dispersion engineering.

6.2 Device Characterization and Second-Harmonic Generation

In order to demonstrate broadband SPDC, an *X*-cut LN microdisk resonator is employed, which exhibits rich mode species. As shown in Fig. 6.3(a), the microdisk has a radius of 45 μm and a thickness of 300 nm, sitting on a 2- μm -high silica pedestal. The device exhibits very low optical losses, with two typical examples presented in Fig. 6.3(b)(c), showing intrinsic optical Q 's of 1.7×10^6 in the telecom band and 6.4×10^5 in the NIR.

The experimental setup is similar to that in SHG experiments discussed in previous chapters (see Figs. 3.5 and 5.2), while here light is launched into the suspended LN microdisk by evanescent coupling via a tapered optical fiber, instead of butt coupling by lensed fibers. In the *X*-cut LN microdisk, a cavity mode at 1549.32 nm in the telecom band is selected as

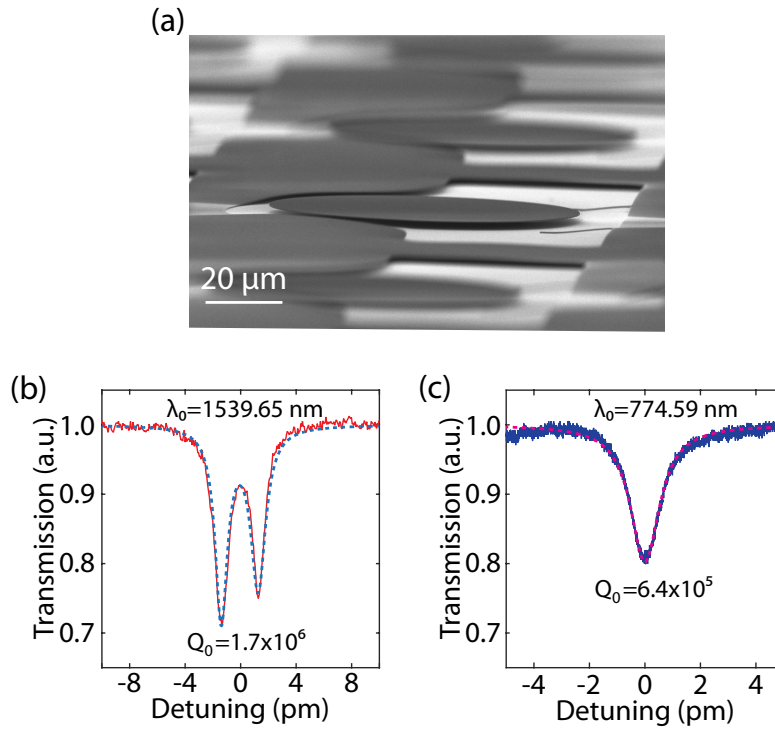


Figure 6.3: (a) Scanning electron microscopy image of a fabricated X-cut LN microdisk. (b),(c) Transmission spectra of typical high- Q cavity resonances of an employed microdisk in the telecom and NIR bands, respectively. Both modes are quasi-TM polarized. Experimental data are shown in solid curves, and theoretical fittings are shown in dashed curves.

the pump mode, which is quasi-TE polarized. The pump mode exhibits a loaded optical Q of 1.2×10^5 , with its transmission spectrum shown in Fig. 6.4(a)(c). When optical power is launched into this mode, coherent radiation is produced at the SH whose emission spectrum is shown in Fig. 6.5(a). Optical mode characterization of the passive cavity around the SH frequency [see Fig. 6.4(b)(d)] shows that the SH corresponds to a quasi-TM mode at 774.66 nm with a loaded optical Q of 2.2×10^5 . Figure 6.5(b)

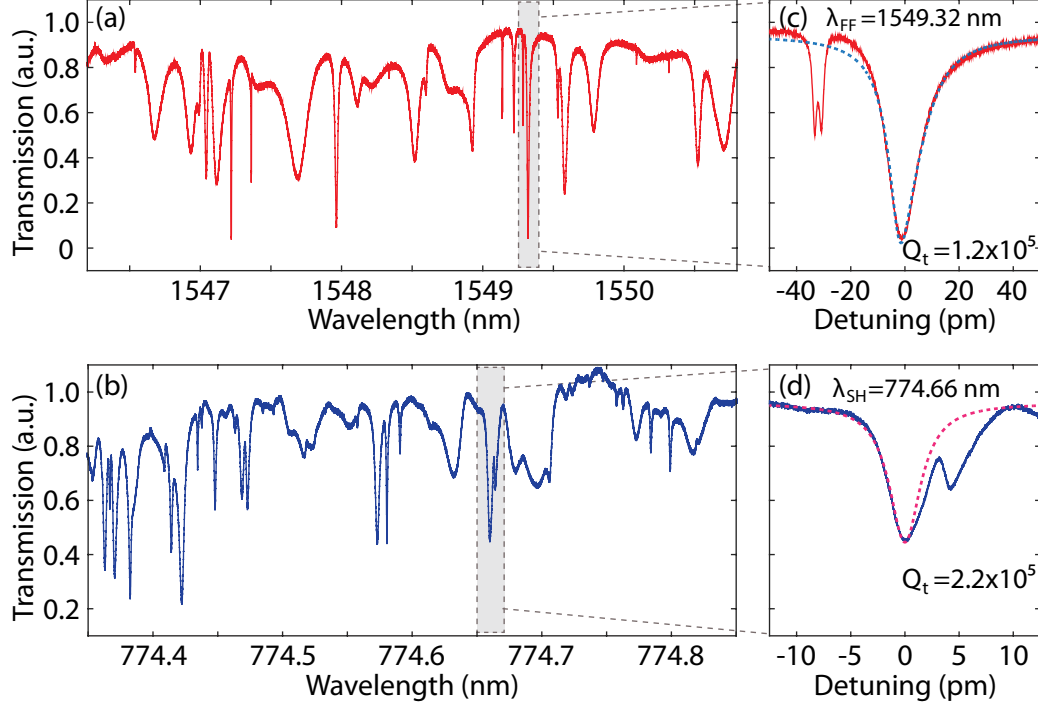


Figure 6.4: Optical mode characterization of an *X*-cut LN microdisk. (a) Transmission spectrum of the device in the telecom band, for the TE polarization. (b) Transmission spectrum of the device in the NIR band, for the TM polarization. (c) Detailed transmission spectrum of the FF mode used for SHG. (d) Detailed transmission spectrum of the SH mode. In (c) and (d), experimental data are shown in solid curves, and theoretical fittings are shown in dashed curves.

plots the power dependence of the SHG signal, exhibiting a clear quadratic relation, which further confirms that the NIR signal is produced by SHG originating from the $\chi^{(2)}$ nonlinearity. From Fig. 6.5(b), an SHG efficiency of $0.36\% \text{ W}^{-1}$ is obtained. This value is lower than those obtained in other cyclic-phase-matched LN microresonators [82], due to the mode overlap, the phase-matching condition, and the light extraction that are not optimized

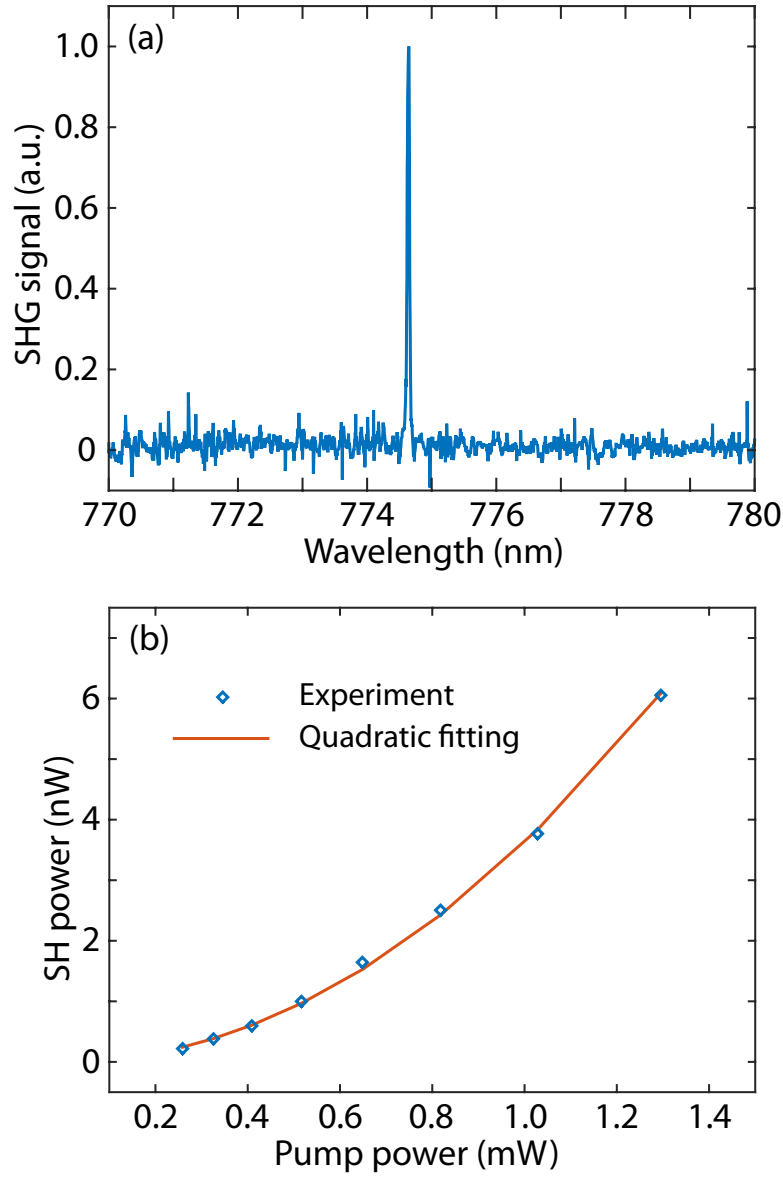


Figure 6.5: Measurement of SHG from an *X*-cut LN microdisk. (a) Recorded spectrum of the SHG signal in the NIR, produced by pump light in the telecom band. (b) Power dependence of the SHG signal on the FF pump.

in this proof-of-concept device. In fact, in this experiment, both modes are high-order radial modes, due to the low effective index of the tapered fiber that makes it difficult to effectively couple higher-index cavity modes. To increase the conversion efficiency, SHG can be designed to occur between two target modes in a waveguide-coupled microring resonator, similar to what is discussed in chapter 5.

6.3 Spontaneous Parametric Down-Conversion

The demonstration of SHG readily implies the potential of the device for SPDC, which forms the foundation for photonic quantum information processing [25, 26, 146–149]. To explore the characteristics of SPDC, an experiment is conducted, with the setup schematically shown in Fig. 6.6. Pump light in the NIR is launched into the cavity mode at 774.66 nm. The down-converted light coupled out of the microdisk by the tapered fiber is separated from the pump by a 780/1550 WDM and a long-pass filter (with a cutoff wavelength of 1100 nm), and is then sent to a spectrometer for spectral analysis. Figure 6.7 shows the recorded photoluminescence (PL) spectrum of the down-converted signal. The SPDC spectrum exhibits a large number of emission lines over a broad spectral range, spanning from below 1400 nm to nearly 1600 nm.

The physical reason for the broadband SPDC lies in the intrinsic nature of energy conservation among interacting photons, since a photon at frequency

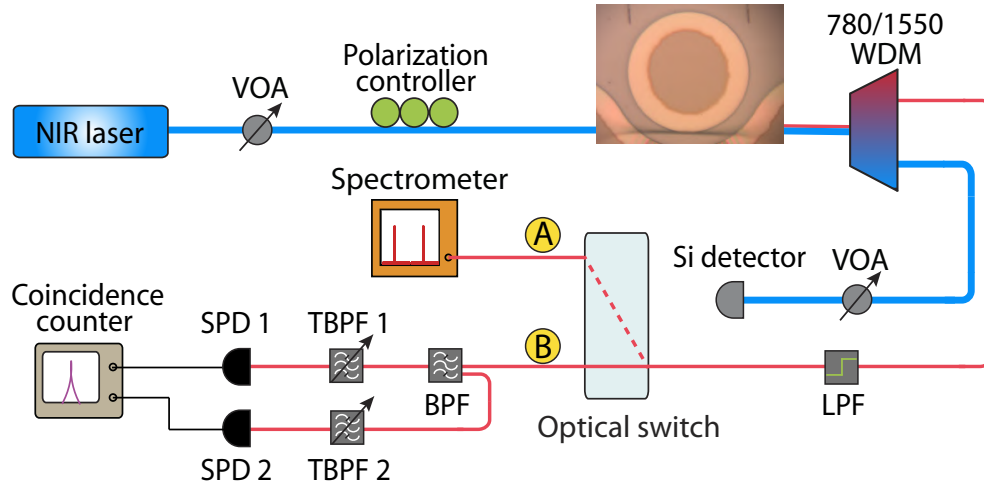


Figure 6.6: Schematic of the experimental setup for SPDC. VOA: variable optical attenuator; WDM: wavelength-division multiplexer; OSA: optical spectrum analyzer; LPF: long-pass filter; BPF: band-pass filter; TBPF: tunable band-pass filter. SPD: single-photon detector.

2ω can be converted into two photons at frequencies $\omega + \Delta\omega$ and $\omega - \Delta\omega$, where $\Delta\omega$ can be nonzero as long as the phase-matching condition is satisfied. This is particularly the case in the *X*-cut LN microdisk that exhibits rich mode species and cyclic phase matching, as illustrated in Fig. 6.2. Therefore, the device is able to induce nonlinear optical interactions between the NIR pump mode and many telecom-band mode pairs from multiple mode families, resulting in SPDC with a remarkably large bandwidth. This phenomenon is in contrast to the generally narrow-band SPDC supported by a *Z*-cut LN optical resonator [148, 150].

The detected SPDC spectrum spans from below 1400 nm all the way up to 1590 nm, the cutoff wavelength of the spectrometer. Because of energy conservation, a signal photon detected around 1370 nm implies an idler

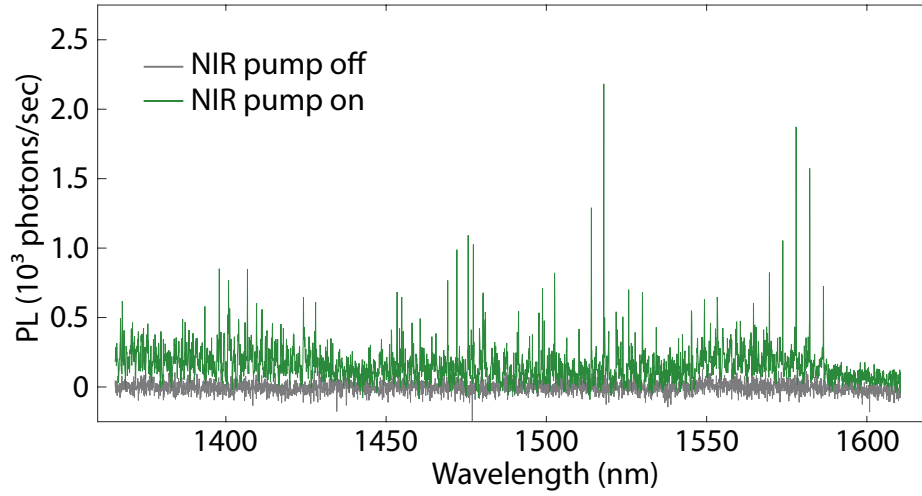


Figure 6.7: Broadband spectrum of SPDC, generated by a pump wave at 774.66 nm with an on-chip power of $115 \mu\text{W}$. The spectrometer used for recording the spectrum has a cutoff wavelength around 1590 nm.

photon produced around 1783 nm (which is beyond the spectral response of the spectrometer). This infers a total SPDC bandwidth of about 400 nm. Such a large bandwidth of SPDC is significantly beyond those obtained by conventional phase-matching approaches that generally exhibit bandwidths of about tens of nanometers [140, 142, 149, 151, 152]. Only chirped nonlinear gratings [153] are able to provide an SPDC bandwidth comparable to that demonstrated by the current device.

To study the generated conjugate photon pairs, the down-conversion spectrum is investigated in detail. Figure 6.8(a) shows a section of the spectrum with a total bandwidth of about 100 nm around the FF mode at 1549.32 nm, which corresponds to half of the pump frequency. Two groups of equally spaced strong PL peaks (indicated by blue and red arrows) are located

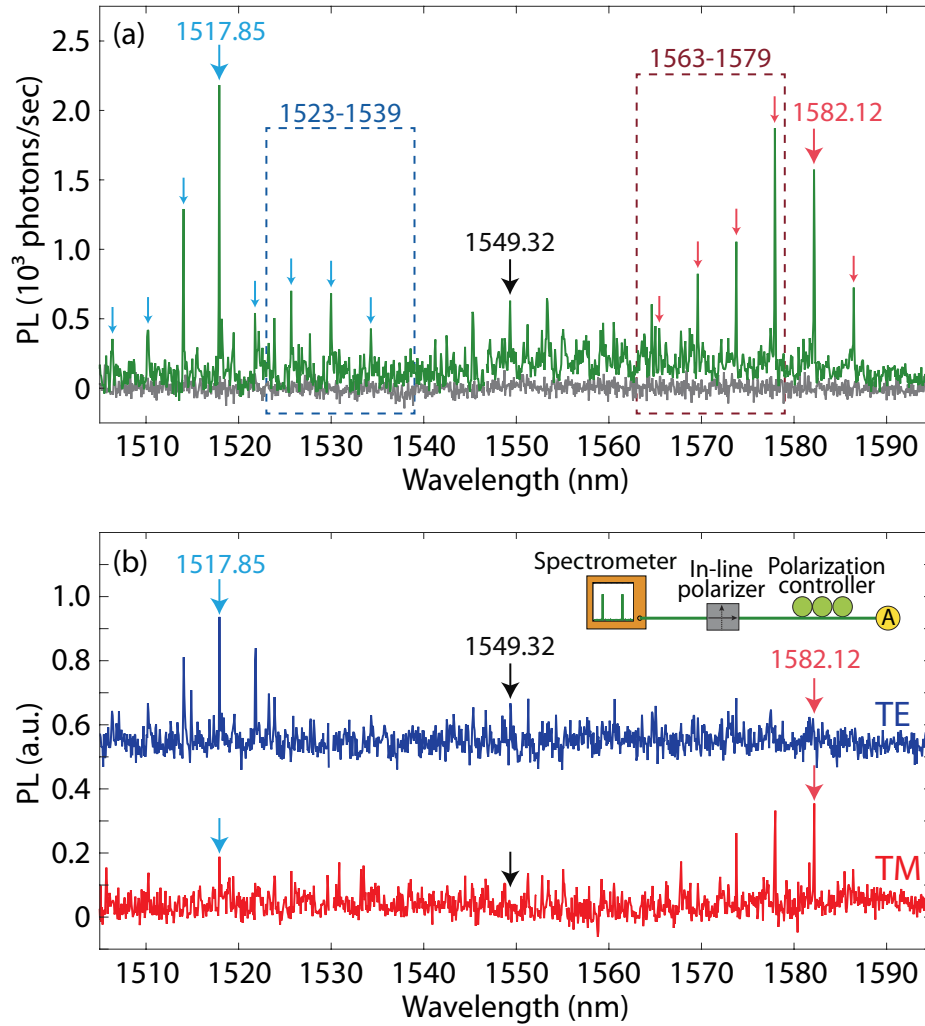


Figure 6.8: (a) Detailed SPDC spectrum in the wavelength range of 1505-1595 nm, showing multiple pairs of emission lines symmetrically located around the degenerate SPDC signal at 1549.32 nm. Blue and red arrows highlight two conjugate mode families. Two big arrows indicate a pair of strong peaks, one at 1517.85 nm and the other at 1582.12 nm. Blue and red dashed boxes indicate two coarse WDM channels. (b) PL spectra for the TE and TM polarizations, with the spectrum for the TE polarization shifted along the vertical axis for a better comparison. The inset shows the experimental setup for polarization analysis, from point A in Fig. 6.6.

symmetrically around the FF mode, with many small peaks covering the whole wavelength range due to the dense modes of the microdisk. Detailed polarization analysis shows that the two groups of strong peaks originate from two mode families with different polarizations. As shown in Fig. 6.8(b), the strong peaks at wavelengths below 1549.32 nm belong to a quasi-TE mode family which shares the same polarization with the FF mode at 1549.32 nm. In contrast, those with wavelengths longer than 1549.32 nm come from a quasi-TM mode family that has the same polarization as the pump mode at 774.66 nm. This indicates that SPDC involving these two telecom mode families is a type-II process, with a quasi-TM NIR pump photon producing a pair of photons exhibiting orthogonal polarizations.

During an SPDC event, two down-converted photons are simultaneously created and thus exhibit strong temporal correlation. To investigate such correlation, coincidence counting experiments are set up, as schematically shown in Fig. 6.6. The down-converted photons are directed to two InGaAs single-photon detectors (SPDs), whose output signals are analyzed to obtain the temporal correlation of the generated photon pairs. First, two strong emission lines at wavelengths of 1517.85 and 1582.12 nm [see Fig. 6.8(a)] are selected, whose sum frequency is equal to the pump frequency, indicating that they are a pair of signal and idler photon modes produced via SPDC. As shown in Fig. 6.9(a), this pair of photon modes are separated by a band-pass filter (BPF), after which one photon mode is directed to the through port (with a pass-band of 1503-1519 nm) and the other to the reflection port.

The two separated photon modes are further filtered by a pair of tunable band-pass filters (TBPFs) before they reach the SPDs.

Figure 6.9(b) plots the coincidence counting result, which clearly shows the dominant true coincidence of the signal and idler photons that overlap in time. The measured coincidence-to-accidental ratio (CAR) is 20.1, which is comparable to those obtained in other state-of-the-art nanophotonic devices [154], demonstrating the strong temporal correlation of the produced photon pairs. The current CAR value is primarily limited by accidental coincidence counts [see Fig. 6.9(b), inset], in which detector dark counts play a fundamental role, since the generated photon pairs exhibit a relatively low flux. Future reduction of detector dark counts (e.g., with superconducting SPDs) and improvement in the nonlinear conversion efficiency of the device is expected to significantly improve the CAR.

Moreover, coincidence counting is also performed for a set of multiple mode pairs to further study the property of the broadband SPDC. As shown in Fig. 6.9(c), the down-converted photons are separated by a coarse WDM into two wavebands, one centered at 1531 nm and the other at 1571 nm [see dashed boxes in Fig. 6.8(a)], and each of them has a bandwidth of 16 nm covering multiple emission lines. The two bands of signal and idler modes are then directed to the SPDs for photon counting. Figure 6.9(d) shows the result, which, again, exhibits a dominant true coincidence when the time separation is zero. The measured CAR is 43.1, showing the strong correlation of photon pairs within the two selected wavebands.

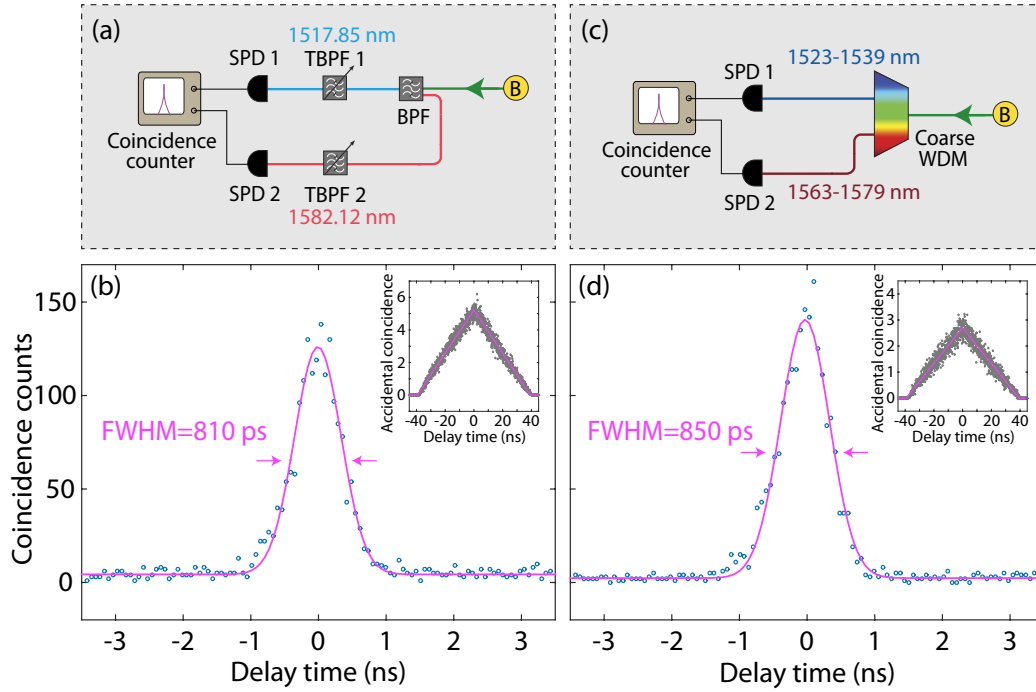


Figure 6.9: Temporal correlation analysis of the conjugate photon pairs. (a) Experimental setup, and (b) coincidence counts as a function of relative time delay, for a single mode pair at 1517.85 and 1582.12 nm. Experimental data are shown in blue circles and a Gaussian fitting is plotted in a magenta curve. The single photons are detected by two InGaAs SPDs, with a gated time window of 40 ns, a gate frequency of 2.5 MHz, a quantum efficiency of 15%, and a data acquisition time of 6 hours. No background subtraction is performed. The FWHM of over 800 ps is due to the detector timing jitter. (c) Experimental setup, and (d) coincidence counts as a function of relative time delay, for broadband multiple mode pairs. The two selected wavebands have an identical bandwidth of 16 nm, with one centered at 1531 nm and the other at 1571 nm. In (a) and (c), SPDC signals are from point B shown in Fig. 6.6. The insets of (b) and (d) present accidental coincidence counts as functions of relative time delay, with experimental data shown in gray dots and theoretical fittings in magenta curves.

The CAR value of SPDC involving multiple mode pairs is higher than that obtained from the single mode pair, resulting mainly from two factors. First, the multiple mode pairs exhibit a higher overall flux of correlated photon pairs than the single mode pair, which leads to a larger number of true coincidence counts [see Fig. 6.9(b)(d)]. Second, the recorded accidental coincidence counts are fewer when filtering is achieved by the coarse WDM [see Fig. 6.9(b)(d), insets]. In the case of a single mode pair, the TBPFs used in the experiment are Fabry-Perot based, with a free-spectral range (FSR) of 60 nm and a linewidth of 1.4 nm. In the broadband SPDC signal at the reflection port of the BPF, besides the target idler photons at $\lambda_{idler} = 1582.12$ nm that matches the center of one pass-band of TBPF 2, some unwanted photons (at wavelengths around $\lambda_j = \lambda_{idler} + j \times \text{FSR}$, where $j = \pm 1, \pm 2, \dots$) also went through TBPF 2 and reached SPD 2, making an extra contribution to accidental coincidence counts. In fact, for the single mode pair, these extra accidental coincidence counts come from the broadband SPDC, and can be easily suppressed by a filter with a smaller linewidth to further increase the CAR. These measured CAR values directly confirm that the broadband photon pairs observed in Figs. 6.7 and 6.8 are indeed produced by SPDC.

6.4 Conclusion

In conclusion, SHG and SPDC are demonstrated in a high- Q X -cut LN microdisk resonator through cyclic phase matching. The SPDC spectrum

exhibits an extremely large bandwidth of up to 400 nm in the telecom band, which is significantly beyond those obtained in conventional devices. Coincidence photon counting is performed to characterize the intrinsic temporal correlation between the down-converted signal and idler mode pairs, with a CAR of 43.1 that directly confirms the physical origin of the down-converted photon pairs. The ultra-broadband SPDC observed in the device is promising for integrated quantum photonic applications that could utilize frequency as a degree of freedom for information coding and processing.

In chapters 3-6, discussions are focused on quadratic nonlinear optics in LN nanophotonic devices induced by $\chi^{(2)}$. In fact, LN also has a large cubic nonlinear susceptibility $\chi^{(3)}$. In the next chapter, Kerr frequency comb generation induced by $\chi^{(3)}$ is demonstrated in an LN microring resonator.

Chapter 7

Kerr Frequency Comb Generation in an LN Microring

On-chip generation of optical frequency combs via the optical Kerr nonlinearity has attracted enormous attention in the recent decade [17, 67, 155–175]. Kerr frequency combs not only exhibit interesting nonlinear optical dynamics and physics [161, 164, 166, 168, 170], but also have potential for a large number of applications such as frequency metrology [160], optical frequency synthesis [174], precision spectroscopy [67, 173], photonic signal processing [157], optical telecommunications [169], and range measurement [171, 172]. To date, Kerr frequency combs have been produced on a variety of device platforms, from silica [155, 164], silicon nitride [156, 157, 166, 167], silicon [163], aluminum nitride [17, 175], diamond [162], to aluminum gallium arsenide [165].

LN exhibits outstanding material properties [6, 7, 12, 176], and has attracted considerable interest in the past few years to the development of electro-optic and nonlinear photonic functionalities on chip-scale platforms [28–38, 74, 82–87, 91, 92, 107, 129, 177–185]. Although attention in nonlinear

optics using LN has been focused on the quadratic nonlinearity $\chi^{(2)}$, LN also exhibits a large Kerr nonlinearity n_2 , originating from $\chi^{(3)}$. Particularly, by harnessing the material birefringence and the waveguide geometry, flexible engineering of group-velocity dispersion (GVD) can be achieved in high- Q LN microring resonators [186], constituting a crucial foundation for Kerr frequency comb generation in such devices.

In this chapter, Kerr frequency comb generation is demonstrated in a high- Q LN microring resonator. The Kerr frequency comb is produced by an on-chip pump power as low as 33 mW, and the comb spectrum spans over 200 nm around the telecom band. In addition, an up-converted comb in the NIR is also observed, resulting from the quadratic optical nonlinearity. The demonstration of Kerr frequency comb generation represents an important step toward a large variety of comb applications on the LN integrated photonic platform.

7.1 Kerr Effect in Cavities

The dynamics of an optical wave inside a driven cavity with a Kerr nonlinearity is described by the generalized Lugiato-Lefever equation [187–189]

$$\begin{aligned} t_R \frac{\partial E}{\partial t} = & \left(-\frac{\Gamma_t}{2} - i\Delta_0\right)E + \sum_{m=2}^{\infty} \frac{i^{m+1}\beta_m}{m!} \frac{\partial^m E}{\partial \tau^m} \\ & + i\gamma_K L \left(1 + \frac{i}{\omega_l} \frac{\partial}{\partial \tau}\right) (|E|^2 E) + \sqrt{\Gamma_e} A_0, \end{aligned} \quad (7.1)$$

where $E(t, \tau)$ is the amplitude of the intracavity field, with t and τ representing the slow and fast times, respectively. A_0 is the field amplitude of the input CW pump launched at a frequency of ω_l , with a laser-cavity detuning of $\Delta_0 \approx (\omega_0 - \omega_l)t_R$, where ω_0 is the center frequency of the cavity resonance closest to the pump field and t_R is the round-trip time. The small influence of the nonzero GVD on Δ_0 has been neglected. The cavity has a round-trip length of L , and a nonlinear parameter of $\gamma_K = \omega_0 n_2 / (c A'_{\text{eff}})$, where n_2 and A'_{eff} are the Kerr nonlinear coefficient and the effective mode area, respectively, and c is the speed of light in vacuum. To describe the optical wave with a broadband spectrum, Eq. (7.1) includes the self-steepening effect and high-order dispersion with β_m denoting the m -th order dispersion coefficient at the pump mode. $\Gamma_t = \Gamma_0 + \Gamma_e$ represents the total power loss per round trip, with Γ_0 and Γ_e describing the intrinsic loss and the external coupling loss, respectively.

When the spectrum is much narrower than octave-spanning, self-steepening and high-order dispersion are not important, and Eq. (7.1) can be simplified to

$$t_R \frac{\partial E}{\partial t} = \left(-\frac{\Gamma_t}{2} - i\Delta_0\right)E - \frac{i\beta_2}{2} \frac{\partial^2 E}{\partial \tau^2} + i\gamma_K L |E|^2 E + \sqrt{\Gamma_e} A_0. \quad (7.2)$$

From Eq. (7.2), a physical picture behind Kerr comb generation can be clearly seen. The dispersion term $-\frac{i\beta_2}{2} \frac{\partial^2 E}{\partial \tau^2}$ is proportional to $\frac{i\beta_2}{2} \omega^2 E$, which has a similar form to the nonlinear term $i\gamma_K L |E|^2 E$. In a nontrivial solution

of a temporal pulse, the phase matching of a four-wave mixing process requires the dispersion term, $-\frac{i\beta_2}{2}\frac{\partial^2 E}{\partial \tau^2}$, to compensate for the nonlinear phase introduced by self-/cross-phase modulation, described by $i\gamma_K L|E|^2 E$. Since most materials have a positive Kerr nonlinearity, i.e., $n_2 > 0$, Eq. (7.2) requires a negative β_2 , or an anomalous GVD, for Kerr frequency comb generation based on cascaded four-wave mixing. Furthermore, with selected physical parameters that enable phase locking, Eq. (7.2) can describe the generation of cavity solitons, which stay in the form of solitary waves upon propagation [189, 190].

7.2 Device and Characterization

As discussed above, in order for Kerr frequency comb generation in a common medium with a positive Kerr nonlinearity, it is required to have an anomalous GVD, which is the opposite of the nature of most optical materials. Fortunately, nanophotonic devices offer a large flexibility in the geometry for dispersion engineering, making broadband anomalous GVD achievable in LN high- Q microresonators [186]. Microrings are fabricated on a Z -cut LNOI wafer with a 580-nm-thick LN thin film sitting on 3- μm -thick buried silicon oxide. The cross section is similar to devices discussed in chapters 3 and 5, with an etched thickness of 400 nm and a waveguide top width of 2.0 μm , which are chosen for the desired GVD. The microring has a radius of 100 μm , and is side-coupled to a pulley bus waveguide for

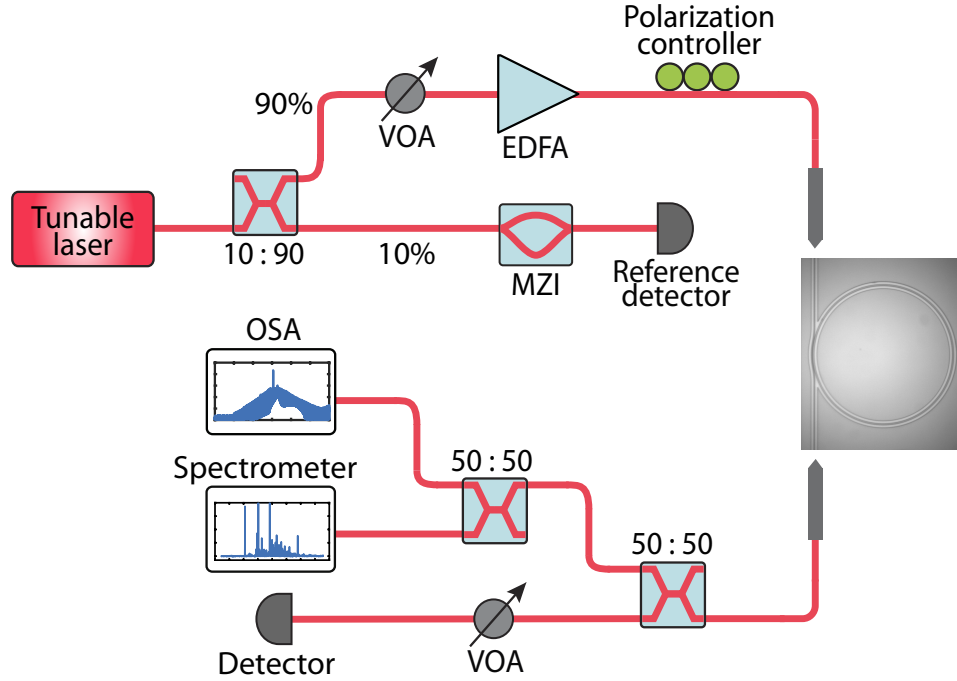


Figure 7.1: Schematic of the experimental setup. The optical microscopy image shows a fabricated LN microring resonator with a radius of $100\ \mu\text{m}$. VOA: variable optical attenuator; MZI: Mach-Zehnder interferometer; EDFA: erbium-doped fiber amplifier; OSA: optical spectrum analyzer.

broadband coupling. The bus waveguide width is about $1.4\ \mu\text{m}$, the coupling length is about $40\ \mu\text{m}$, and the gap (measured at the top surface of the LN thin film) is about $0.7\ \mu\text{m}$, which are chosen for good external coupling. These device parameters are optimized to improve the optical quality and dispersion properties for the purpose of Kerr frequency comb generation, as discussed in the following.

For device characterization and Kerr comb generation, an experimental setup schematically shown in Fig. 7.1 is employed. Optical power from a

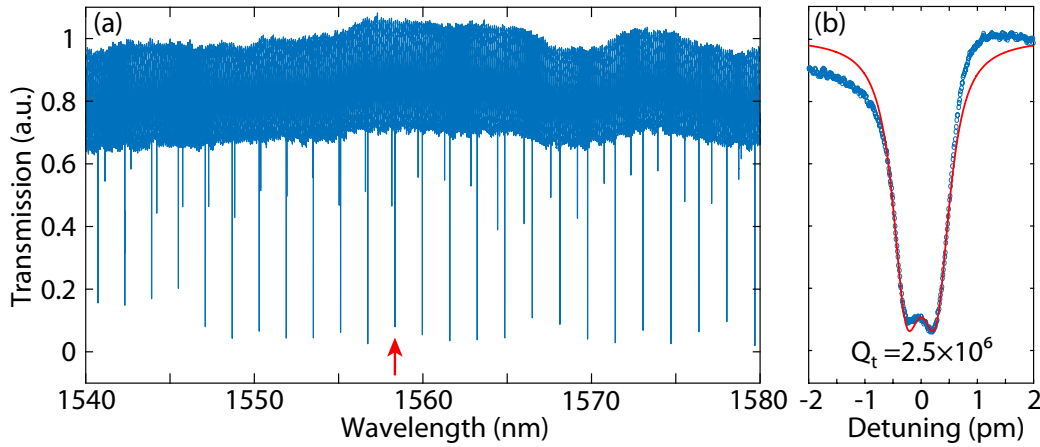


Figure 7.2: (a) Transmission spectrum of an LN microring resonator with a radius of $100\ \mu\text{m}$. (b) Detailed transmission spectrum of a TE_{00} mode at $1558.35\ \text{nm}$, indicated by the red arrow in (a). Blue circles are experimental data, and the red curve is a theoretical fitting.

CW laser is launched via a lensed fiber into an on-chip bus waveguide, which couples the laser power into the microring. The output from the microring is collected by the same waveguide and delivered to a second lensed fiber for detection. The laser wavelength is calibrated by a fiber-based Mach-Zehnder interferometer. To produce the frequency comb, the laser power is boosted by an erbium-doped fiber amplifier before launched to the resonator. The generated frequency comb is recorded by an OSA that covers the telecom band and a spectrometer that covers the visible/NIR spectral region. The fiber-to-chip coupling loss is measured to be about $4.5\ \text{dB/facet}$.

By scanning the laser wavelength, the transmission spectrum is obtained, as shown in Fig. 7.2(a) for the quasi-TE polarization (with the electric field dominantly lying in the device plane). The TE_{00} mode family is clearly

shown, with an FSR of about 1.6 nm (or 200 GHz) and good external coupling over a broad wavelength range. The transmission spectrum shows a background of sinusoidally oscillatory ripples, which is due to the Fabry-Perot effect introduced by back reflection at the polished chip facets. These ripples can be suppressed in the future by optimizing waveguide tapers at facets, to reduce the back reflection. Detailed characterization shows that the TE_{00} mode family exhibits a high optical quality. Figure 7.2(b) presents the detailed transmission spectrum of a TE_{00} mode at 1558.35 nm [indicated by the red arrow in Fig. 7.2(a)], which exhibits a split doublet resonance, with a coupling depth of about 90%. By fitting the doublet resonance, an intrinsic optical Q of 7.0×10^6 and a loaded optical Q of 2.5×10^6 are obtained. The same level of optical Q is observed in the whole TE_{00} mode family over a wide spectral range. This optical quality is considerably higher than the devices discussed in chapters 3 and 5, mainly due to a larger cross section of the current microring that reduces the scattering loss. This optical Q is also comparable to a recent report [91]. Such a high optical Q is ready for Kerr frequency comb generation.

On the other hand, Kerr frequency comb generation depends critically on the GVD of the device, which is generally described by the Taylor expansion of the cavity resonance frequency ω_μ around a reference resonance ω_0 , i.e., $\omega_\mu = \omega_0 + \mu D_1 + \frac{1}{2}\mu^2 D_2 + \frac{1}{6}\mu^3 D_3 + \dots$, where μ is the relative mode number, $D_1/(2\pi)$ represents the FSR around ω_0 , D_2 is related to the GVD β_2 as $D_2 = -\frac{c}{n} D_1^2 \beta_2$, where c is the speed of light in vacuum,

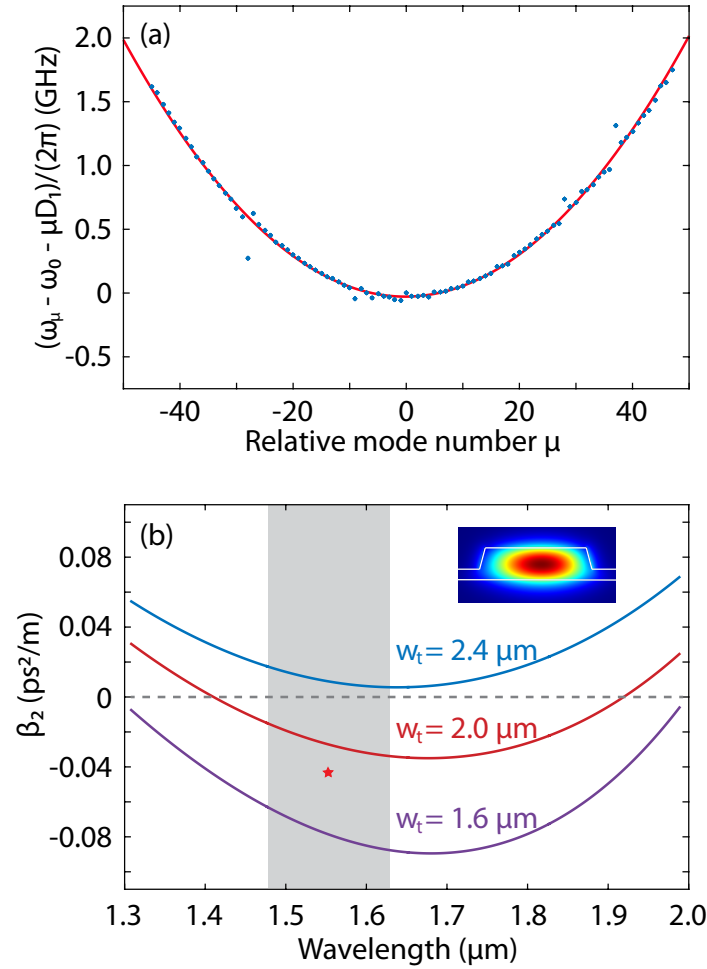


Figure 7.3: (a) Measured dispersion of the TE_{00} mode family as a function of relative mode number for the 2.0-μm-wide microring, with experimental data shown in blue dots and a theoretical fitting shown in a red curve with $D_1/(2\pi) = 199.716$ GHz and $D_2/(2\pi) = 1.623$ MHz, corresponding to $\beta_2 = -0.043$ ps²/m. The mode with $\mu = 0$ is located around 1553.5 nm. (b) Simulated GVD of straight waveguides with $h_1 = 400$ nm, $h_2 = 180$ nm, $\theta = 75^\circ$, and different top widths w_t [see Fig. 5.1(a), inset]. The gray shaded area indicates the wavelength range of the dispersion measurement, and the red star indicates the fitted β_2 in (a). The inset shows the TE_{00} mode profile.

n is the effective refractive index, and D_j ($j \geq 3$) accounts for high-order dispersion [164]. To characterize the GVD of the device, the laser wavelength is scanned across a broad spectral range, from 1480 to 1630 nm, to measure the resonance frequencies of the TE_{00} mode family. Figure 7.3(a) shows the recorded dispersion profile, exhibiting a clear quadratic dependence on the mode number. Dispersion deviations around certain mode numbers are introduced by mode crossing with higher-order cavity modes. By fitting the data, $D_1/(2\pi) = 199.716$ GHz and $D_2/(2\pi) = 1.623$ MHz are obtained, corresponding to an anomalous GVD of $\beta_2 = -0.043$ ps²/m. The experimentally measured GVD [see Fig. 7.3(b), red star] agrees with the expectation by numerical simulations [see Fig. 7.3(b), $w_t = 2.0$ μm], and a slight discrepancy is introduced by fabrication imperfections. This small value of GVD enables a large bandwidth of a generated Kerr frequency comb, as shown in the following.

7.3 Kerr Frequency Comb Generation

To produce a frequency comb, the pump laser is launched into a cavity mode located at 1558.35 nm [indicated by the red arrow in Fig. 7.2(a)]. When the pump power is increased to a certain level, a clear frequency comb is observed. Figure 7.4(a) shows the comb spectrum, with a pump power of about 33 mW in the on-chip bus waveguide. The generated comb spectrum spans over a bandwidth of about 230 nm, from 1450 nm to 1680 nm.

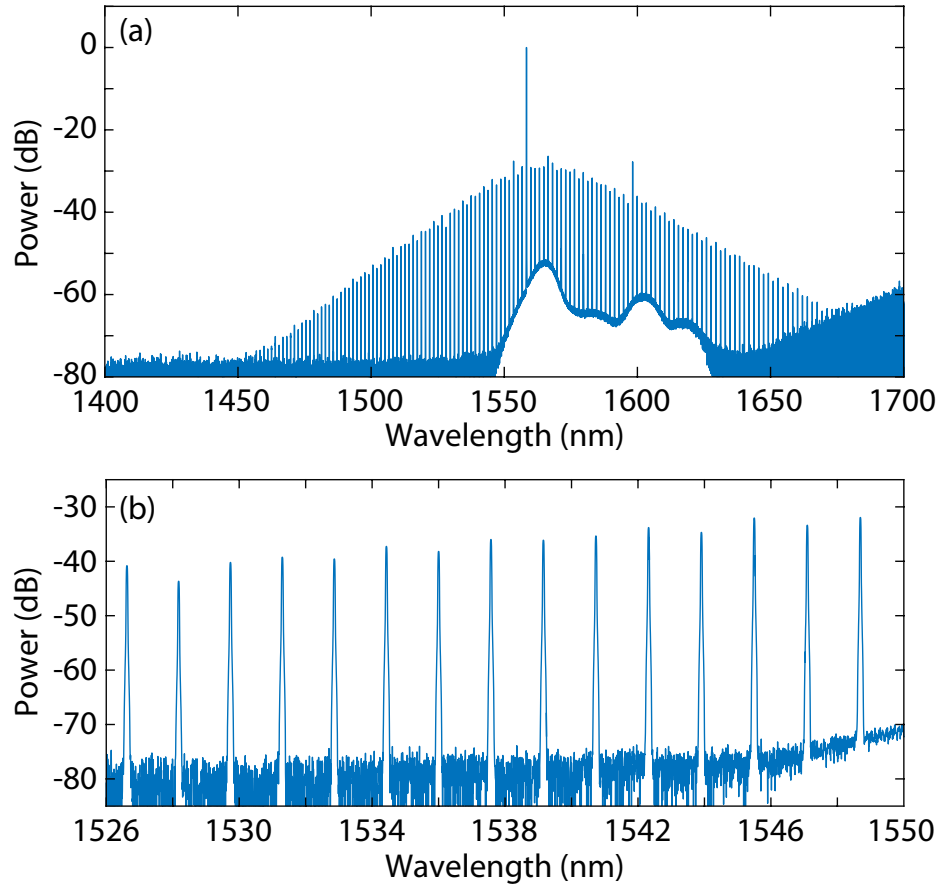


Figure 7.4: (a) Optical spectrum of a Kerr frequency comb by pumping a cavity resonance at 1558.35 nm, with an on-chip pump power of about 33 mW. (b) Zoom-in of the comb spectrum. The spectrum is recorded with a resolution of 0.05 nm.

Interestingly, the spectrum exhibits an envelope close to a sech^2 -function, the signature of cavity solitons. The spectral envelope exhibits a peak at a wavelength longer than the pump mode, implying a potential impact from Raman scattering [166]. Figure 7.4(b) shows a section of the spectrum in detail. The comb lines are equally spaced, with a spacing of about 1.6 nm

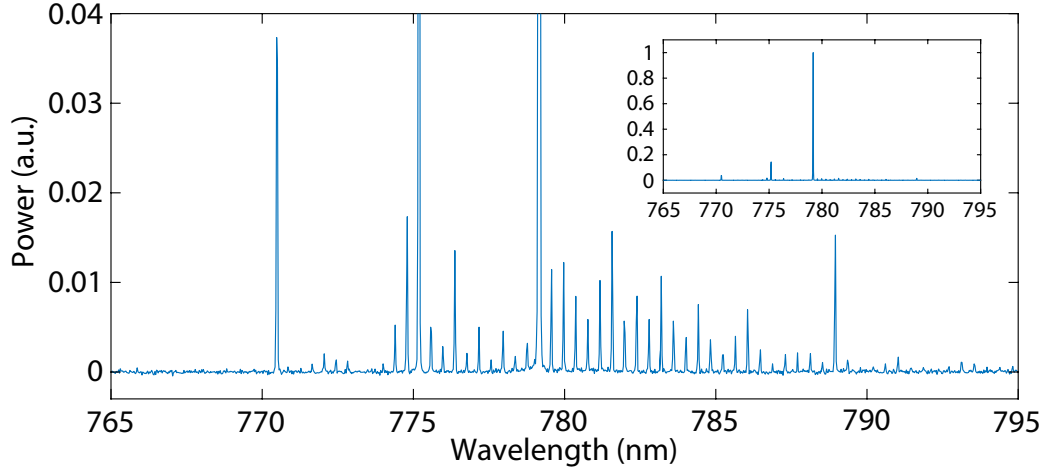


Figure 7.5: Optical spectrum of an up-converted comb around 780 nm. The inset presents the spectrum in a larger scale, showing the relative power between the strong SH of the pump and the weak comb lines nearby.

between adjacent lines, corresponding to the FSR of the TE_{00} mode family. Meanwhile, the presented comb lines have fairly uniform power levels well above the noise floor of the OSA.

To study the dynamics of Kerr frequency comb generation, numerical simulations can be performed based on Eq. (7.1), with the knowledge of the nonlinear parameter γ_K . The value of γ_K can be experimentally obtained by characterizing optical parametric oscillation [164] or supercontinuum generation [191]. Also, under certain conditions, Kerr frequency combs correspond to temporal solitons, whose properties can be characterized by electrical spectrum analyzers and frequency-resolved optical gratings [161]. These studies will be left for future exploration to gain a better understanding of the physical system.

LN exhibits a large quadratic optical nonlinearity, which has the potential for up-conversion of the telecom-band Kerr comb. To explore this potential, a spectrometer that covers the visible/NIR spectral range is employed. As shown in Fig. 7.5, a frequency comb in the range of 770-790 nm is indeed observed. The spacing between adjacent peaks is about 0.4 nm (or 200 GHz), indicating their origin of $\chi^{(2)}$ processes, including SHG and sum-frequency generation from the pump laser and the Kerr frequency comb in the telecom band. The spectral line at 779.17 nm is significantly stronger than other lines nearby, since it corresponds to the SH of the strong pump laser. In general, the observed up-converted comb is fairly weak, mainly due to the unoptimized phase matching in the $\chi^{(2)}$ processes, as well as the poor coupling between the microresonator and the bus waveguide in the NIR, since the device is not specifically designed for this up-conversion purpose. Future optimization of the device dispersion and external coupling is expected to significantly enhance the up-converted comb [175, 192].

7.4 Conclusion

In conclusion, Kerr frequency comb generation is demonstrated in an LN microring resonator. The microring exhibits a loaded optical Q of 2.5 million and an anomalous GVD of $\beta_2 = -0.043 \text{ ps}^2/\text{m}$ in the telecom band. The high optical Q and the small anomalous GVD enable an on-chip pump power of only 33 mW to produce a Kerr frequency comb with a broad spectral range,

from 1450 nm to 1680 nm. In addition, an up-converted frequency comb in the NIR is also observed.

Apart from the cubic optical nonlinearity that is closely related to Kerr frequency combs, LN also exhibits a large quadratic nonlinearity and an outstanding electro-optic effect. By harnessing these optical properties on the same monolithic platform, LN microresonators are expected to live up to the potential in f - $2f$ self-locked frequency metrology, high-resolution spectroscopy, and massively parallel optical telecommunications.

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